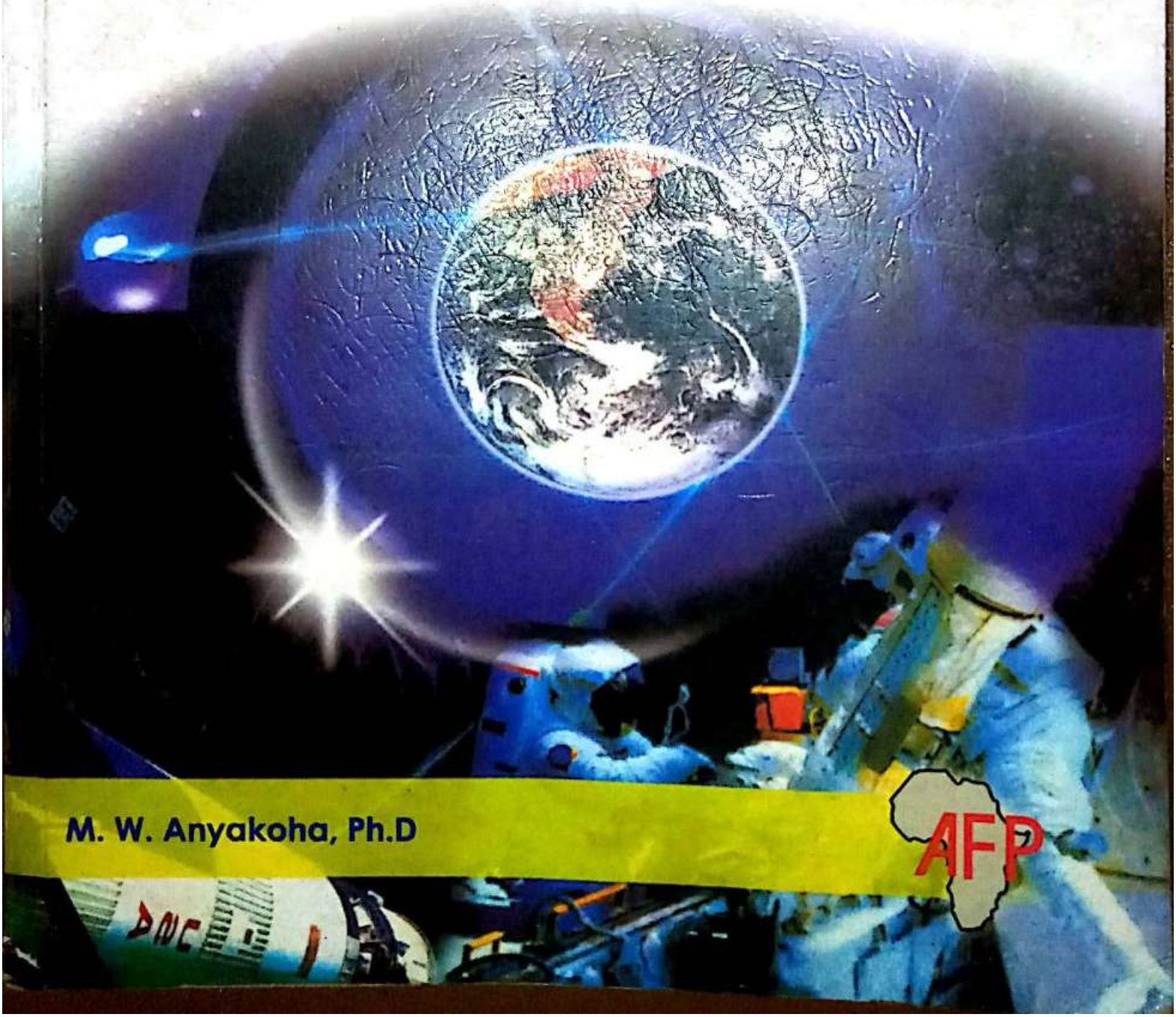
2016 EDITION

NEW SCHOOL PHYSICS

for Senior Secondary Schools

(BASED ON THE NEW NERDC CURRICULUM FOR SENIOR SECONDARY SCHOOLS)



M. W. Anyakoha, Ph.D

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New School PHYSICS

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(BASED ON THE NEW NERDC CURRICULUM FOR SENIOR SECONDARY SCHOOLS)

M. W. ANYAKOHA, Ph.D

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Awka

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PREFACE

New School Physics for Senior Secondary Schools covers the entire current Senior Secondary School Physics curriculum as well as the examination syllabus of the West African Examinations Council (WAEC). It also covers all topics required for the entrance examinations conducted by the Joint Admissions and Matriculation Board (JAMB) for universities, polytechnics and colleges of education.

The book aims at providing a comprehensive and coherent introduction to the principles of Physics. Recognizing that many students have often found Physics a daunting subject, the author has attempted to simplify the treatment of each topic using very simple English. The language level and depth of coverage of each topic reflect the student's year of study.

The text has been organised in three parts corresponding to the three years of the Senior Secondary School Physics curriculum. *Book 1* is for use by students in the first year of the senior secondary school (SS1). *Book 2* is for the second year (SS2) and *Book 3* is for the third and final year (SS3). Each part follows closely the subject sequence required by the Senior Secondary School Physics curriculum. This has been so designed to enhance the teaching/learning process and to ensure comprehensive coverage of the subject.

Each chapter of the text begins with a list of learning objectives which intimate the learner with what to expect from the chapter. The chapter is then developed to cover the contents of each topic. Each chapter also includes some laboratory based student activities to aid the appreciation of the topics by the students. Some of these activities, especially in the third part of the book, are modelled after the Practical Physics Examinations conducted by the WAEC for the Senior School Certificate. Each chapter ends with a *Summary* that focuses on the essential concepts. It recalls the terms that one should be able to define and the relationships one should be able to apply in solving problems. It also facilitates a quick revision of the topics. The *Exercises* following the *Summary* include multiple-choice and essay type questions designed to enable the student to test his comprehension of the topics and to provide exercises in developing problem-solving skills and to guide the student in reviewing the large amount of subject matter. Many of these exercises are actual school certificate and university/college entrance examination questions and are aimed at helping the student to test his/her readiness for these examinations. Answers are provided to multiple-choice questions and numerical problems.

Essential mathematics has been employed throughout the text to clarify the Physics concepts or principles under consideration. The mathematics has however been kept at a level that is comprehensible to the student. Several *worked examples* are given in each chapter to illustrate the applications of the principles discussed and also to illustrate how typical numerical problems are solved.

The text is attractively and highly illustrated with well-labelled diagrams and photographs.

Prof M. W. Anyakoha

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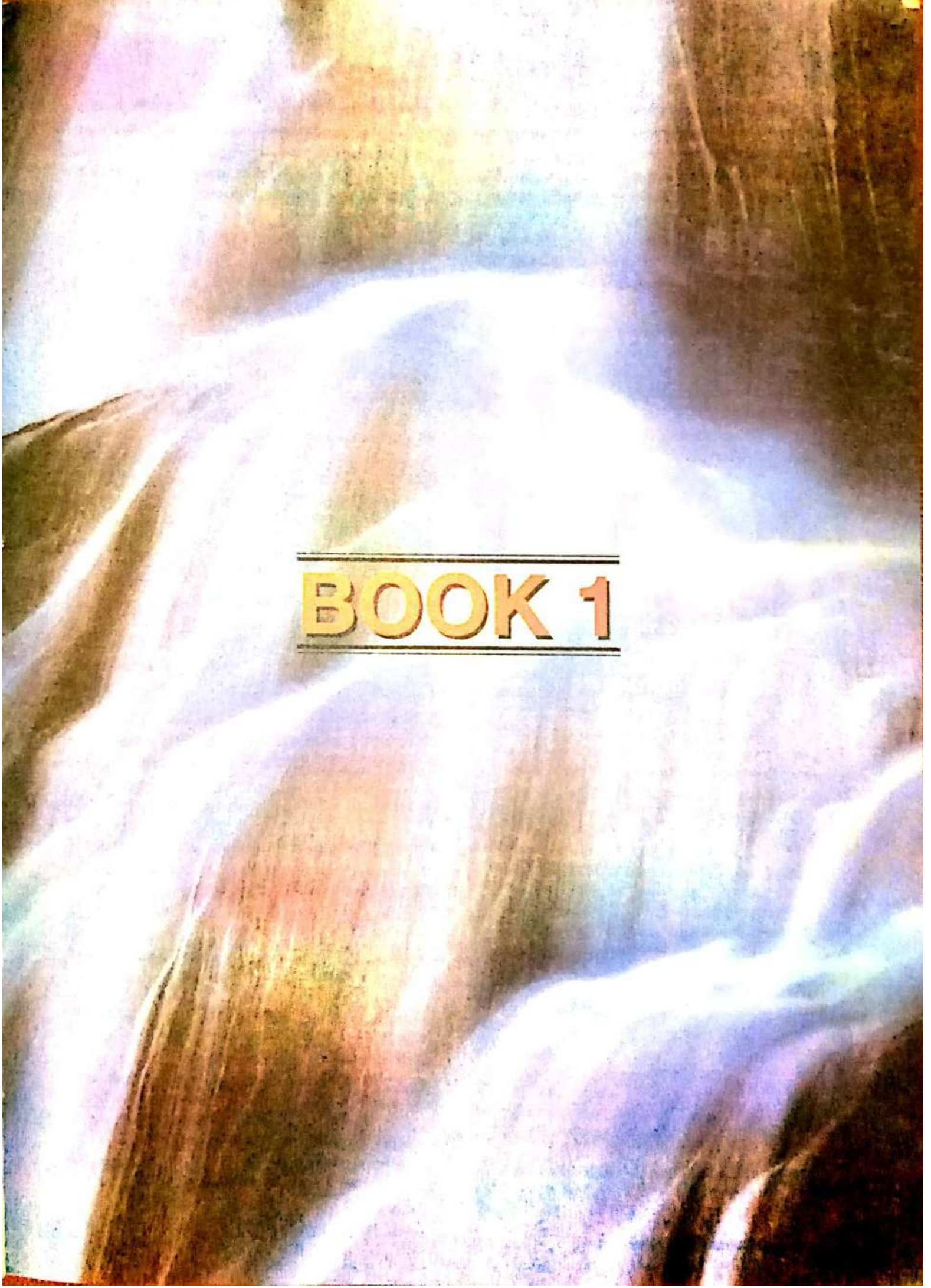
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1 UNITS AND MEASUREMENTS

Measurement is a very important aspect of Physics and other sciences. No fact in science is accepted, no law is established, unless it can be exactly measured and quantified. As Physics is based on exact measurements, every such measurement requires two things: first a number or quantity, and secondly a unit, e.g. 10 metres as the length of a room.

Measurements are basically a means of communication and are used by scientists and engineers for understanding natural phenomenon, by the society for transacting business, and by the engineers for practical ends.

A unit or standard of measurement should be such that (a) it must be possible to define it unambiguously. (b) It is easily reproducible. (c) It does not vary with time and place. (d) It should be possible to multiply or divide each one of the standard.

There are different instruments used by physicists for measuring physical quantities and different units in which these quantities are expressed. There are also relationships between quantities and some uncertainties associated with the measuring devices.

At the end of this chapter the student should be able to:

1 Distinguish between *fundamental* and *derived* quantities and units.
2 List and handle various measuring instruments common in a good senior secondary school physics laboratory, and used for the measurement of length, mass and time.
3 Estimate the uncertainties associated with these instruments.
4 Be familiar with the concept and use of *significant* figures.

1.1 Fundamental and Derived Units

The three most important basic quantities in physics are *length*, *mass* and *time*.

Length may be defined as the extent of space or distance extended.

Mass is commonly defined as the quantity of matter or material substance.

Time is defined as that in which events are distinguishable with reference to before or after.

Length, Mass and Time are three of the most *fundamental quantities*.

Fundamental quantities are the basic quantities that are independent of others and cannot be defined in terms of other quantities or derived from them. They are the basic quantities upon which most (though not all) quantities depend.

Fundamental units are the basic units upon which other units depend. They are the units of the fundamental quantities.

Examples of fundamental quantities and their units are as in Table 1.1

Table 1.1: Fundamental quantities and units

Quantity	Unit	Unit abbreviation
Length	metre	m
Time	second	s
Mass	kilogram	kg
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol.

The units in the above table are in the *SI* (Systeme International) units. The *SI* is the most important system today. It used to be called the MKS (metre-kilogramme-second) system. The *SI* unit is based on the metre as the unit of length, the kilogramme as the unit of mass, the second as the unit of time, ampere as the unit of electric current and the kelvin as the unit of temperature. Another metric system is the c.g.s. system in which the centimetre, gram and second are the standard units of length, mass, and time.

The *SI* units are the principal system of units used in scientific work today.

2. Units used in Industries

Units used in industry and business sometimes differ from those used in laboratory, because, in these areas, the quantities involved are very large. For example, power measured in watts in the laboratory may be measured in horsepower in industry. One horsepower (1hp) equals 0.746 killoWatts. In oil industries, the barrel is used as a unit of measurement; 1 barrel of oil is equal to between 120 and 159 litres. Oil prices are usually quoted in dollars per barrel. An hectare is a unit for measuring the area of land. An hectare (h) equals 10,000 square metres.

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Derived quantities and units

Derived quantities and units are those obtained by some simple combination of the fundamental quantities and units. They are thus dependent on the fundamental quantities and units. Examples of derived quantities , their derivation and their units are summarized in Table 1.2.

Table 1.2: Derived quantities and units

Derived quantity	Derivation	Derived unit
Area (A)	length x breath	m2
Volume (V)	length x breath x height	m3
Density	$\frac{\text{mass}}{\text{volume}}$	kg.m^{-3}
Velocity (v)	$\frac{\text{displacement}}{\text{time}}$	m.s^{-1}
Acceleration (a)	$\frac{\text{change in velocity}}{\text{time}}$	m.s^{-2}

Derived quantity	Derivation	Derived unit
Force (F)	mass × acceleration	Newton, N
Energy or work (W)	force × distance	Joule (Nm)
Power (P)	$\frac{\text{work}}{\text{time}}$	J.s^{-1} (or watt, W)
Momentum (p)	mass × velocity	kg.m.s^{-1}; Ns
Pressure (P)	$\frac{\text{force}}{\text{area}}$	N.m^{-2} (pascal, Pa)
Frequency (f)	$\frac{\text{number of oscillations}}{\text{time}}$	per second or s^{-1} (hertz, Hz)
Electric charge		C (coulomb)
Electric potential difference		V (volt)
Electric resistance		Ohm
Electric capacitance		farad (F)

We can always obtain multiples or submultiples of the basic units using the following prefixes (Table 1.3):

Table 1.3

Submultiples	Example	Prefixes
0.1 or 10^{-1}	decimetre = 10^{-1} m	deci- (d)
0.01 or 10^{-2}	centimetre = 10^{-2} m	centi- (c)
0.001 or 10^{-3}	millimetre = 10^{-3} m	milli- (m)
0.000001 or 10^{-6}	micro-farad = 10^{-6} F	micro (μ)
Multiple		
10^1	decametre = 10^1 m	deca (da)
10^2	hectometre = 10^2 m	hecto (h)
1000 (10^3)	kilometre = 10^3 m	kilo (k)
1,000,000 or 10^6	megawatts = 10^6 W	Mega (M)

1.2 Measurements

Measurements play a crucial role in physics, but they can never be perfectly precise. We usually note the reading accuracy of the measuring instrument and specify the measurement only in the correct number of *significant figures*.

In giving the result of a measurement, it is good practice to state the *precision* or *estimated uncertainty* in the measurement. For example, we can write the length of a wire as 51.3 ± 0.1 cm, where 0.1 represents the *estimated uncertainty* in the measurement. This means that the length is probably between 51.2 and 51.4 cm.

The number of reliably known digits in a number is called the number of *significant figures*. They are usually taken as all the figures from the first non-zero digit to the last non-zero digit together with the final zeros if they occur after the decimal points. Thus there are four significant figures in 51.35 and two in 0.053 cm. We specify measurements only in the correct number of significant figures. Thus the length of the wire is recorded as 51.3 cm not 51.30 cm. The figure 51.30 implies an uncertainty of 0.01 cm. Significant figures indicate the precision with which a particular measurement has been made.

1.2.1 Length

In the secondary school physics laboratory, we can measure length with a metre rule, callipers, vernier callipers, and micrometer screw gauge.

a. The metre rule

The metre rule is graduated in centimetres and millimetres. The smallest graduation of the metre rule is 1 mm or 0.1 cm. This is its *reading accuracy*. Measurements can be estimated up to half this smallest graduation, i.e. 0.5 mm or 0.05 cm. This is the *estimated uncertainty*. Thus, we can record a length as 21.55 ± 0.05 cm.

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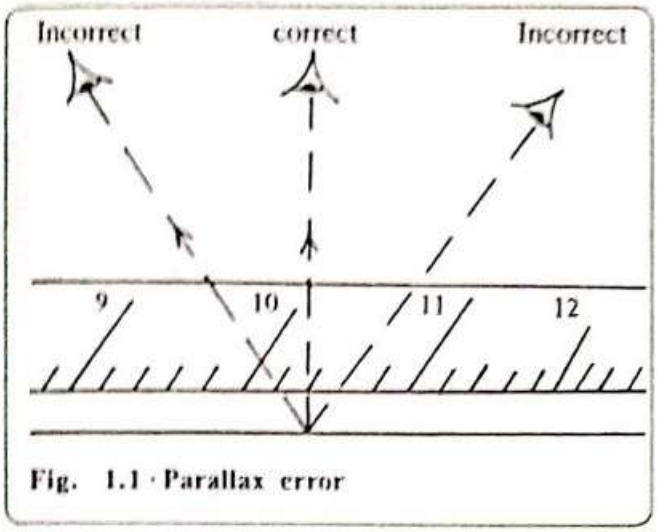


Fig. 1.1 · Parallax error

In measuring with a metre rule, a necessary precaution is to avoid the error of parallax by looking vertically downwards on the markings instead of slantingly.

Large distances such as length or width of a football field can be measured with steel tapes graduated in metres.

The callipers

Callipers are used to measure lengths on solid objects that are in a cylindrical form. The jaws of the callipers are adjusted until they touch the object in the correct positions. The length between the jaws is then measured on a graduated scale (fig. 1.2b).

b. The vernier callipers

These can measure lengths more accurately than the metre rule. To measure small lengths, to the nearest 0.1 mm, e.g. the thickness of a metre rule, the internal and external diameters of a tube, or the diameter of a rod, we use the vernier callipers. This device gives a more accurate determination of such lengths.

The instrument has two sets of jaws and two scales, *the main* and *the vernier scales*. The vernier scale which slides on the main scale is a smaller scale which enables us to measure a fraction of the smallest division on the main scale. The main scale is graduated in centimetres and millimetres. The vernier scale is constructed by dividing a 9 mm length into 10 equal intervals such that each vernier division has a length of 0.9 mm or 0.09 cm. The difference in length between a vernier scale division and a main scale division is thus 0.1 mm or 0.01 cm.

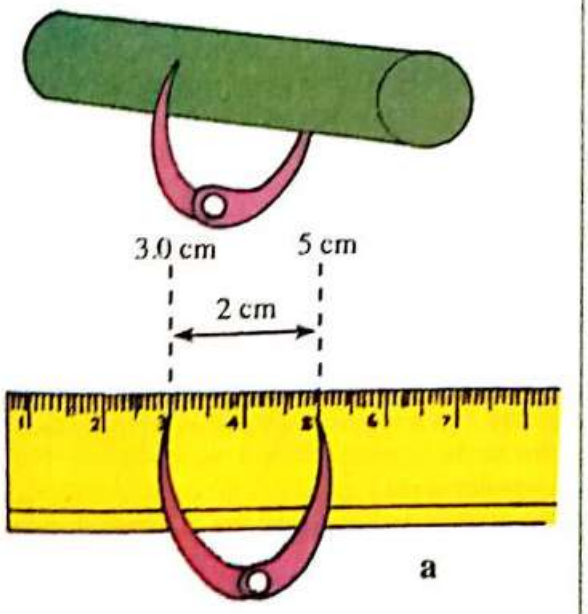


Fig. 1.2 The callipers

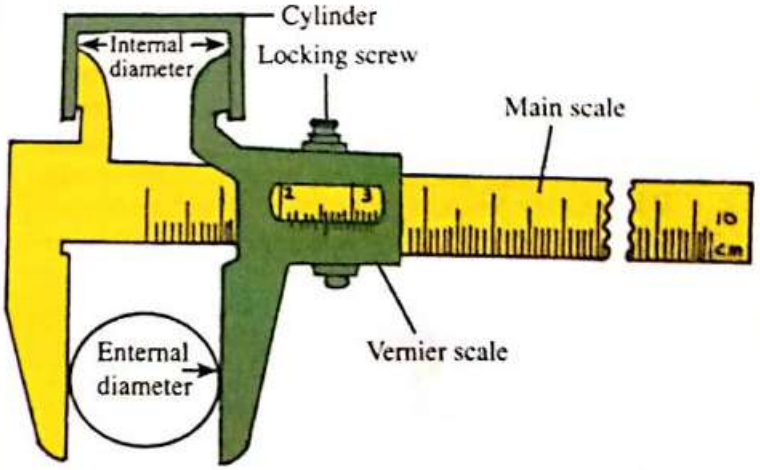


Fig. 1.3 Vernier Calliper

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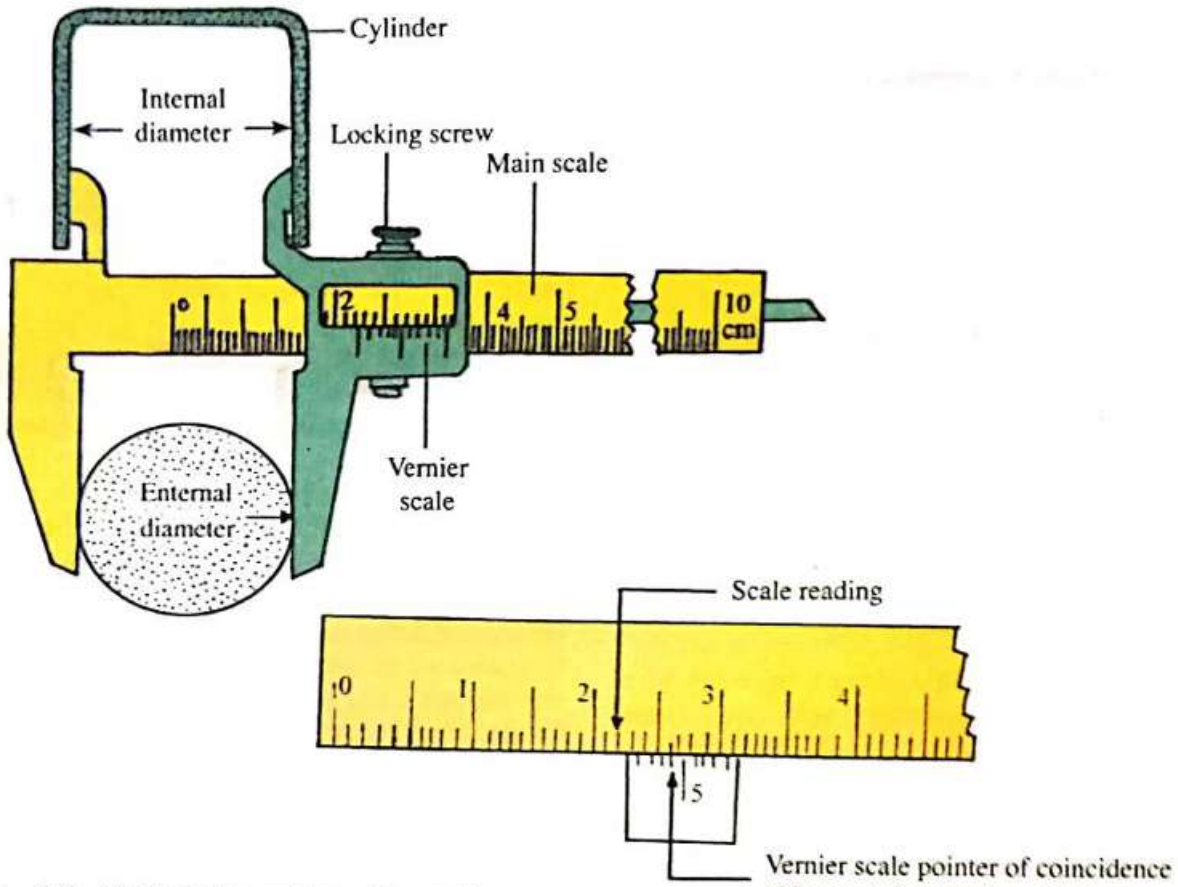


Fig. 1.4 Main scale and vernier scale

For any measurement, the readings on both the main and the vernier scales are added together to obtain the total reading of the length. For example in fig. 1.4 the reading is 2.24 cm. This is obtained by recording the reading on the main scale immediately preceding the zero mark on the vernier scale; we then record the number of the graduation on the vernier which comes most directly in line with the graduation on the main scale. This gives the decimal place in millimetres (mm) or the second decimal place in centimetres (cm). The two readings are then added up (22 mm + 0.4 mm).

The vernier callipers can usually measure to an accuracy of 0.01 cm. The necessary precaution in using this instrument is always to note if it has any zero error. This is done by setting the zeros of the main scale and the vernier to coincide. If they do not coincide, any difference is noted as the 'zero error'. This is added or subtracted in all subsequent readings as the case may be.

The reading accuracy of the vernier callipers is 0.01 cm.

c. The micrometer screw gauge

For accurate measurements of still smaller lengths such as the diameter of a wire, the diameter of a small ball (e.g. a pendulum bob) or the thickness of a piece of paper, we use the micrometer screw gauge. It can measure to an accuracy of 0.001 cm.

Before any measurements are made with the screw gauge, its *anvil* and *spindle* should be closed to check for zero error.

Like the vernier callipers, the micrometer screw gauge has a main scale and a vernier scale. The main scale is graduated in millimeters. The circular vernier scale consists of 50 equal divisions. When the screw head carrying the circular vernier scale is turned round once, it moves a distance of 0.5 mm along the main scale. Thus one division on the vernier scale equals $\frac{0.5}{50}$ or 0.01 mm on the main scale.

To measure with the instrument, the object is placed between the jaws (i.e. the anvil and the spindle) which are then closed using the *ratchet* until the object is firmly gripped by the jaws. The main scale reading is then noted to the nearest.

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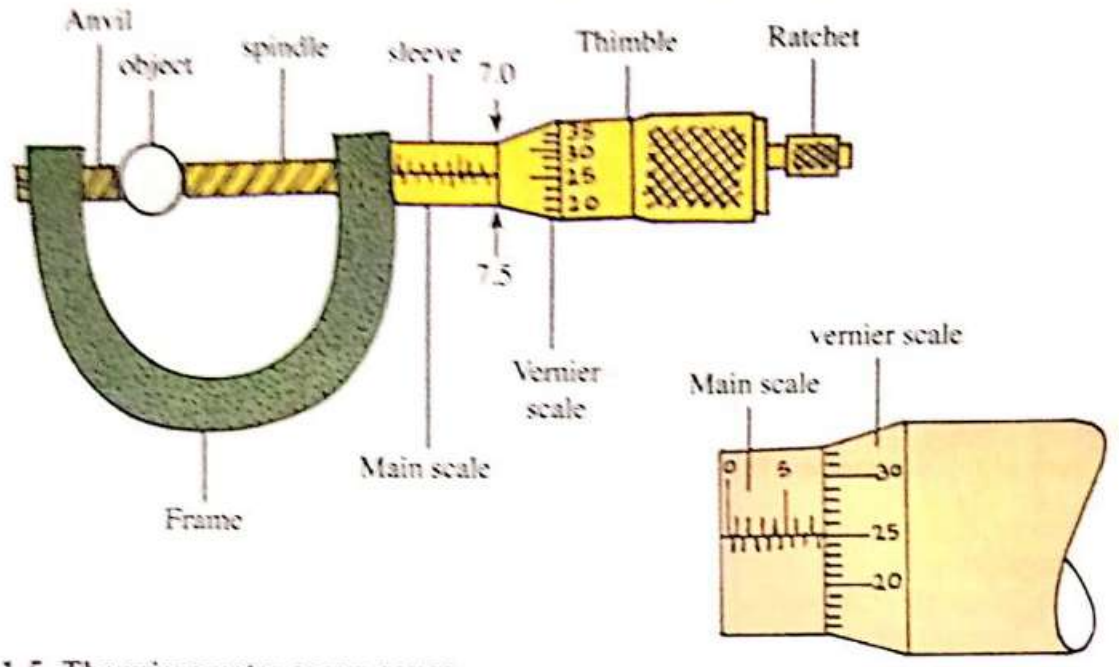


Fig. 1.5 The micrometer screw gauge

graduation (7.5 mm in fig. 1.5a). We then note the reading on the circular vernier scale (25 divisions in fig. 1.5b, i.e. 0.25 mm). The readings on the two scales are then added giving 7.5 + 0.25 = 7.75 mm or 0.775 cm. If the instrument has any zero error, this is then added or subtracted as the case may be.

1.2.2 Measurement of volume

a. Rectangular block

We can obtain the volume of a rectangular block by measuring its length (l), breadth (b) and height (h). The volume is then given by

$$V = l \times b \times h \qquad 1.1$$

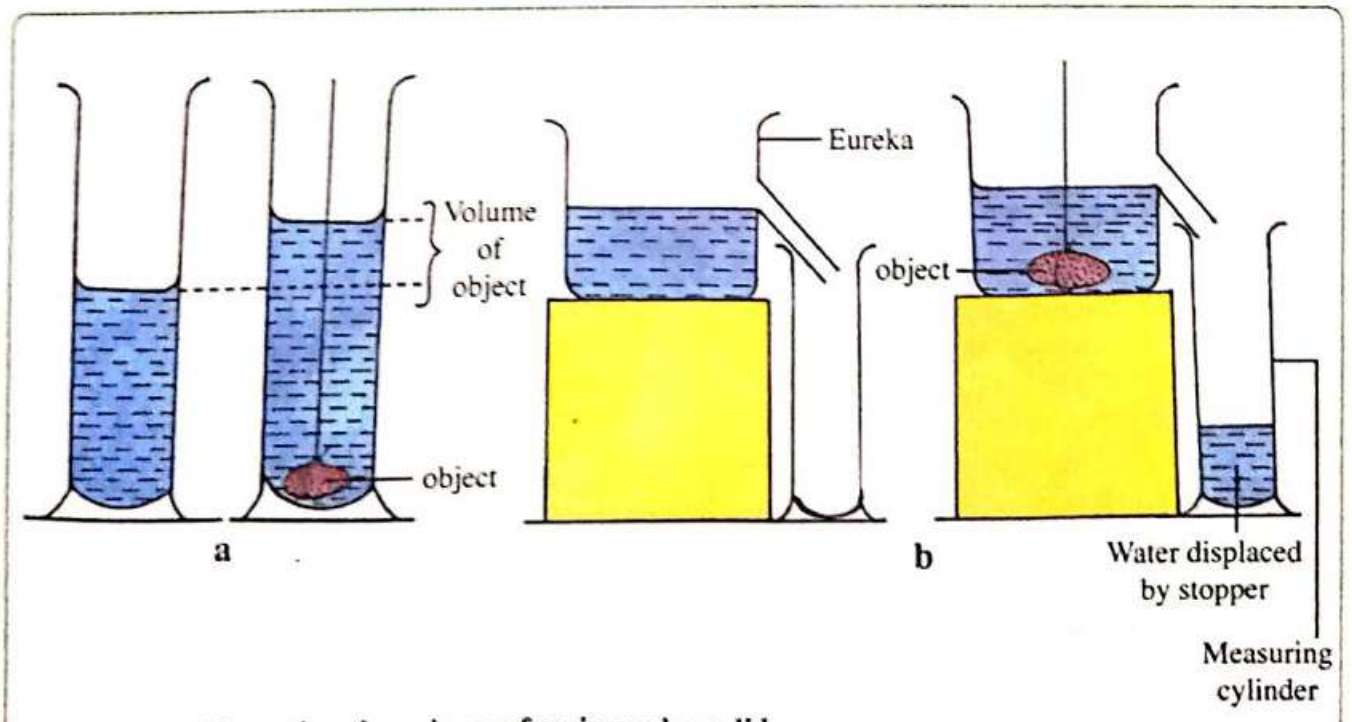


Fig. 1.6 Measuring the volume of an irregular solid

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b. Irregular solid

The volume of an irregular solid (e.g. a glass stopper) is obtained by immersing it completely in a measuring cylinder containing a liquid in which the solid is insoluble. The volume of the liquid displaced, gives the volume of the solid.

c. A sphere

We measure the diameter (d) with a micrometer screw gauge and obtain the volume by

$$V = \frac{4}{3}\pi r^3$$

where $r = \frac{d}{2}$. For greater accuracy, we obtain three different values of the diameter and find the mean value.

d. Cylindrical wire

We measure the length (l) of the wire using a metre rule, and the diameter (d) of the wire using a micrometer screw gauge. We measure the diameter at different points of the wire and calculate the mean diameter.

$$\text{Volume} = \pi r^2 l \text{ or } \pi\left(\frac{d}{2}\right)^2 l = \pi\frac{d^2 l}{4}$$

e. Liquids

The volume of a liquid can be measured with a measuring cylinder, a pipette or a burette.

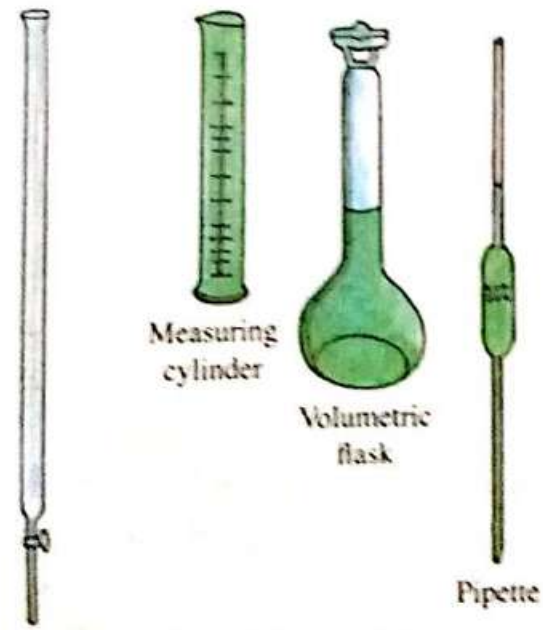


Fig. 1.7a Graduated glass containers

The correct liquid level is read at the bottom of the *meniscus* (or curved surface of the liquid).

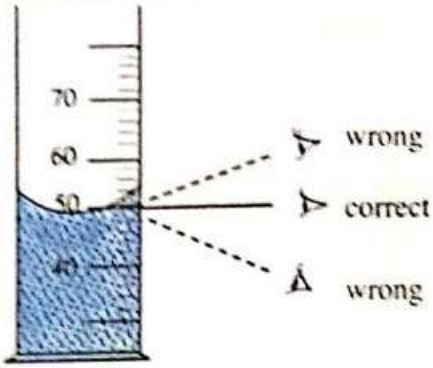


Fig. 1.7b Correct position of eye

1.2.3 Mass and Weight

Mass is defined as the quantity of matter or 'stuff' contained in a body; while *weight* is the force or pull with which the earth attracts the body towards the centre of the earth.

The mass of a body is measured with a balance of which there are various types — a beam or chemical balance, a lever balance as shown in fig. 1.8.

Different types of balance

The mass of a body is usually measured by comparing it with standard masses.

To measure with a beam balance, the object is placed on the left-hand scale pan and standard masses are placed on the right-hand scale pan. When raised for weighing, the beam of the balance should swing freely about a pivot in the centre of the beam. Standard masses are added until the pointer swings equal number of divisions on each side of the central mark or the pointer balances on the central mark of the scale. When this happens the masses in the two scale pans are equal.

The mass of the object is then obtained by adding the standard masses on the right scale pan.

The working of the beam balance is based on the principle of moments' and its reading accuracy could be up to 0.001 grams.

Precautions to be taken when using beam balance

For more accurate results, the following precautions should be observed:

a. forceps (not bare hands) are used for picking up the standard masses,
b. objects are wiped dry before being placed on the scale pan,
c. hot objects are allowed to cool before weighing,

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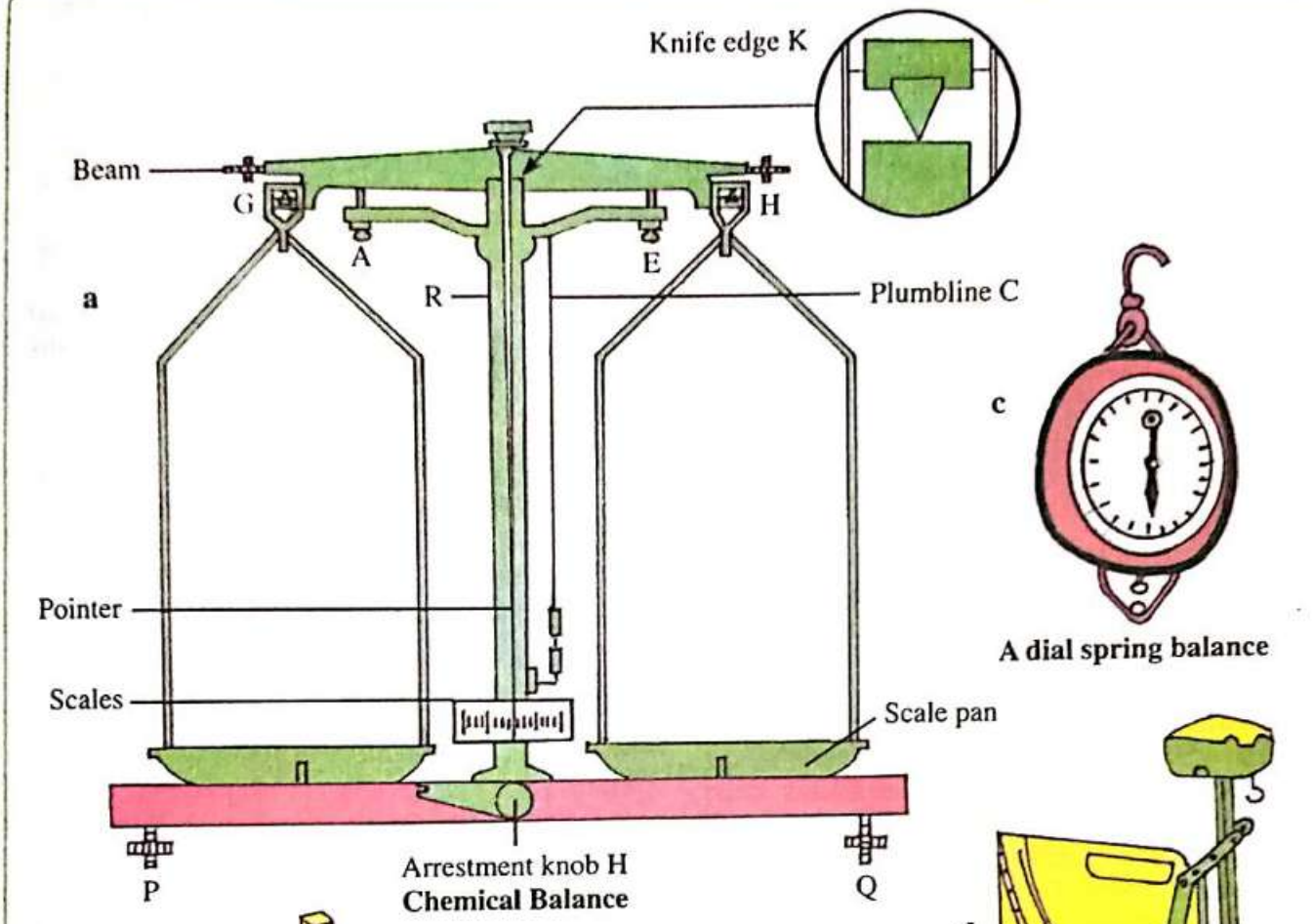


Fig. 1.8 Another type of beam balance

Fig. 1.9 The Spring balance

d. known masses are not added or removed from the scale pan while the beam is in the raised position.

The *weight* of an object is found with a *spring balance*. The balance is calibrated using *Hooke's law* which states that the extension of the spring of the balance is proportional to the applied force as long as the elastic limit of the spring is not exceeded. The balance, therefore, has a uniform scale and measures the weight directly.

The object whose weight is required is suspended from the hook of the spring balance. The weight causes the spring to stretch moving the pointer which directly indicates the weight on the graduated scale.

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Precautions to be taken when using spring balance

For accurate results, we should observe the following precautions:

a. The balance is held still and vertically while taking the reading. It is better to suspend it from a firm support or hook.
b. The spring is stretched lightly with hand and released before taking readings. This is to remove possible kinks in the spring.
c. The eye is placed horizontally with the pointer while taking reading. This is to avoid the error due to parallax.

1.2.4 Time

We define time as that in which events are distinguishable with reference to before and after. The most natural time unit is the *solar day* which is manifested by the passing of day and night. It takes the earth one solar day to complete one revolution about its axis. For astronomical purposes, a different time scale known as *sidereal* time is used. This is the time interval between two successive passages of a star over the meridian. At the present period, the essential mechanisms of the devices used for marking the passage of time are masses so mounted or suspended that when set in motion under the influence of forces, they repeat with great *regularity* fixed cycles of their motion. The pendulum of a clock and the balance wheel of a watch are two of such mechanisms.

Hence nowadays we measure and indicate time with clocks. In the laboratory, time is measured with a stop-clock or stop-watch.

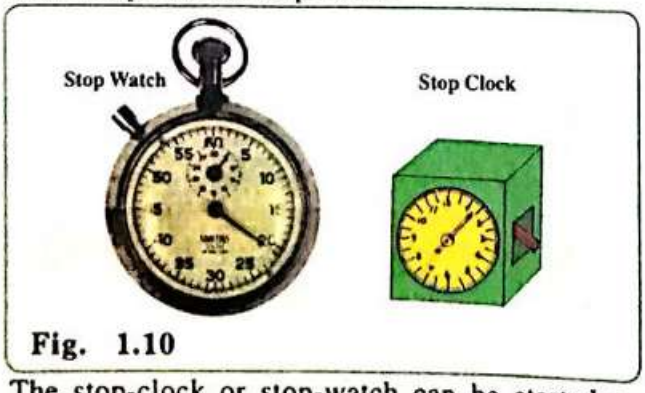


Fig. 1.10

The stop-clock or stop-watch can be started or stopped with the use of a knob. The knob can be pressed or pushed. The stop watch enables us to measure small intervals of time very accurately. Stop watches can measure time to 0.1 second.

The unit of time is the *second* with its multiple units as

60 seconds = 1 minute
60 minutes = 1 hour
24 hours = 1 day

The most accurate clocks of all are the quartz crystal clock and the atomic clock.

1.3 Dimensions of Physical quantities

Any quantity which can be measured in units of lengths is said to have the *dimension*, of length (L). So also any quantity which can be measured in units of time is said to have a dimension of Time (T) and any quantity which can be measured In unit of mass is said to have a dimension of Mass (M).

In Physics, the word *dimension* has a special meaning. It usually denotes the physical nature of a quantity. The symbols that are used to specify length, mass and time are [L], [M], [T], respectively. They are usually enclosed in bracket.

The dimension of velocity (V) are written

$$V = \frac{[Length]}{[Time]} = \frac{[L]}{[T]}$$

Or $V = [L][T^{-1}]$

Dimension for area, A = *length* x *breadth*

$= [L] \times [L] = [L^2]$

For volume, $V = [L] \times [L] \times [L] = [L^3]$

Example

Deduce the dimension of (a) force (b) pressure (c) work

Solution

(a) *Force = mass x acceleration*

Dimension of Force = Dimension of Mass x Dimension of Acceleration

$= [M] \times [LT^{-2}]$

$= [MLT^{-2}]$

(b) $\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{\text{Dimension of Force}}{\text{Dimension of Area}}$

$$\text{Dimension of Presure} = \frac{[MLT^{-2}]}{[L^2]} = [M^{1}T^{-2}]$$

(c) *Dimension of Work = Dimension of Force x Dimension of distance*

$[MLT^{-2}] \times [L] = [ML^2T^{-2}]$

We can use the method of dimension or dimensional analysis to verify whether a physical equation is correct or not. For example, consider the formula, $V = u + at$. The dimension of each term on each side of the equation must be the same.

$$u = \frac{[L]}{[T]} = [LT^{-1}]$$

$$a = \frac{[L]}{[T^2]} = [LT^{-2}]$$

$$t = T$$

hence,

$V = u + at$

$[LT^{-1}] = [LT^{-1}] + [LT^{-2}] \times [T]$

$= [LT^{-1}] + [LT^{-1}]$

Therefore, the equation is correct dimensionally

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Summary

1 **Fundamental quantities** are the basic quantities that are not dependent upon others and cannot be defined in terms of other quantities. Their units are fundamental units.

2 **Derived quantities and units** are those obtained by some simple combination of the fundamental quantities and units.

3 We measure lengths by the metre-rule, diameters by vernier callipers and micrometer screw gauge, mass by beam or chemical balance, weight by spring balance and time by stop clocks or watches.

Activities

1 Use a metre rule to measure out a string of length six metres. Use this six-metre string to measure the length and breath of your school's assembly hall, football field, and your dormitory.

2 Use a stopwatch to measure the number of beats your pulse makes in 'a minute. Collect such number of beats for 10 students in your class. Find the average pulse rate for the 11 of you.
Ask the students to jump up and down twenty times. Repeat the experiment and compare their pulse rates with their previous values. Obtain the new average pulse rate.

3 Use a metre-rule to cut out a length of string 90 cm. Tie a piece of pebble to one end of the string and suspend the other end from a firm support. Displace the pebble and string from their rest position. Using a stopwatch, find the time it takes the pebble to move to and fro, thirty times.
Repeat the experiment using strings of length 70 cm, 50 cm and 40 cm. Compare the times for thirty complete movements of the string and pebble.
This is the simple pendulum experiment, the theory of which will be discussed in a later chapter.

4 Suspend a spring balance from a firm support. Suspend 109, 20 g, 30 g, and 50 g slots of weight from its hook and note the extension produced for each slot of weight. Remove the weights in turn and note the new positions of the pointer, and the new extensions corresponding to each weight. Compare the extensions for load increasing with those for load decreasing.

Exercise 1

1 Which of the following are fundamental units?
A kg B kg.m-3 C s
D m3 E Nm-2.

2 What are the reading accuracies of
A metre rule B vernier callipers
C micrometer screw gauge D stopwatch.

3 Which of the units of the following physical quantities are derived?
1 Area 2 Thrust
3 Pressure 4 Mass (SSCE)

4 What are the best Instruments for measuring
a. length of a simple pendulum
b. diameter of a metal rod
c. thickness of the central portion of a thin converging lens
d. internal diameter of test-tube
e. diameter of a spherical metal bob?

5 Name the instruments you would use to make measurements on a given length of wire in order to calculate the volume of the wire. Which of the measurements should be repeated several times in order to make the calculation very accurate, and why? (SC/GCE)

6 How many seconds are there in a
1 year 2 day 3 month of march?

7 Distinguish between fundamental and derived quantities. Give two examples of each.

8 Distinguish between *mass* and *weight*. Name the instruments for measuring each.

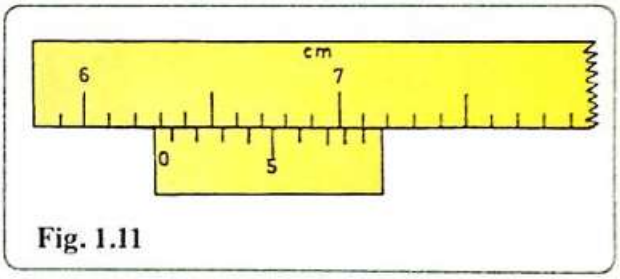
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Fig. 1.11

The diagram above represents a portion of a vernier calliper. What is its reading?
A 7.10 cm B 7.09 em C 6.90 em
D 6.34 em E 6.29 cm. (SSCE)

10 The internationally agreed system of units (S1) for physical measurements are
(a) lb, ft, sec (b) g, m, sec (c) kg, m, sec
(d) cm, g, sec (e) kg, cm, sec (Jamb)

11 Which of the following is NOT a fundamental S.I Unit?
(a) Metre (b) Ampere (c) Kelvin
(d) Second (e) Radian (Jamb)

12 Which of the following is a fundamental unit?
(a) Newton (b) Watt (c) Joule
(d) Second (Jamb)

13 Which of the following statements about a spring balance and/or a chemical balance is NOT corrects?
(a) The Chemical balance operates on the principal of Moments
(b) The spring balance operates on Hooke's law.
(c) Either may be used to measure the weight of a substance
(d) The reading of a spring balance changes

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14 Which of the following statements about *mass* and *weight* are correct?

(i) Weight is the force of gravity on a body.

(ii) The mass of a body is the quantity of matter in the body.

(iii) The weight of a body is greatest at the equator.

(iv) The mass of a body is greatest at the poles.

A (i) and (ii) only B (i) and (iii) only

C (ii) and (iv) only D (iii) and (iv) only

E (i) and (ii) and (iii) only

15 What are the S.I. units of K so that the equation Velocity = K x density is dimensionally correct? Give your answer in terms of the basic units. ($Kg^{-1}m45^{-1}$)

16 Which of the following is a fundamental quantity?

A Heat capacity B Electric current

C Torque D Reactance

E Density

17 Which of the following is a fundamental quantity?

A Speed B Density

C Length D Impulse

E Energy

18 Which of the following is a derived unit?

A Ampere B Kilogram

C Second D Ohm

E Kelvin

19 A sphere of radius r moving with a velocity v under streamline conditions in a viscous fluid experiences a retarding force, F, given by $F = Krv$ where K is a constant.

Derive the S.I. units of K in terms of the base units.

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2 MOTION

Motion involves a change of position of a body with time. It also involves how things move and what makes them to move. *Kinematics* is the description of how objects move without regard to forces causing their motion, and *Dynamics* deals with why objects move as they do.

At the end of this chapter the student should be able to:

1 List and explain the different types of motion: random, rotational, translational and oscillatory.
2 Identify for a given motion, the two bodies between which there is a relative separation.
3 Identify the forces that cause a given body to move, to slow down and to stop moving.
4 Identify a force resisting the motion between two surfaces in contact and moving relative to each other.
5 Reduce friction in a given situation.
6 Identify circular motion.
7 Distinguish between distance and displacement, speed and velocity, and be able to plot distance – time graph and velocity – time graph and to interprete their gradients or slopes.
8 Explain the concept of uniform motion, instantaneous speed/velocity, uniform acceleration.
9 Solve simple problems on speed, velocity and acceleration.

2.1 Types of motion

We distinguish between four types of motion:

a. Random motion

Objects move irregularly or at random or haphazardly or disorderly with no preferred direction or orientation. As a result of such random movement, the particles collide with one another. An example of such motion is the *Brownian motion* – an irregular motion of particles of various kinds suspended in water, or of smoke particles suspended in air, or the motion of gas particles.

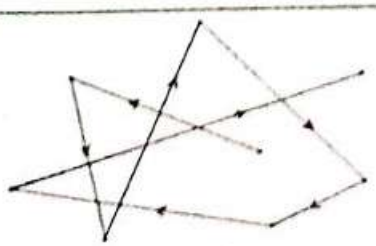


Fig. 2.1 Random motion

b. Translational motion

When *rigid* objects move from one point in space to another without rotating, the motion is said to be translational. Each part of an object undergoing pure translational motion follows the same path. An example is the diver shown in fig. 2.2, a bee flying from one flower to another, a car travelling from Enugu to Lagos, or from one station to another.

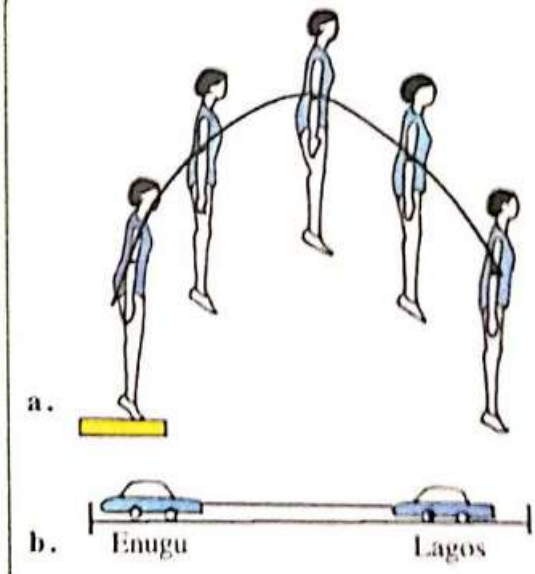


Fig. 2.2 Translational motion

In translational motion, every point in the body remains relatively fixed to one another.

c. Rotational motion

By rotational motion we mean that all points in the body move in concentric circles, such as the point P in a rotating wheel (see fig. 2.3a) and that the centre of these circles all lie on a line called the *axis of rotation*. Examples are the earth rotating about its axis, a cricket ball spinning about its axis, the wheels of a moving car, the rotation of the blades of an electric fan.

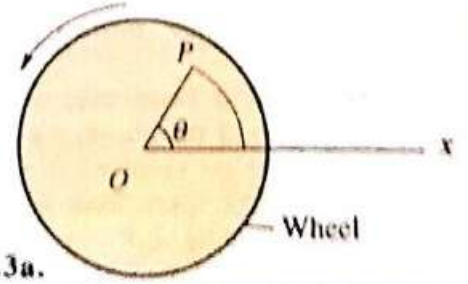
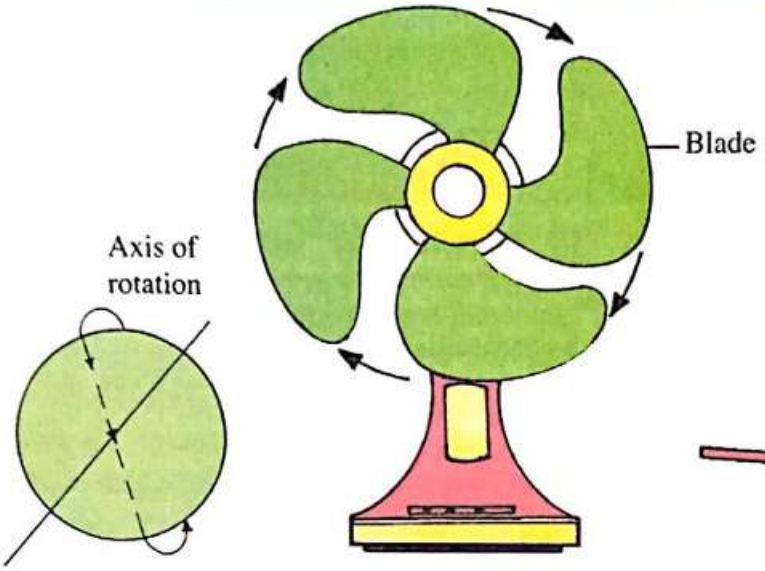


Fig. 2.3a.

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b. Electric fan

Fig. 2.3 Rotational motion

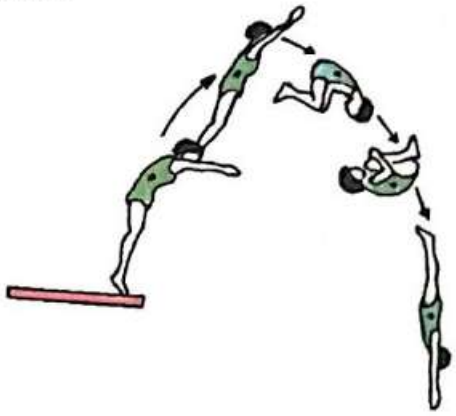
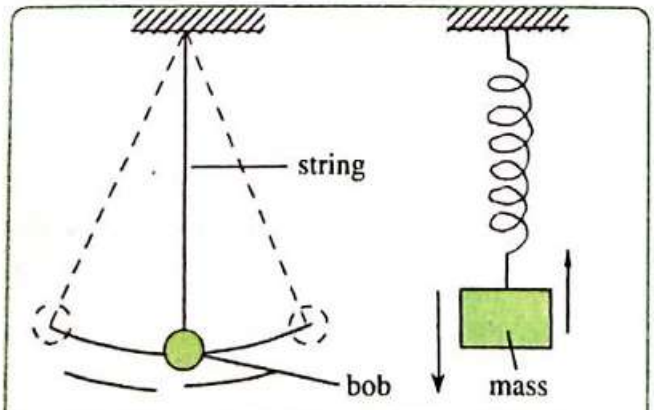


Fig. 2.5 Combination of translation and rotation

d. Oscillatory (or Vibratory) motion

When a body moves to and fro, reversing the direction of its motion and returning regularly to its original position, it is said to have a *vibratory* or *oscillatory* motion. Such a motion is *periodic*, it repeats itself.

Examples are the motion of a rocking chair, the motion of a pendulum as it swings back and forth, a diving board, the strings of a plucked guitar, the prongs of a tuning fork struck with a hard object, and the vertical movement of a disturbed mass on a spiral spring.



a. Simple pendulum b. Vibrating mass on a spring

Fig. 2.4 Oscillatory motion

The motion of a body may consist of a combination of two of the above, e.g. translation and rotation motion are involved in a ping-pong ball spinning through space from one player to another, or the diver in fig. 2.5.

2.2 Relative Motion

We may define motion as a change of position. It will however be impossible to determine the change of position without a *point* or *frame of reference* which we shall have assumed to be at rest. Thus every measurement must be made with respect to such points or frames of reference.

If you are sitting in a train travelling past a railway platform on which your friend is standing, there is said to be relative motion or separation between you and your friend.

A person sitting still on a back seat in a moving bus, has a speed with respect to the earth which is the same as that of the bus. But if he now walks towards the driver of the bus, he has a speed relative to the earth which is more than that of the bus. Suppose the bus is moving at 100 $kmhr^{-1}$ and the person walks at 5 $kmhr^{-1}$ towards the driver, his forward speed relative to the earth is $100 + 5 = 105$ $kmhr^{-1}$. But when he walks back from the driver to his back seat at the speed of 5 $kmhr^{-1}$, his speed relative to the earth is now $100 - 5 = 95$ $kmhr^{-1}$. Relative to the bus, the person's speed is the same in each instance but the directions of motion are opposite. Relative to the earth, the speeds of the person are different in the two occasions. In this example, the points of reference are the earth and the bus.

If a student travelling in a school bus looks down on the floor of the bus, the bus will appear to him to be stationary. This is because the bus and the student are moving with the same speed and hence there is zero or no relative motion between them. But if the student looks out of the window, the trees outside the bus will appear to be moving

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in opposite direction as the bus. Trees do not move. The illusion of moving trees is due to the relative motion between the student in the moving bus and the stationary trees.

2.3 Position, Distance and Displacement

The position of a point in space, is determined by its *distance* and *direction* from other points. The statement of position is accomplished by means of a frame of reference or a point of reference, which we shall call the *origin*.

In order to locate the position of a point in a plane (such as a blackboard, the wall of a room, or a drawing paper), it is customary to draw two lines intersecting at an origin O and perpendicular to each other.

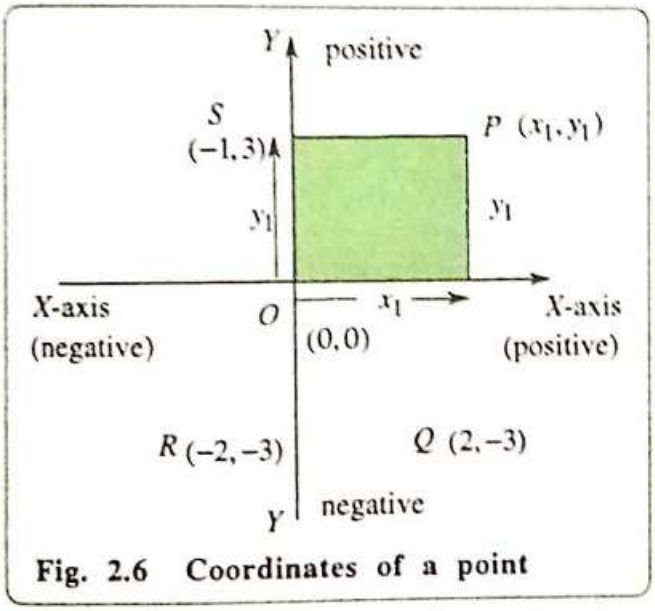


Fig. 2.6 Coordinates of a point

The horizontal line going across from left to right of the page is called the X-axis and the vertical line going up the page is the Y-axis. By convention, points to the right of O are positive, points to the left are negative, points above O are positive, points below O are negative. The position of any point such as P is specified by two quantities or *coordinates*, the x and the y coordinates. If the position P is specified by the coordinates (x_1, y_1) it means that the distance of P from O in the X-direction is x_1 and in the Y direction, it is y_1.

The x-coordinate is called the *abscissa*, the y-coordinate is called the *ordinate*. The positions of Q, R, and S are as specified in fig. 2.6. The coordinates of Q, R, S mean that Q is 2 units in the $+x$-direction, and 3 units in the $-y$-direction; R is 2 units in the $-x$-direction and 3 units in the $-y$-direction; S is 1 unit in the $-x$-direction and 3 units in the $+y$-direction.

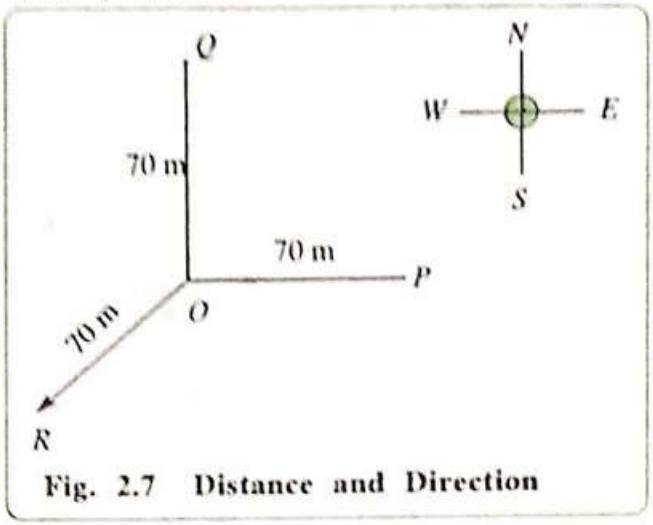


Fig. 2.7 Distance and Direction

When an object moves, say 70 m from a point O to another point P, we say that the *distance* covered by that object is 70 m (fig. 2.7). The distance indicates how far the object has moved. However knowing the distance alone is not sufficient to tell us where or in what direction the object has moved. Points R and S are also 70 m from O, and we cannot tell which point is the final position of the object. In the study of motion, it is necessary to specify the *directions* in which a body is moving. If the body in fig. 2.7 travelled 70 m eastward, its final position is at P, and if northward, the position is Q, if it moved in the south-west direction, its final position is R. Therefore, with the direction specified, we know the final position of the object. We say that the *displacement* of the object is 70 m northwards if the final position is at Q.

Displacement is defined as distance travelled in a specified direction.

It is a change in the position of an object. It requires distance and direction to specify it.

To see further the distinction between total distance and displacement, imagine a person walking 500 m to the east, and then turning around and walking back (west) a distance of 300 m, the total distance travelled is 800 m but the *displacement* is only 200 m since he is now only 200 m from the starting position.

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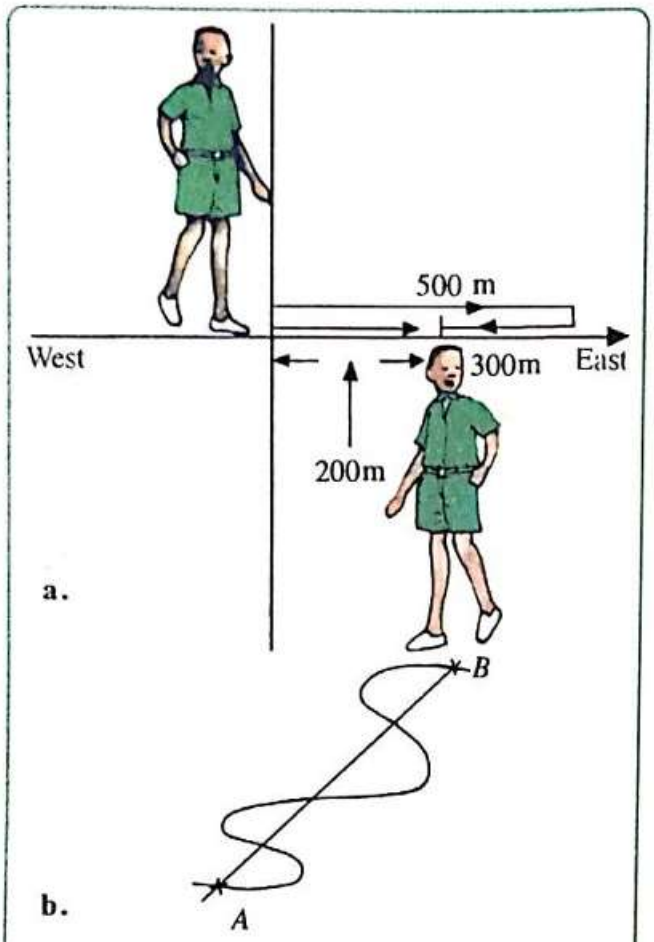


Fig. 2.8 Displacement

A man travels from a town A to another B through a winding road, covering a distance of 300 km. If the shortest distance between A and B is 240 km, represented by a straight line AB, then the displacement is 240 km in the direction of the line AB.

If a boy walks 30 m eastward and then 40 m northward, his displacement is 50 m in a northeast direction.

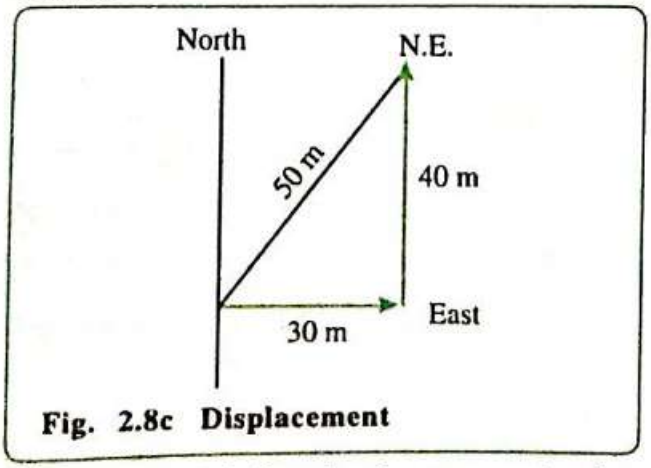


Fig. 2.8c Displacement

The direction is indicated using compass bearing, e.g. N, E, W, S, N 60° E or 60° East of North, South 45° West or 45° West of South.

Speed and Velocity

Speed

In describing the motion of a body, we note both the distance and time it takes to cover that distance. The rate at which a body covers a distance is called the *speed* of the body. Thus

$$\text{Speed} = \frac{\text{distance}}{\text{time}} \qquad 2.1a$$

This is indeed an average speed ($\bar{v}$) over the distance s. Thus

$$\bar{v} = \frac{s}{t} \qquad 2.1b$$

The SI unit of distance is metre, that of time is second. Thus the SI unit of speed is metre per second (ms^{-1}). Other multiples of that unit are *centimetres per second* and *kilometres per hour.* If a bus covers a distance of 200 km in 2 hours, its average speed is then $\frac{200\text{ km}}{2\text{ hr}} = 100\ kmh^{-1}$.

Uniform Speed

When a body covers equal distances in equal time intervals, no matter how small the time interval may be, the speed is said to be a *uniform speed* or *constant speed.*

Velocity

In everyday language, the words velocity and speed are often used interchangeably. In the study of motion, it is necessary to distinguish between the two. In speed, no direction is specified, but in velocity, it is necessary to specify direction.

Since *displacement* refers to the distance covered in a specified direction, we define velocity in terms of *displacement.*

Velocity is the rate of change of *displacement.*

$$\text{Velocity} = \frac{\text{displacement}}{\text{time}} \qquad 2.2a$$

$$v = \frac{s}{t} \qquad 2.2b$$

Uniform Velocity

When a body moves with equal displacements in equal time intervals, no matter how small the time intervals may be, the velocity is said to be a *uniform velocity* or *constant velocity.* Its unit is metre per second (ms^{-1}).

If the body moves round a circular path at constant speed, then it is said to move with *non-*

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uniform velocity, because its direction of motion is constantly changing.

Example 2.1

A car travels at an average speed of 100 kmh^{-1}, what distance does it cover in 5 minutes?

Solution

100 km per hour $= \frac{100}{60}$ km per min.

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{distance} = \text{speed} \times \text{time}$$

$$= \frac{100}{60} \times 5$$

$$= \underline{8.3 \text{ km}}$$

Distance (or displacement) – time graph

If we measure the distance covered by a moving object at known time intervals (say 2 seconds), and we plot the values of distance on the Y-axis and time on the X-axis of a graph, on joining the points, we obtain the distance (or displacement) – time graph.

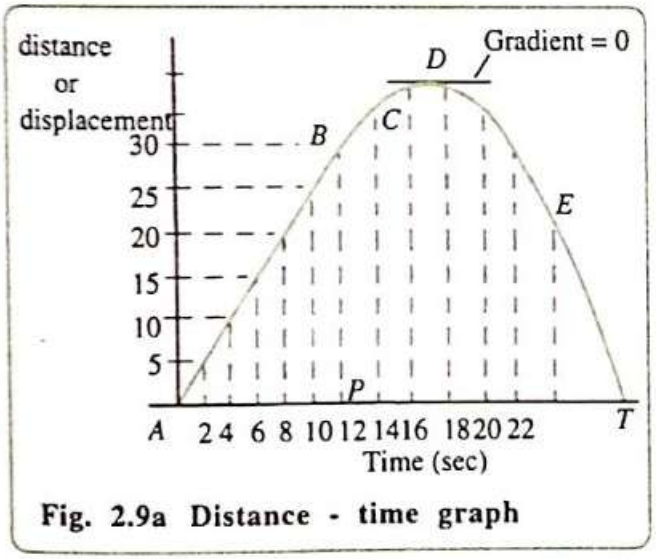


Fig. 2.9a Distance - time graph

Such graphs provide useful information about the motion of the object. If the graph is a straight line such as AB of fig. 2.9a, the speed (or velocity) is uniform since the body covers equal distances in equal times.

$$\text{Speed (velocity)} = \frac{\text{distance (displacement)}}{\text{time}}$$

$$= \frac{AB}{AP} = \text{Gradient or slope of line } AB.$$

Velocity = Gradient of Displacement – Time Graph

The slope or gradient is the same or constant all along AB.

Beyond B, the graph is no more straight but curved along $CDET$. The speed (velocity) between C and T is no more uniform but variable.

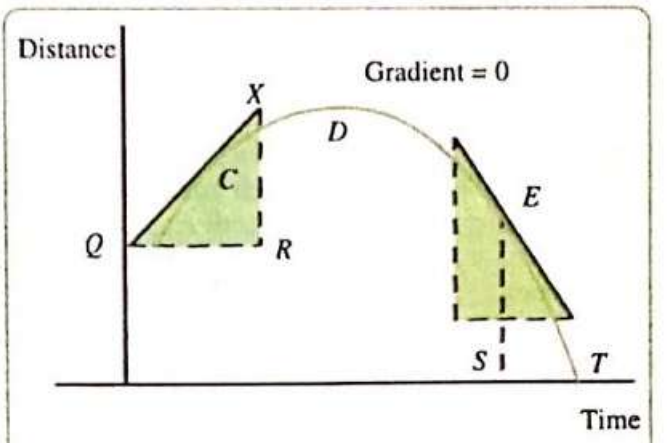


Fig. 2.9b Distance - time graph: Variable velocity

The velocity of the body at any instant such as C, D, E is now given by the gradient of the tangent to the curve at C, D, E; velocity at C = gradient of the tangent at $C = \frac{XR}{QR}$

This velocity is called *instantaneous velocity* at C.

Instantaneous velocity is defined as the velocity at any instant of time.

The speedometers of moving vehicles indicate instantaneous velocity.

Acceleration

Acceleration is defined as the rate of change of velocity with time. Its unit is metre-per-second-per-second. (ms^{-2}).

$$\text{Acceleration } (a) = \frac{\text{velocity change}}{\text{time taken for change}}$$

When the velocity increases with time, the rate of change of velocity with time is termed *acceleration* but when the velocity decreases with time, it is called *retardation*. Retardation is negative acceleration.

$$\text{Acceleration } (a) = \frac{\text{final velocity} - \text{initial velocity}}{\text{time}}$$

$$a = \frac{v - u}{t}$$

or $v = u + at$ 2.3

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Thus, suppose a car moves from a velocity of 20 ms^{-1} to a velocity of 30 ms^{-1} in 5 seconds, its average acceleration is given by

$$a = \frac{30 - 20}{5} = 2 \text{ metre per sec per sec}$$
$$= 2\ ms^{-2}.$$

This means that the velocity of the car increases by 2 ms^{-1} per second.

Uniform acceleration: When the velocity of a moving body increases by equal amounts in equal intervals of time, no matter how small the time intervals may be, it is said to move with uniform acceleration.

Experiment shows that a body falling freely under gravity has a uniform or constant acceleration of about 10 ms^{-2}. If a body is thrown up into the air, it experiences a uniform but negative acceleration or retardation of about 10 ms^{-2} due to the influence of the earth's gravitational attraction.

Velocity-time graph

Let us suppose that an object travelling along a straight path, has the following velocities for the first 12 seconds of its motion.

Velocity (ms^{-1})	0.0	5.0	10.0	15.0	15.0	7.5	0.0
time (s)	0	2	4	6	8	10	12

If we plot a graph of velocity against time, we obtain a velocity-time graph which describes this motion.

Along OA, the velocity increases with equal amounts in each second interval for the first six seconds. The uniform acceleration along OA is given by

$$a = \frac{\text{velocity change}}{\text{time taken for change}} = \frac{AP}{OP}$$

Acceleration = gradient of the line OA,

$$= \frac{15}{6} = 2.5\ ms^{-2}.$$

Acceleration = Gradient of Velocity - Time Graph

Along AB between the time intervals 6 and 8 seconds, the velocity does not change. Acceleration during this period is zero. (Gradient of $AB = 0$). Along BCD, the velocity of the object is decreasing with time. We have a retardation or deceleration. It is a uniform retardation.

$$\text{Retardation} = \text{gradient along } BD$$
$$= \frac{BQ}{QD}$$
$$= \frac{15}{4} = 3.75\ ms^{-2}.$$

For a non-uniform motion, the acceleration at any time is given by the gradient of the tangent to the graph at this point.

Example 2.2

A car starting from rest accelerates uniformly and attains a speed of 80 m per second in 30 seconds. It maintains this steady speed for another 30 seconds. It then slows down uniformly until it comes to rest in the next 40 seconds. Sketch the velocity-time graph for the motion of this car. Obtain from the

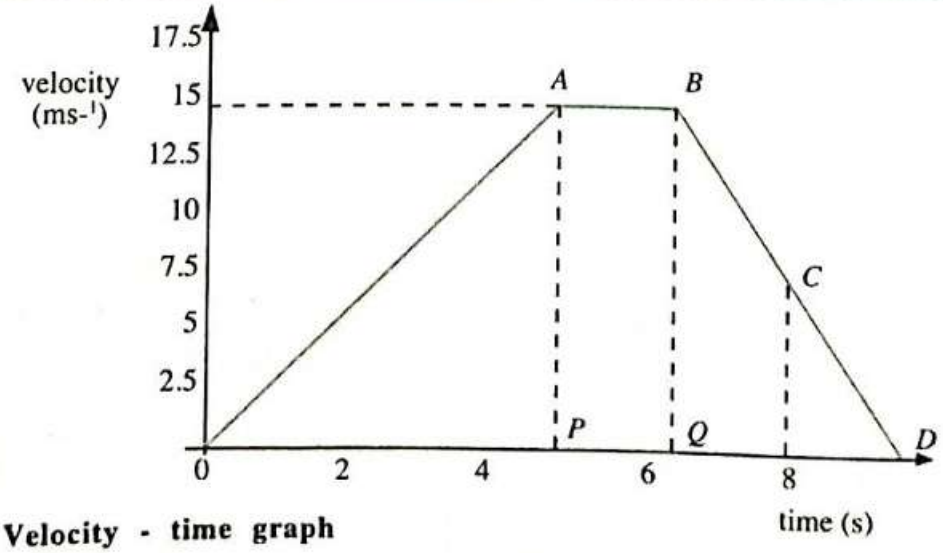


Fig. 2.10 Velocity - time graph

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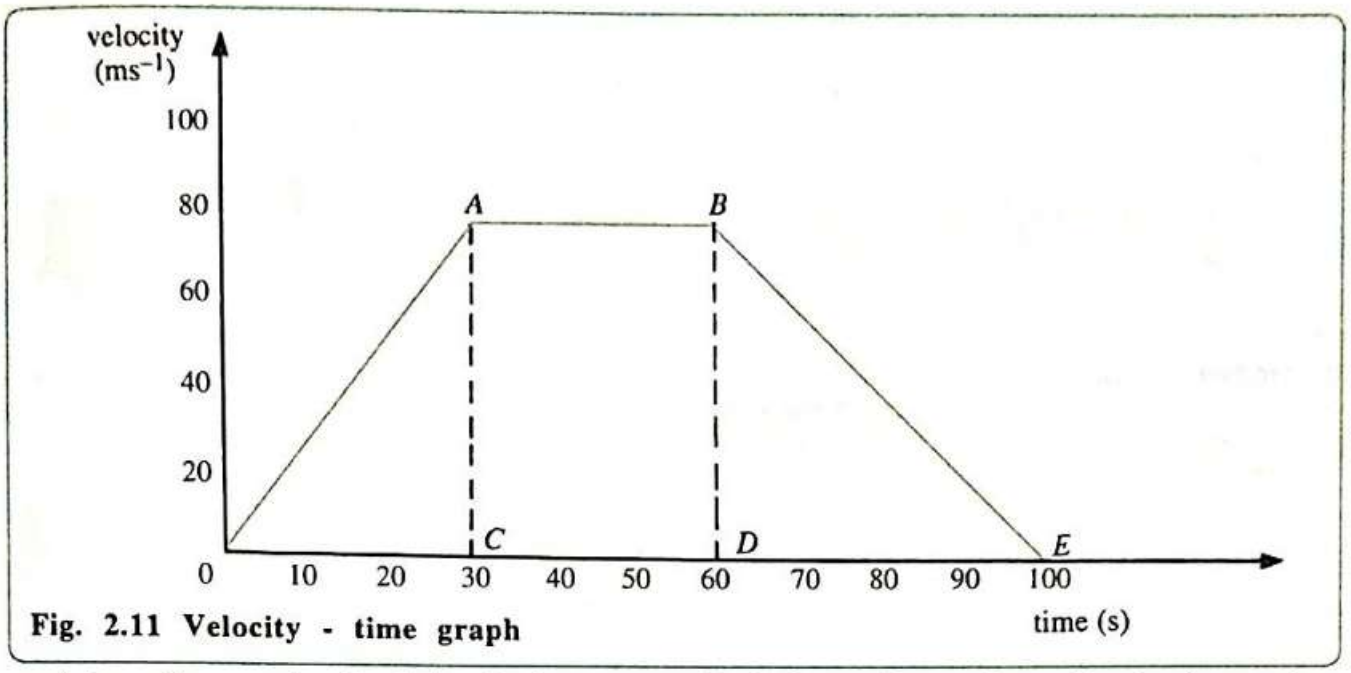


Fig. 2.11 Velocity - time graph

graph the uniform acceleration during the first stage of the journey, and the retardation during the last stage of the journey. For how long did the car travel altogether?

Solution

The velocity-time graph is as shown in the fig. 2.11 above.

Acceleration along OA = gradient of the line OA

$$= \frac{AC}{OC}$$

$$= \frac{80 - 0}{30 - 0} = \frac{80 \text{ m per second}}{30 \text{ seconds}}$$

$$= 2.67 \text{ m per second per second}$$

$$= \underline{2.67 \text{ ms}^{-2}}$$

Retardation = gradient of the line BE

$$= \frac{BD}{DE} = \frac{80 - 0}{100 - 60}$$

$$= \frac{80}{40} = \underline{2.0 \text{ ms}^{-2}}$$

Total time of travel = $(30 + 30 + 40)$ seconds

$$= \underline{100 \text{ seconds}}$$

2.4 Cause of Motion: Force

We have been describing the motion of a body without regard to what causes the motion. Hitherto motion has merely been discussed in terms of displacement, velocity and acceleration. Now we want to ask, why does a body at rest remain at rest, and a body moving with uniform velocity continue its uniform motion? What will make the body at rest to move, and a body moving with uniform velocity to change its state of motion?

A block of wood resting on a table will remain at rest until it is pushed or pulled by an agent. Similarly a trolley moving uniformly along a plane smooth straight surface will continue this state of uniform motion in a straight line unless an agent pushes it or pulls on it.

Such an agent that changes or tends to change the state of rest or of uniform motion in a straight line of a body is called *force*.

Force is the term used in everyday discussion. Though it may be difficult for us to describe force, it is however easy for us to see the effect of force, as we now illustrate in the following activities:

Activity 2.1

1 Hold the two ends of a strip of rubber band and try to stretch it. Allow it to return to its original length. Do this several times. What do you feel? Repeat this activity using a spiral spring.

2 Hold the two ends of a metal rod about 40 cm long and try to bend the rod through these ends. What did you feel? Gradually return the rod to its original position and note your feelings in your muscles.

3 Use a hammer and knock on the sides of an empty metal can. What happens to the shape of the can?

4 Place some iron nails on a table, Now bring a magnet near these nails. What happens?

In each of these activities we have either felt or seen the presence or the effect of force. As we pulled the rubber and spiral spring to stretch them, they also pulled on our hands as they tried to return to their original length. As we hit the empty metal

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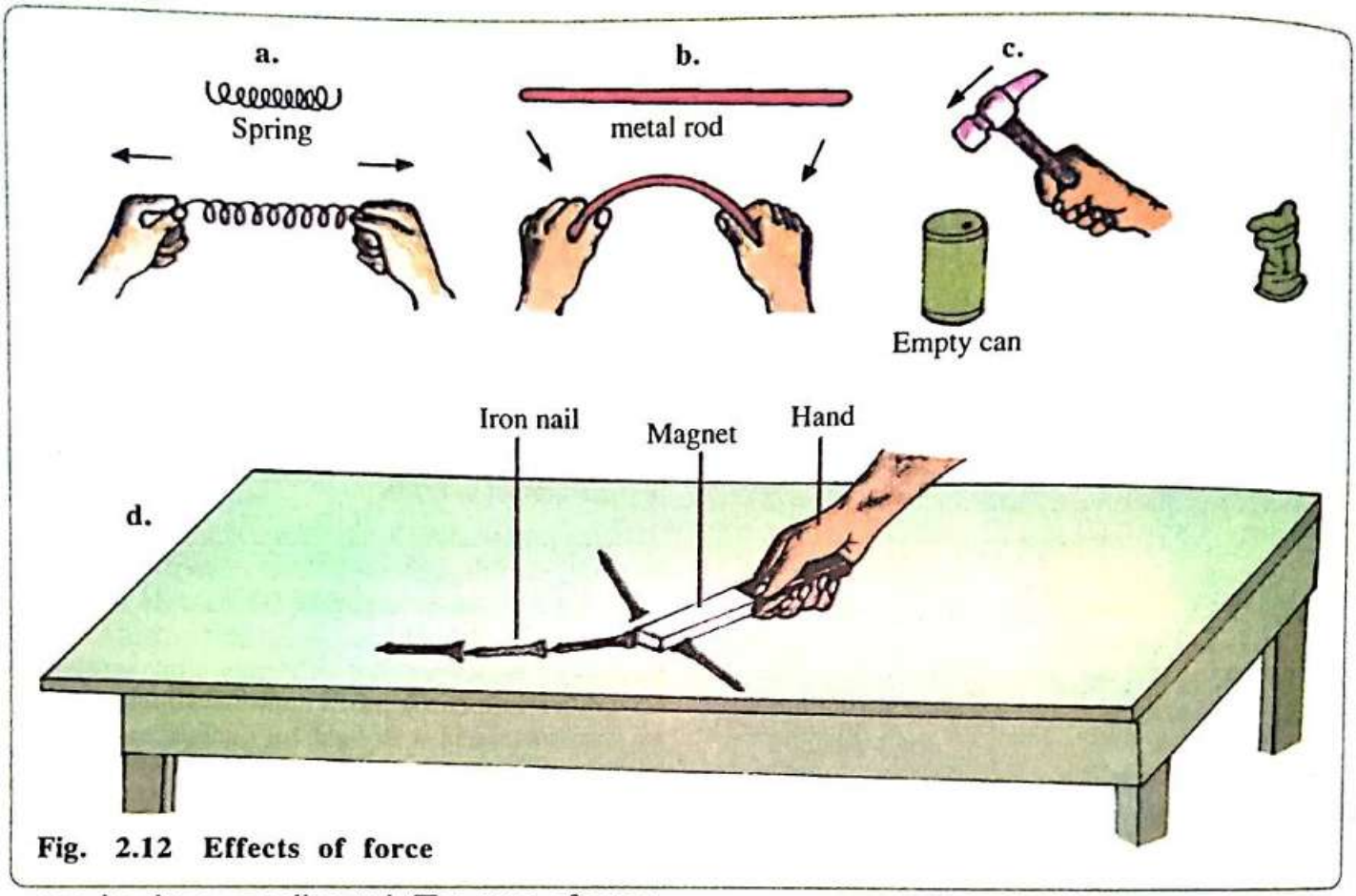


Fig. 2.12 Effects of force

can, its shape was distorted. The cause of every motion is a force.

A force can move a body, change its direction of motion, make it to slow down or stop moving altogether, distort or change the shape of a body to which it is applied. Fig 2.12 shows that force can increase the length of a spring, bend a rod, distort the shape of an empty can or move a body at rest.

Types of forces

There are two types of forces (a) *contact forces* (b) *force fields.*

a. Contact forces are forces which are in contact or in touch with the body to which they are applied. Examples are forces of push, pull, tension, reaction and frictional forces.

b. Force fields: These are forces whose sources do not require contact with the body to which they are applied. Examples are *gravitational* force, *electrical* (electrostatic) force, *magnetic* force.

Gravitational force is the force with which the earth attracts objects towards its centre. Electrostatic force is the force that exists round charged bodies. Magnetic force exists around a bar magnet. The effects of such forces are felt by bodies which are not in contact with them. Their actions are felt at distance (e.g. fig 2.12d).

2.5 Friction

When a solid is in contact with another, the roughness of their surfaces prevents them from sliding freely over one another. Such action is brought about by the effect of friction.

Friction is defined as a force which acts at the surface of separation between two objects in contact and tends to oppose the motion of one over the other.

Frictional forces act tangential to the surface of separation between two bodies in contact. Friction is a force of opposition. It does not appear unless there is relative motion, or a force tending to produce motion, then it always appears to oppose the motion or the force tending to produce motion. Some frictional force has to be overcome before an object can move over another, frictional force also continues to act as long as one body moves over another. When the force producing the motion is withdrawn, friction slows down the moving body until it eventually comes to rest.

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Static and dynamic friction

We can observe the effect of friction by attaching a spring balance to a block of wood and pulling the balance along a smooth board (fig 2.13).

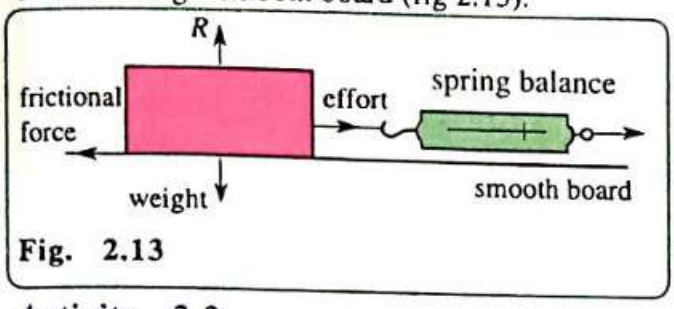


Fig. 2.13

Activity 2.2

Obtain a rectangular wooden block with broad and narrow sides from your carpentry workshop. One broad side and one narrow side of the block should be made as smooth as possible, while the sides opposite to these should have rough surfaces. Place one of the broad but smooth sides on a smooth board and attach a spring balance to the end of the block. Pull lightly on the spring balance and note the effect on the block. Gradually increase the pull on the block until the block just begins to move. Record the maximum force required to start the movement of the block by taking the reading of the spring balance just as the block starts to move. This maximum force is called the *limiting frictional force*. Also record the force necessary to keep the block sliding uniformly along the board.

Repeat the procedure using one of the rough surfaces and compare the forces required in both sets of experiments.

Place the narrow side of the block in contact with the board and repeat the above set of experiments. Are there any changes in the magnitude of the forces involved?

Static or limiting friction is the maximum force that must be overcome before a body can just start to move over another.

When the block is pulled along so that it moves on the surface of the board with uniform speed, the frictional force opposing motion is known as *sliding*, *kinetic* or *dynamic* friction. Sliding friction is less than static friction.

Kinetic or dynamic friction is the force that must be overcome so that a body can move with uniform speed over another body.

Results of experiments lead to the following *laws of solid friction.*

1 Friction opposes the relative motion between two surfaces in contact. It acts in a direction opposite to that of the motion.
2 The force of friction increases to the same extent as the force which tends to start the motion.
3 Frictional force depends on the nature of the two surfaces in contact – the nature of the substance and the conditions of the surfaces such as rough, smooth, polished, wet, dry, etc.
4 It is independent of the area of the surfaces in contact.
5 It varies directly with the normal force pressing the surfaces together, i.e. it is proportional to the normal reaction R.

Results of experiments show that the frictional force F is roughly proportional to the normal reaction R between the two surfaces in contact.

i.e. $F \propto R$

or $F = \mu R$ 2.4a

The constant μ is known as the *coefficient of friction.*

$$\mu = \frac{F}{R} \qquad 2.4b$$

Coefficient of friction (μ) is defined as the ratio.

$$\mu = \frac{\text{frictional force } F}{\text{normal reaction between the two surfaces in contact } R}$$

For static friction, the frictional force F required to determine μ in equation 2.4 is the maximum force required to start the body moving and μ is then called the *coefficient of static or limiting friction.* For kinetic friction, the frictional force used is the force required to keep the body in uniform motion, and μ in equation 2.4 is known as *coefficient of dynamic* or *kinetic friction.* The coefficient of static friction is greater than the coefficient of dynamic friction, for it is a matter of common experience that more force is required to start a body sliding along a surface than is required to keep it in motion once it is started.

Experimental determination of coefficient of static friction

A level board equipped with a pulley at one end is used (fig 2.14). A block of wood B is first weighed with a spring balance and its weight recorded (W_B). The block of wood is placed on the board and connected to a cord adjusted in such a way that it pulls parallel to the surface of the board. The cord passes over a pulley and is attached to a scale pan to which known weights can be added. Known

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weights are suspended in the scale pan until the block is just about to move.

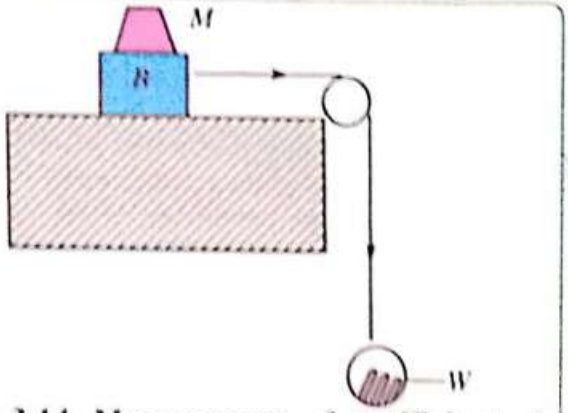


Fig. 2.14 Measurement of coefficient of friction

The block is tapped gently to prevent sticking each time a weight is added into the pan. A known weight M is placed on the block B and the weight on the scale pan is increased until the block is just about to move again after gentle tapping. The experiment is repeated five more times by varying M and noting the corresponding values of W in the scale pan. The total weight pulling the block also includes the weight of the scale pan (w_0).

A table is drawn as shown

M	$R = W_B + M$	$F = W + w_0$

A graph of force $(W + w_0)$ against normal reaction $R = (W_B + M)$ is then plotted and the slope gives the coefficient of static friction.

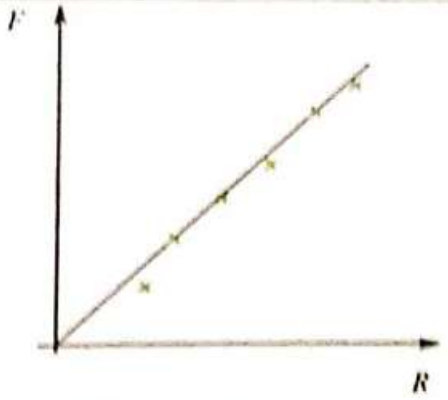


Fig. 2.15 Graph of F against R

Example 2.3

A block of wood weighing 4 N is placed on a horizontal table. It is then pulled by means of a spring balance attached to one of its ends. The block just begins to move when the spring balance records a horizontal force of 2.5 N.

(i) What is the coefficient of static friction?

(ii) What is the frictional force when a horizontal force of 2.0 N is recorded by the spring balance?

Solution

(i) Limiting frictional force $F = 2.5$ N

Normal reaction $R = 4$ N

$$\mu = \frac{F}{R} = \frac{2.5\ \text{N}}{4\ \text{N}} = 0.625.$$

(ii) When the applied force is 2.0 N the block will not move because 2.0 N is less than the limiting friction of 2.5 N. The applied force less than limiting frictional force is always balanced by opposing frictional force.

Hence frictional force = 2.0 N

Advantages of friction and its uses

Friction is desirable for a number of reasons:

(i) Without friction, it would be impossible to walk or stop walking after getting started. We walk only with the help of friction between our feet and the ground. If the ground is slippery (i.e. little friction) and cannot provide enough friction, our feet will not experience the reaction force which moves us forward. So we slip and fall.

(ii) Friction enables the automobile tyres to make a firm grip with the roadway, thus aiding their movement. This is made possible by the pattern of the threading on vehicle tyres which are designed to give the tyres a good grip on the road even when it is wet.



Fig. 2.16 Automobile tyre

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It is the frictional force between the tyres and the road which enables a car to move forward.

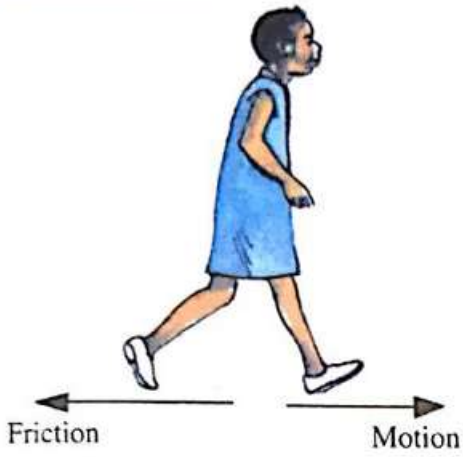


Fig. 2.17 The role of friction in walking

(iii) Friction is utilised in fan belts used over wheels or pulleys in machinery. The belting that drives the wheels in machines depends on friction to make the wheels to turn without slipping.

(iv) Friction enables a screw or a nail to remain in place after being screwed into position. Such nails thus hold pieces of wood together because of the friction between the nails and the wood.

(v) Friction enables the brakes to stop the car, and the locomotive to pull the train.

(vi) Friction is used in the grindstone to sharpen knives and chisels. The grindstone has a very rough surface which wears away the metal surface to sharpen them.

Disadvantages of friction

1 Friction leads to much wear and tear on the moving parts of machines.

2 Much useless work is done by machine in overcoming friction which opposes their motion. Thus friction causes loss of energy in machines and, therefore, reduces their efficiency.

3 Friction causes the heating of engines. This is because the energy which must be used to overcome friction is converted into heat. Consequently, the bearings of machines get hot when they are running. Sometimes the energy is partly converted to sound, hence the squeaks and groans we hear when we ride an old rusty bicycle.

Methods of reducing friction

Three chief ways of reducing friction are:

(i) The use of lubricants like oil, grease, air, and graphite.

(ii) The use of ball or roller bearings, and

(iii) The streamlining of body shapes of moving objects.

Lubricants

Friction can be reduced by using lubricants such as oil. More effective in reducing friction between two surfaces is to maintain a layer of air or other gas between them. The purpose of lubricants is to keep the two metal surfaces from coming into direct contact and thus lessen the friction between them.

Balls or roller-bearings

Where a wheel is turning on an axle, friction is better reduced through the use of ball or roller bearings. These are like wheels within wheels and are precisely shaped and have specially hardened surfaces.

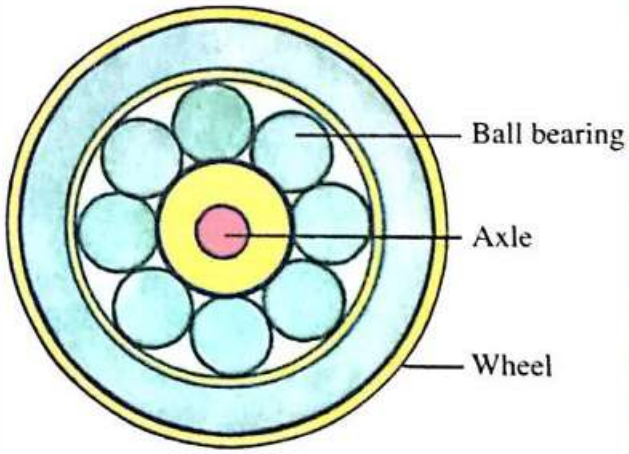


Fig. 2.18 The use of ball-bearings in rotating parts reduces friction

In such roller bearings the surfaces do not slide but roll over the others, and rolling friction is much less than sliding friction.

Streamlining

By shaping a body to the streamlines of the fluid through which it is moving, the retarding force or frictional force may be greatly reduced. Hence modern cars, ships and aeroplanes are made in such shapes as to lessen the friction between them and the fluid or the medium through which they move.

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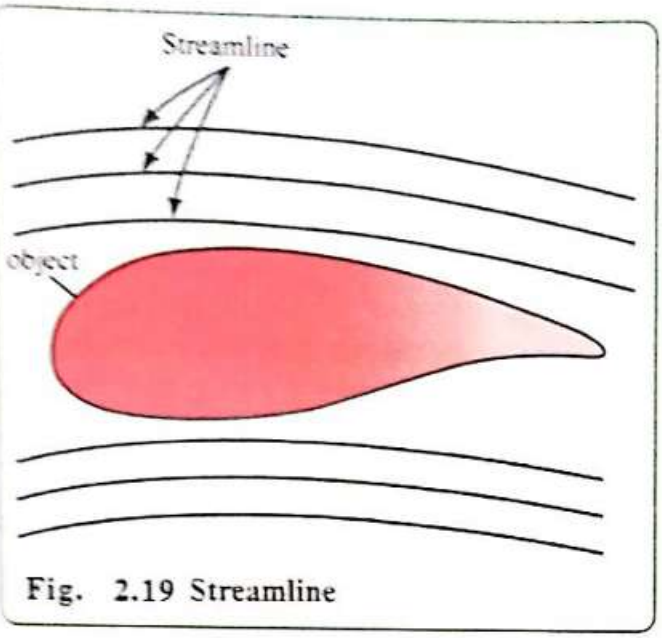


Fig. 2.19 Streamline

2.6 Simple Idea of Circular Motion

An object moving with a constant speed along a circular path is said to have a uniform circular motion. Examples of such motion are

(i) a stone tied to a string which is whirled in a horizontal or vertical circle;
(ii) a satellite circling around the earth;
(iii) the earth moving round the sun;
(iv) the moon circling the earth;
(v) the planets moving round the sun.

The speed of the object is constant in magnitude but its direction is changing in a uniform manner. The velocity of the object is, therefore, changing. Since acceleration is defined as the rate of change of velocity, a change in the direction of a velocity constitutes an acceleration just as does a change in magnitude.

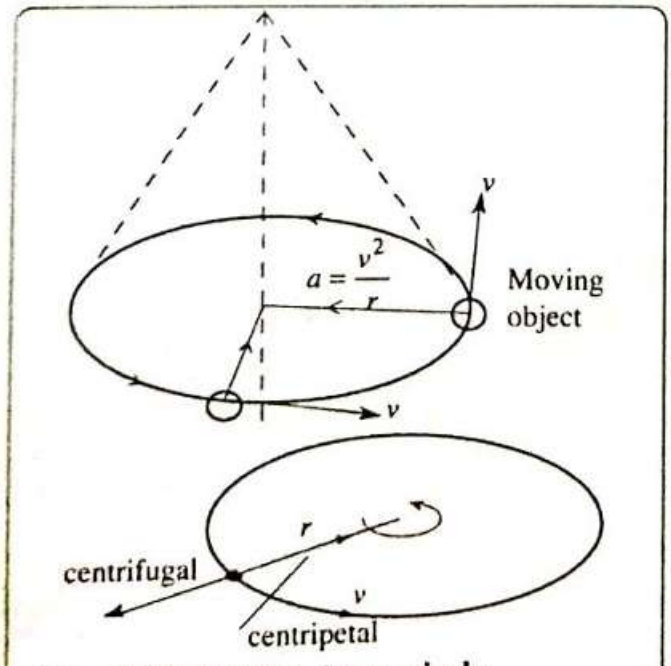


Fig. 2.20 Motion in a circle

This acceleration is directed towards the centre of the circular path and is known as *centripetal acceleration*. Its magnitude a is given by

$$a = \frac{v^2}{r} \qquad 2.5$$

where v is the uniform speed and r is the radius of the circular path. While v is always directed along the tangent to the circle at any point of the path, a is always directed towards the centre of the circle. This inwardly directed acceleration leads to a force known as *centripetal force* which is also directed towards the centre of the circle.

Centripetal force F_T is defined as that inward force required to keep an object moving with a constant speed in a circular path.

The centripetal force is given by

$$F_T = \frac{mv^2}{r} \qquad 2.6$$

where m is the mass of the object moving with a uniform velocity v in a circular path or radius r. When m is in kilogram, v in metre per second, and r is in metres then the derived unit of F_T is in Newtons.

The direction of the centripetal force or centripetal acceleration is perpendicular to the direction of the velocity. Mud or water clings to an automobile tyre until the speed becomes high enough and then the mud or water flies off on a tangent as shown in fig. 2.21.

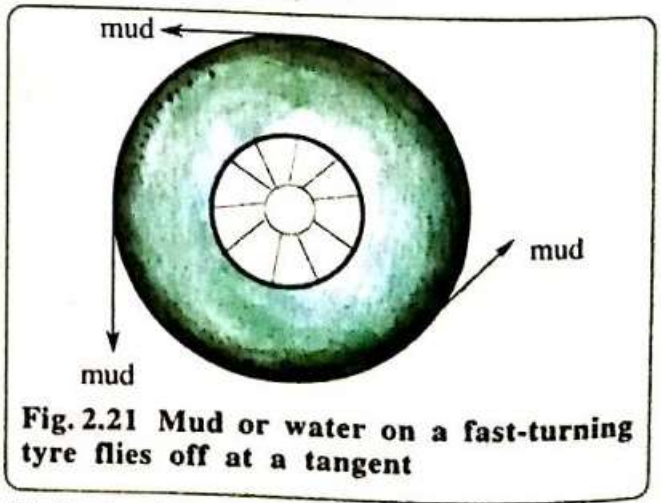


Fig. 2.21 Mud or water on a fast-turning tyre flies off at a tangent

Similarly when a string tied to a stone which is whirled in a circular path is suddenly cut, the stone flies off tangential to the circular path.

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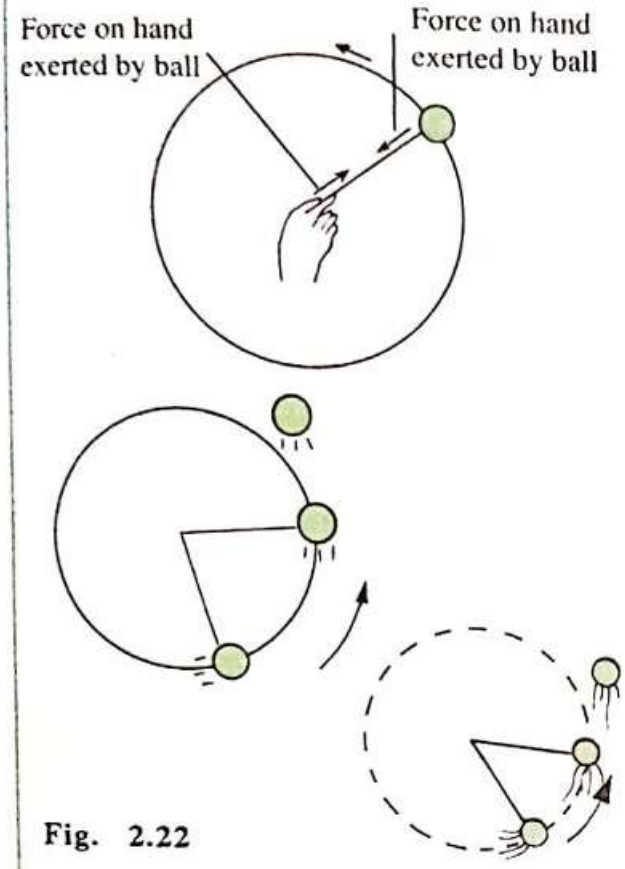


Fig. 2.22

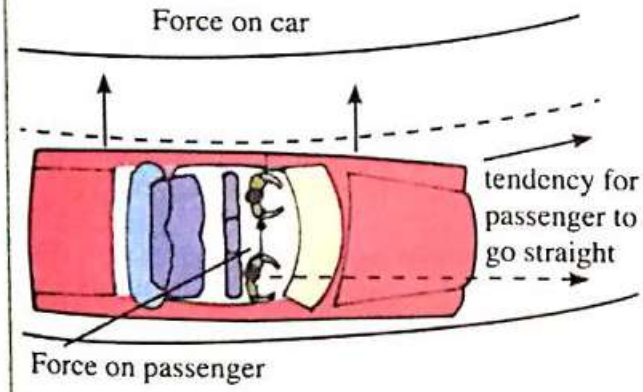


Fig. 2.23 Car rounding a curve

For the stone tied to a piece of string and whirled in a circular path the tension in the string provides the centripetal force. For a car moving along a curved path, friction between the tyres and the road provides the centripetal force needed by the car to make the turn. If the frictional force is not great enough (as on wet tarmac) sufficient force cannot be supplied and the car will skid out of a circular path into a more nearly straight path (fig. 2.23).

Banking of curves on roads can reduce the chance of skidding because the normal force of the road will have a component towards the centre of the circle (fig. 2.24) thus reducing the need for friction. When road curves are 'banked', they are constructed with a slope towards the inward edge.

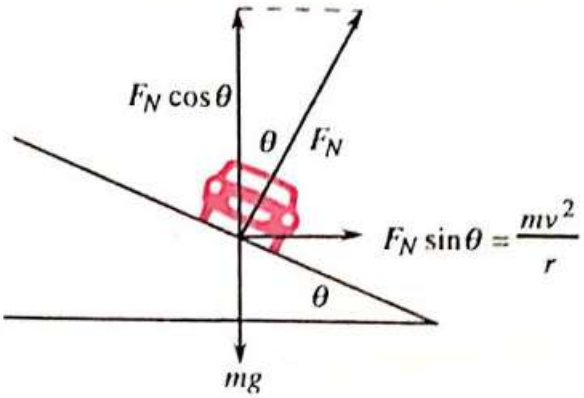


Fig. 2.24 Normal force on a car rounding a banked curve, resolved into its horizontal and vertical components

Centrifugal force

Since a *centripetal force* must be applied to a body to restrain or keep it moving in a circular path, there must be according to Newton's third law of motion an equal and oppositely directed force. This force is known as *centrifugal force*. When a stone tied to a string is whirled round in a circular path, the centripetal force is the inward force exerted by the string on the stone, but according to Newton's third law (Action and reaction are equal and opposite), the stone exerts an equal but opposite force on the string. This is the *centrifugal force*.

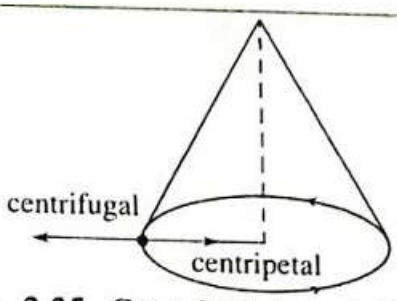


Fig. 2.25 Centripetal and Centrifugal force

Examples and applications of centrifugal force

The effects of centrifugal force is felt by a person riding in a car as it turns a corner.

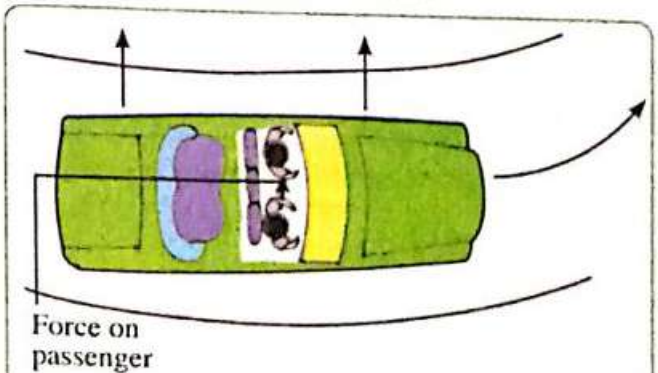


Fig. 2.26 Motion of car round a bend

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Motion of Motorcyclist Round a Curved Track

When going round a curved track, a cyclist has to lean inwards. Why? Fig. 2.27 show a motorcyclist trying to go round a bend without leaning inwards.

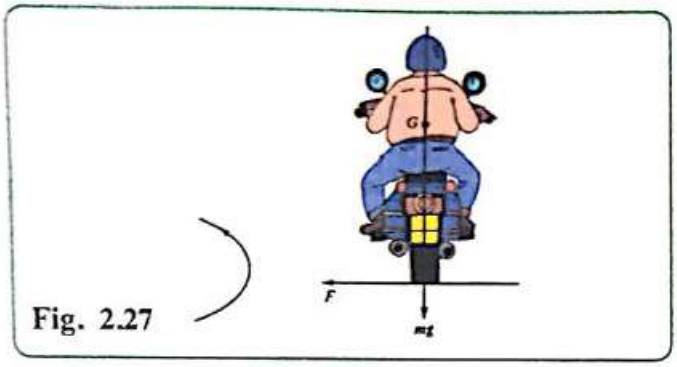


Fig. 2.27

Besides the weight, mg and the normal reaction, R, (R = mg); there is a sideway frictional force between the tyres and the road which provides the centripetal force F.

If the motorcyclist tries to round the bend without leaning inwards, as shown in Fig. 2.27 the frictional force F provides a torque on him. This torque would rotate the motorcyclist in the clockwise direction about G and cause him to topple.

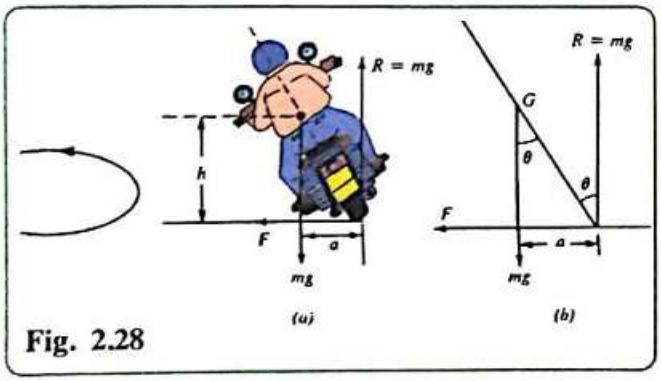


Fig. 2.28

When the motorcyclist leans inwards (Fig 2.28) the weight mg and the normal reaction R form a couple. This couple balances the torque due to the force F. Therefore, he can round the bend safely.

$$\text{Torque due to } F = \text{moments of couple}$$

$$Fh = mga$$

$$\frac{mv^2}{r}h = mga$$

$$\frac{m}{h} = \frac{v^2}{rg}$$

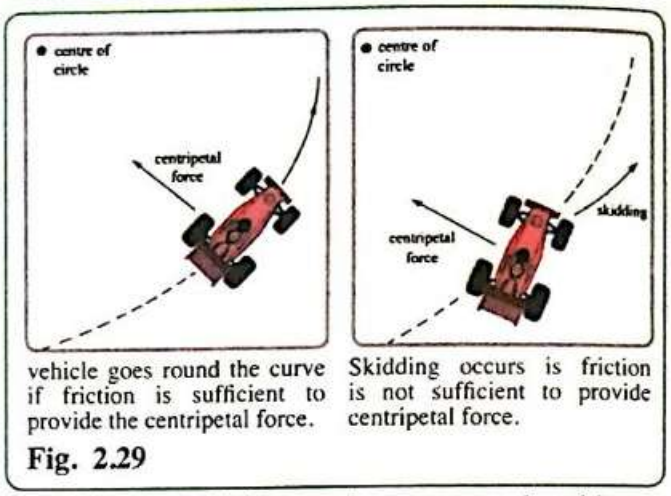
If θ = angle the motorcyclist leans from the vertical,

$$\tan\theta = \frac{a}{h} = \frac{v^2}{rg}$$

$$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$$

That is, the angle θ has to increase if (a) the speed v is higher, (b) the radius r is smaller so that the motorcyclist rounds the bend safely.

Vehicle Round a Curved Track



vehicle goes round the curve if friction is sufficient to provide the centripetal force.

Skidding occurs is friction is not sufficient to provide centripetal force.

Fig. 2.29

When a vehicle rounds a corner, the sideway friction between the tyres and the ground provides the centripetal force that enables the vehicle to follow the curved track. If the centripetal force required is large, such as when the vehicle goes round the track at a high speed, the sideway friction is not sufficient to provide the centripetal force. The vehicle fails to follow the curved track and the tyres slide sideways. The car skids.

An application of the effect of centrifugal force is the use of the *centrifuge* to separate particles from liquid in which they are suspended, or to separate two liquids of slightly different densities, e.g. cream and milk. The centrifuge is used in the medical field to separate the various solid constituents of blood which have different mass densities. In the *centrifugal drier*, the centrifugal force is used to separate water from a wet cloth.

Angular speed and velocity

Consider the speed of a stone that is whirled on the end of a string. The stone moves in a circular path as shown.

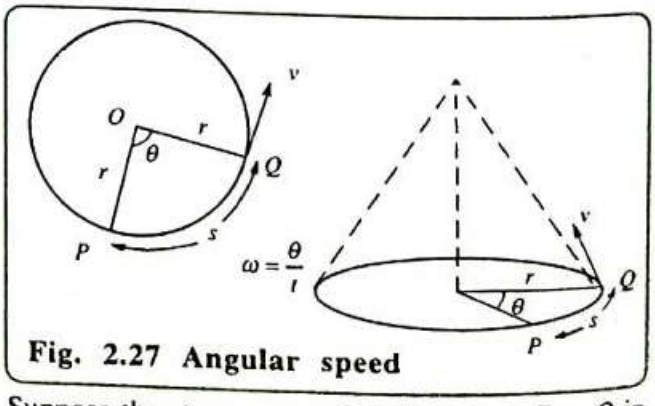


Fig. 2.27 Angular speed

Suppose the stone moves from the point P to Q in t seconds so that the radius OP sweeps through an angle θ in the same time, i.e. angle $P\hat{O}Q = \theta^\circ$, the angular velocity of motion, ω is defined by

$$\omega = \frac{\theta}{t} \qquad 2.7$$

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Angular velocity (ω) is defined as the angle turned through, divided by the elapsed time.

We can compare this with the corresponding definition of linear velocity v given by

$$v = \frac{s}{t} \qquad 2.8$$

where s is the length of the arc PQ.

We can see from equation 2.7, that the magnitude of angular velocity is defined in analogy with ordinary linear velocity. Instead of linear displacement, we use the angular displacement θ. If the angle θ is measured in *radians*, then it can be shown that

$$\theta = \frac{s}{r} \qquad 2.9$$

The *radian* is a unit of angular measure just as the centimetre is a unit of linear measure.

The radian is defined as the angle subtended by the arc of a circle whose length is equal to the radius of the same circle.

From equation 2.9, we have $s = r\theta$. Putting this in equation 2.8, we have

$$v = \frac{s}{t} = \frac{r\theta}{t} = r \cdot \frac{\theta}{t}$$
$$= r\omega.$$

Thus linear velocity v is related to angular velocity ω by

$$v = r\omega \qquad 2.10$$

Units: s is in metres, r is in metres, v is in metre per second (ms^{-1}), ω is in radians per second ($rad.s^{-1}$).

Example 2.4

A stone whirled at the end of a rope 30 cm long, makes 10 complete revolutions in 2 seconds. Find

(i) the angular velocity in radians per second;
(ii) the linear speed;
(iii) the distance covered in 5 seconds.

Solution

(i) The angle covered in 1 revolution is given by
$\theta = 360^\circ$ or 2π radians.
Angle covered in 10 revs. $= 2\pi \times 10$ radians
$= 20\pi$ radians
Time for 10 revs. $= 2$ seconds
Angular velocity $\omega = \frac{\theta}{t} = \frac{20\pi}{2}$
$= 10\pi$ radians per second
$= 10 \times 3.14$
$= \underline{31.4\ rad.s^{-1}}$

(ii) Linear speed is given by
$v = r\omega$
$= 30\ cm \times 31.4\ rad\ s^{-1}$
$= 942\ cm.s^{-1}$
$= \underline{0.942\ ms^{-1}}$

Note that the radian as a unit has no dimension, it is a ratio between two lengths, so that $cm \times rad.s^{-1} = cm.s^{-1}$.

(iii) Distance covered in 5 seconds is given by
$s = vt$
$= 0.942 \times 5$
$= \underline{4.71\ m}$

Example 2.5

A mass of 10 kg is moving in a circular path of radius 2 m with a uniform speed of 50 ms^{-1}. Find the centripetal acceleration and the corresponding centripetal force.

Solution

Centripetal acceleration is given by

$$a = \frac{v^2}{r}$$
$$= \frac{50 \times 50}{2}$$
$$= 1250\ ms^{-2}.$$

Centripetal force is given by
F = mass × acceleration
$= 10 \times 1250$
$= 12500\ kg.m.s^{-2}$
$= \underline{12500}$ newtons.

Activity 2.3

1 Obtain a pillow made of foam rubber. Press down hard with your two hands on the surface of the pillow. Then withdraw your hands. Describe what happened to its shape while you pushed hard on the pillow and when you withdrew your pushing. Repeat this activity on a piece of rectangular block of wood and a plasticine. Note the different effects.

2 Spread some iron fillings on a board and bring a piece of magnet near them but not touching them. Note the effect of the magnet on the iron fillings.

3 Tie a piece of cord to a stone and whirl the stone in a vertical circle. Note the effect of the tension in the cord as felt by your hand.

4 Attach a spring balance to a block placed on a wooden board. Pull gently on the spring balance and note the reading when the block just begins to move. Place another wooden block on top of the first and repeat the experiment. Note the new force required to start the block moving.
Repeat the experiment

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a. with chalk powder dusted on the block,
b. with the surfaces of the block and the board wet,
c. with the block resting on top of some small spherical beads spread on the board.

Note the force required to start the block moving in each instance.

1 We can distinguish between four types of motion –
(i) **random motion** in which a body moves irregularly with no preferred direction,
(ii) **translational motion** in which every point in a moving body remains fixed in relation to other points,
(iii) **rotational motion** where every point in a body moves in circles, and
(iv) **oscillatory or vibratory motion** in which a body moves to and fro, reversing its direction, and returning regularly to its position.

2 **Relative motion** of a body A with respect to another body B is the motion A as observed from B. B is the frame of reference. In general, all motion involves the separation of an object relative to an observer (or point of reference).

3 **Displacement** is the distance travelled in a specified direction. **Speed** is the distance travelled in unit time. **Velocity** is the rate of change of displacement with time. When a body moves with equal displacement in equal time interval no matter how small the time interval may be, the velocity is said to be a **uniform or constant velocity**

4 The slope of a displacement/distance - time graph is the velocity/speed.

5 **Instantaneous velocity** is the velocity at any instant in time.

6 **Acceleration** is the rate of change of velocity with time.

7 The area under a velocity - time graph gives the total displacement.

8 The cause of every motion is force.

9 **Force** is that which changes the state of rest or of uniform motion of a body. There are two types of forces – the **contact force** and the **force field** Contact forces are in contact with the object of their application. Force fields are forces felt at a distance.

10 **Friction** is the force which acts to oppose motion at the surface separating two objects in contact. **Static or limiting friction** is the maximum force that must be overcome before a body can just start to move over another. **Kinetic or dynamic friction** is the force that must be overcome so that a body can move with uniform speed over another.

11 Coefficient of static friction is the ratio of the frictional force (F) to the normal reaction R.

$$\mu = \frac{F}{R}$$

12 Frictional force is independent of the areas of surfaces in contact.

13 Friction has both desirable and undesirable effects.

14 Friction is reduced by lubrication, ball bearings and by streamlines.

15 In a **circular motion**, there is always a force directed towards the centre of the circular path. This is the **centripetal force**. This force keeps the body moving in a circular path. A **centrifugal force** acts in opposite direction to the centripetal force.

16
$$\omega = \frac{\theta}{t}$$
$$v = r\omega$$
$$v = \frac{s}{t}$$

ω is the angular velocity, θ is the angle covered by the moving object, t is the time of motion, v is the linear velocity.

17 Centripetal acceleration a is given by

$$a = \frac{v^2}{r}$$

where r is the radius of the circular path, and v is the speed of the moving body.

1 List four types of motion and give an example of each.

2 Define **motion, relative motion** Give an example of relative motion. A bus is travelling along a straight road at 100 $kmhr^{-1}$ and the bus conductor walks at 6 $kmhr^{-1}$ on the floor of the bus and in the same direction as the bus. Find the speed of the conductor relative to the road, and relative to the bus. If the bus conductor now walks at the same rate but in opposite direction as the bus, find his new speed relative to the road.

3 Distinguish between
a. *distance* and *displacement*,
b. *speed* and *velocity*.
How can you find speed on a distance - time graph?

4 What is meant by the following: *uniform velocity, average velocity, instantaneous velocity, acceleration, retardation*?

5 A car travels for 3 hours to a distant city 200 km due East. What was its (1) speed, (2) velocity?

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6 A bus travels with an average velocity of 60 kmh^{-1}. How long does it take to cover a distance of 500 km?

7 What causes motion? Distinguish between *contact force* and *force field*. Give two examples of each.

8 Distinguish between *instantaneous speed* and *average speed*. A man drives a car a distance of 200 km at an average speed of 44 kmh^{-1}. What must be his average speed for the next 220 km if he is to cover the total distance in 9 hours?

9 How much longer would it take to travel a distance of 600 km at a speed of 50 $kmhr^{-1}$ than it would at a speed of 60 $kmhr^{-1}$?

10 A stone tied to a string is made to revolve in a horizontal circle of radius 4 m with an angular speed of 2 radians per second. With what tangential velocity will the stone move off the circle if the string cuts?

A 16.0 ms^{-1} B 8.0 ms^{-1}
C 6.0 ms^{-1} D 2.0 ms^{-1}
E 0.5 ms^{-1}? (SSCE)

11 Which of the following is not a consequence of a force field?

A Weight B Surface tension
C Gravitational pull D Magnetic force
E Electrical force? (SSCE)

12 The slope of a straight line displacement-time graph indicates

A distance travelled
B uniform velocity
C uniform acceleration
D acceleration at an instant
E uniform speed. (SSCE)

13 Which of the following correctly gives the relationship between linear speed v and angular speed ω of a body moving uniformly in a circle of radius r?

A $v = \omega r$ B $v = \omega^2 r$ C $v = \omega r^2$
D $v^2 = \omega r$ E $v = \frac{\omega}{r}$. (SSCE)

14 What are the units of (a) displacement (b) velocity (c) acceleration?

15 A car moves with a velocity of 72 $kmhr^{-1}$. It is brought to rest in 10 s. Find (i) the velocity in ms^{-1}, (ii) the retardation.

16 What do you understand by the terms *friction, coefficient of static friction, limiting frictional force, coefficient of sliding friction*? List two methods of reducing friction.

17 Define coefficient of static friction. A rectangular block rests on a horizontal surface. Explain why a minimum force is required to move the block along the surface. State whether this force will be larger, smaller or unchanged when

a. the block is turned on to another side such that a bigger area is in contact with the surface;
b. a film of oil is placed between the block and the surface.

18 a. State the laws of static friction.
b. Describe an experiment to determine the coefficient of static friction between two surfaces.
c. State
(i) two advantages of friction,
(ii) two disadvantages of friction,
(iii) two methods of reducing friction. (SC/GCE)

19 A metal block of mass 5 kg lies on a rough horizontal platform. If a horizontal force of 8 N applied to the block through its centre of mass just slides the block on the platform, then the coefficient of limiting friction between the block and the platform is:

A 0.16 B 0.63 C 0.80
D 1.6 E 2.00. (JAMB)

20 The speed of a bus increases uniformly from 15 ms^{-1} to 60 ms^{-1} in 20 seconds. Calculate

a. the average speed,
b. the acceleration,
c. the distance travelled during the entire period.

21 During the same time interval, it is observed that a train travels the same distance as does a lorry. The two vechicles therefore have the same.

(a) Uniform acceleration
(b) Instantaneous velocity
(c) Initial velocity
(d) Average speed
(e) Average velocity (SSCE)

22 A piece of stone attached to one end of a string is whirled round in a horizontal circle. When the string is suddenly cut the stone will

(a) Move towards the centre of the circle
(b) Stop moving immediately
(c) Move along a circular path of smaller radius
(d) Fly off in a direction tangential to the circular path
(e) Acquire greater centripetal force (SSCE)

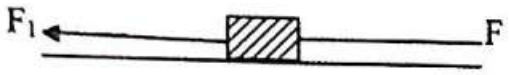
23 A body of mass 25kg, moving at 3 ms^{1} on a rough horizontal floor is brought to rest after sliding through a distance of 2.5m on the floor. Calculate the coefficient of sliding friction [g = 10.0ms-2]

(a) 0.09 (b) 0.18 (c) 0.36 (d) 0.54 (SSCE)

24 Two buses leave a bus station and travel in opposite directions from that same starting point. If the speed of one is twice the speed of the other and they are 240Km apart at the end of one hour, what is the speed of each bus?

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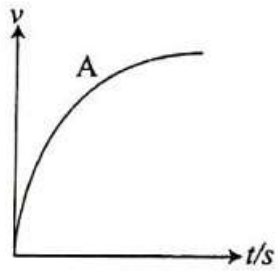
25 (a) Give two examples each of the following
(i) Rotational motion
(ii) Linear motion
(b) Describe a laboratory experiment to determine the density of an irregular shaped solid
(c) State Newton's second law of motion
(d) Explain the term *inertia*
(e)

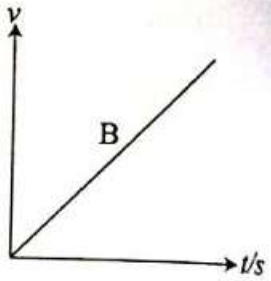


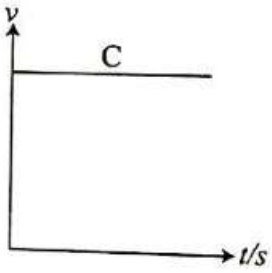
The above diagram illustrates a body of mass 5.0Kg being pulled by a horizontal force *F*. If the body accelerates at 2.0 ms^{-2} and experiences a frictional force of 5 N, calculate the
(i) net force on it
(ii) magnitude of *F*
(iii) coefficient of kinetic friction
[g = 10 ms^{-2}] SSCE

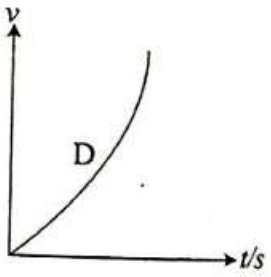
26 If no net force acts on an object the object maintains a state of rest or constant speed in a straight line. The above is a statement of Newton's
A first law
B second law of motion
C third law of motion
D law of universal gravitation

27 Which of the velocity-time graphs below correctly describes the motion of a body pulled by a force that is equal to the frictional force acting on it?









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3 WORK, ENERGY AND POWER

In this chapter, we will learn the meanings of work, energy and power. We will also show that they are closely linked.

At the end of this chapter the student should be able to:

1 Explain work, energy and power and give examples of each.
2 Calculate
   a. the work done, given a force and displacement it produces in its direction.
   b. the gravitational potential energy at a height h above a given reference plane.
3 Calculate the power in watts given an applied force and the time it takes to produce a displacement.
4 Identify the type of energy possessed by a body under given conditions.
5 Distinguish between kinetic and potential energy.
6 Identify energy transformation from one form into another.
7 State the law of conservation of energy.

3.1 Work

Concept of work

Different people have different concepts of **work**. Hence one person may say 'I go to work'; another may say 'I work in the school farm' or 'My machine works in such and such a way'. In every day life, the term *work* is applied to any form of activity that requires the exertion of physical or mental effort.

In physics, (or in mechanics) however, the term is used in a very specific sense. **Work** is said to be done whenever a force is applied to cause a body to move. When you push a car a certain distance, or you lift a load from the floor on to a table, or you stretch or compress an elastic body some distance, you are said to do work. When no movement takes place after you have applied a force to a body, no work is done. If you lift a bag of palm kernels from the ground on to your head, and carry the load standing on the same spot even for several hours, you have only done work by lifting the load from the ground to your head. You have done no work in carrying the bag for so long, standing on the spot.

Work is said to be done whenever a force moves its point of application a distance in the direction of the force.

Measurement of work

If a force F moves a body a distance S in the direction of the force the work done is given by:

$$W = F \times S \qquad 3.1$$

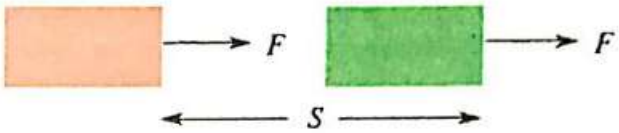


Fig. 3.1 Work = $F \times S$

Work done is defined as the product of the force and the displacement in the direction of the force. $W = F \times S$

The unit of work is the *joule* when the force is measured in *newtons* and the displacement is in *metres*. Thus the work done when a force of 10 newtons displaces an object through 5 metres is given by $W = 10 \times 5 = 50$ joules.

Work	= Force ×	Displacement
(joules)	= newtons ×	metres

Work done in a force field

a. Lifting a body

In chapter 2 (p. 17) we introduced the idea of a *force field*. We also noted that the earth's gravitational field was an example of a force field. In that gravitational field there is always a force pulling a body towards the earth's centre. We defined the weight of a body as the force of attraction on the body due to the earth's gravity. This weight acts downwards. To lift a load through a height h, a pulling force must be applied to overcome the weight of the body. Therefore, when an object is lifted vertically upwards, work is done against the force of gravity or against the weight of the body. The magnitude of the work done is given by

$$\text{Work} = \text{force} \times \text{distance}$$
$$= mg \times h$$
$$= mgh \qquad 3.2$$

where m = mass of the body, g = acceleration due to gravity and h = height. ($g = 10\ \text{ms}^{-2}$)

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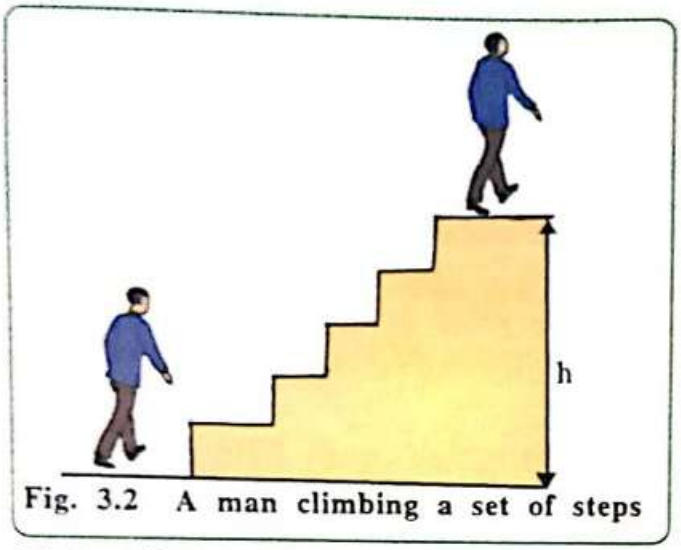


Fig. 3.2 A man climbing a set of steps

If a boy of mass m, moves up a set of steps of total height h, the work done by the boy = his weight (mg) × the height h.

Hence to lift a body against a force field (e.g. gravity) an opposing force is needed. Work done against gravity is equal to the product of the weight of the body (mg) and the vertical upward displacement (h).

$$w = mgh \qquad 3.3$$

Example 3.1

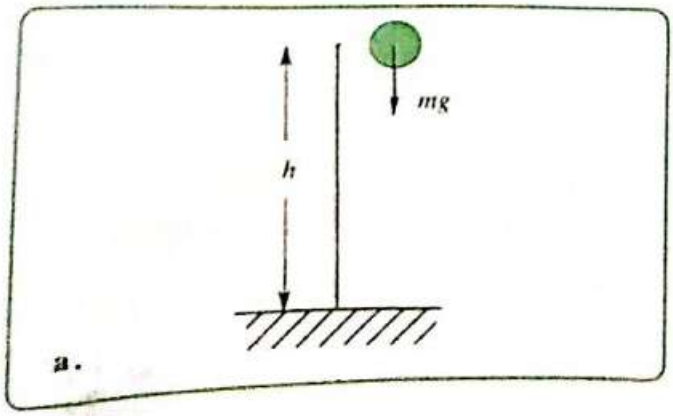
A boy of mass 50 kg runs up a set of steps of total height 3.0 m. Find the work done against gravity.

Solution

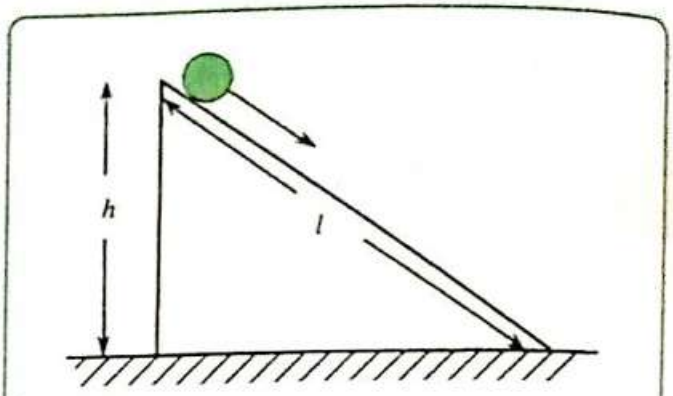
Work done = mgh
= $50 \times 10 \times 3$
= 1500 joules

b. Falling bodies

When a body falls freely in a force field, the force of the field does work on the body. Hence for a body falling freely under gravity the earth's gravitational force does work on the body. If a body of mass m, falls through a vertical height h, the work done by gravity on the body is given by $w = mgh$.



a.



b.

Fig. 3.3 Potential energy is dependent on vertical height

If a body of mass m rolls down a hill of height h and length l, the work done is mgh. It is the vertical height of the hill that is used in the calculation, not the length of the slope l.

Example 3.2

A loaded sack of total mass 100 kg falls down from the floor of a lorry 2.0 m high. Calculate the work done by gravity on the load.

Solution

$W = mgh$
$= 100 \times 10 \times 2$
$= 2000$ J

3.2 Energy

Concept of energy

Energy is defined as the capacity to do work.

Anything that is capable of doing work has energy. A person pushing a car along a road is doing work on the car. The person is said to possess energy which he exercises by making the car move some distance. A student running round the school field has energy. He is capable of moving his weight some distance. A football player, a farm labourer, a palm wine tapper climbing a palm tree and a mango fruit falling from the top of the tree all possess energy.

Work and energy are measured in the same unit, the joule. There are many forms of energy. These include:
(i) mechanical energy, (ii) heat (or thermal) energy, (iii) light energy, (iv) chemical energy, (v) electrical energy, (vi) atomic energy (vii) solar energy, etc.

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Types of mechanical energy

Mechanical energy is classified into two types – *potential energy* and *kinetic energy*.

Potential energy (P.E.) is simply 'stored energy' or the energy possessed by a body by virtue of its position or state.

Such stored energy is used to do work when the body is free to move. A heavy stone on top of a table has potential energy. When allowed to fall on to a glass plate on the floor, it will shatter the plate. The potential energy of the stone due to its position above the floor is expended in shattering the plate.

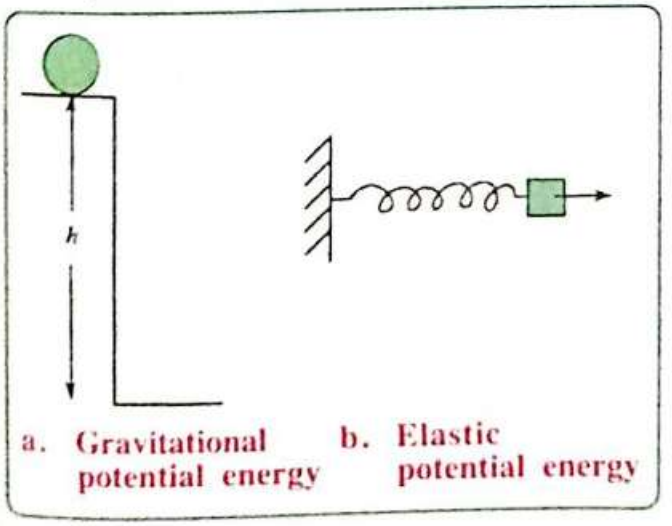
A body may have a potential energy due to its position in a force field. If the force field is the gravitational field, the body is said to possess a gravitational potential energy. The stone resting on top of a table has gravitational potential energy due to its height above the ground level. If the body is of mass m and the height of the table is h, then the gravitational potential energy is given by

$$E_{gp} = mgh$$

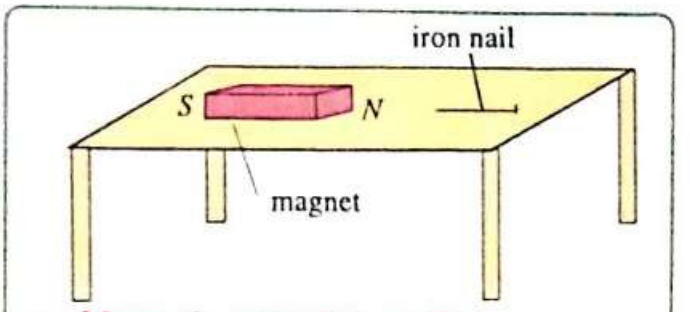
where g is the acceleration due to gravity. If m is in kg, g in ms^{-2} and h in metres, then E_{gp}, the gravitational potential energy, is in joules.

Other examples of potential energy are

(i) a magnet at rest in a magnetic field (*magnetic potential energy*),
(ii) an electric charge at rest in an electric field (*electrical potential energy*),
(iii) a coiled spring when stretched or compressed possesses *elastic potential energy*,
(iv) *chemical potential energy* is released when petrol, wood, and other fuel sources burn.



a. Gravitational potential energy b. Elastic potential energy



c. Magnetic potential energy

Fig. 3.4 Types of potential energy

These forms of energy are of immediate practical use in daily life. For example, an electric motor uses electrical energy to drive an electric train; a steam engine uses heat energy to move the coaches, sound energy causes the diaphragm of a microphone to vibrate; the chemical energy in the food we eat causes us to grow and also provides us with muscular energy for our physical activities.

Kinetic energy (K.E.) is the energy possessed by a body by virtue of its motion.

Examples of kinetic energy are

(i) a student running a race
(ii) an object falling freely under gravity
(iii) wind or air in motion
(iv) electrical charges in motion
(v) a moving bullet or a moving hammer head.

Measurement of energy

a. Potential energy (P.E.)

Gravitational potential energy depends on the weight and the height of an object above the earth's surface. If a hammer head is lifted some metres above the ground, work has been done in lifting the hammer against the weight of the hammer head. The work put into the lifting of the body is stored as potential energy at the height h above the ground. If the mass of the body is m and the height of the body above the reference level is h, then the gravitational potential energy is mgh as in equation 3.3. So to measure gravitational potential energy, we measure the mass of the body, and its height above a reference level h.

b. Kinetic energy (K.E.)

While the gravitational potential energy depends on the mass of the body, its height above a reference level and the acceleration due to gravity, kinetic energy depends only on the mass and velocity of

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the body. The kinetic energy (K.E.) of a body in motion is given by

$$\text{K.E.} = \frac{1}{2}mv^2 \qquad 3.4$$

where m is its mass and v is its velocity. If m is in kg, and v in ms^{-1}, then K.E. is in joules.

Thus if two bodies in motion have the same mass, the faster body has the greater kinetic energy and if they have the same velocity, the body with greater mass has a greater kinetic energy.

Example 3.3

Find the potential energy of a boy of mass 10 kg standing on a building floor 10 metres above the ground level.



Solution

P.E. $= mgh$
$= 10 \times 10 \times 10$
$= \underline{1000 \text{ joules}}$

Example 3.4

An object of mass 5 kg is moving at a constant velocity of 15 ms^{-1}. Calculate its kinetic energy.



Solution

K.E. $= \frac{1}{2}mv^2$
$= \frac{1}{2} \times 5 \times 15^2$ joules
$= \underline{562.5 \text{ joules}}$

3.3 Power

Power is defined as the time rate of doing work.

If two boys of the same weight climb a flight of steps of the same height, the boy that gets to the top first is said to have the greater power. This is because he has done the work of moving his weight through that height at a shorter time. If the work W joules is done in time t seconds, then the power = $\frac{\text{work}}{\text{time}}$.

$$\text{Power} = \frac{\text{work done or energy expended}}{\text{time}}$$

If the work is in joules and the time is in seconds, then the power is measured in *watts*.

1 watt = 1 joule per second

Other commonly used units of power are the kilowatt (kW), the megawatt (MW) and the horsepower (h.p.).

1 kW = 1000 W = 10^3 W
1 MW = 1,000,000 W = 10^6 W
1 h.p. = 746 W

One kilowatt-hour is the work done in one hour by an agent working at the constant rate of one kilowatt.

Example 3.5

A car is moving at a constant speed of 20 ms^{-1}. The force retarding its motion is 500 N. Calculate the engine power of the car required to maintain the motion.



Solution

Engine power = work done by the car per second in overcoming the retarding force
= force × distance moved per second
= 500 N × 20 ms^{-1}
= 10000 Js^{-1}
= $\underline{10^4 \text{ watts or } 10 \text{ kW}}$

Example 3.6

A boy of mass 10 kg climbs up 10 steps each of height 0.2 m in 20 seconds. Calculate the power of the boy.



Solution

Total height climbed = 10×0.2 m = 2 m
Work done = $mgh = 10 \times 10 \times 2$
Time taken = 20 seconds

$$\text{Power} = \frac{\text{work}}{\text{time}} = \frac{10 \times 10 \times 2}{20}$$
$$= \underline{10 \text{ watts}}$$

3.4 Transformation and Conservation of Mechanical Energy

The foregoing discussions have shown that work, potential energy and kinetic energy are all related. By means of suitable machines energy can be changed from one form to another. There is an important law of nature which governs these changes in forms of energy. This law is known as the **law of conservation of energy**.

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The law of conservation of energy states that in an isolated or closed system, the total amount of energy is always constant, although energy may be changed from one form to another.

By an *isolated* or *closed* system, we mean a group of objects that neither receives energy from or gives energy to objects outside the system. For example, the whole universe is an isolated system. Because there is nothing outside it, the universe cannot share energy with things outside it.

If we consider mechanical energy, the law shows that the sum of the potential energy and kinetic energy is always constant for a given body, but the energy may change from potential energy to kinetic or from kinetic to potential. As examples of energy conservation, let us consider the potential and kinetic energy of

(i) a falling body and

(ii) the simple pendulum.

(i) Potential and kinetic energy of a falling body

Consider a body of mass m, at rest at a point P, h metres above the ground level. At this point the total energy of the body is all potential energy and is equal to mgh. There is no kinetic energy since the body has no velocity.

Hence at P, P.E. = mgh

K.E. = 0

Total energy = P.E. + K.E. = mgh

At Q, the body has acquired some velocity, v_Q. Hence it has kinetic energy ($\frac{1}{2}mv_Q^2$). It is also at some height above the ground level, and therefore has potential energy. Hence the total energy at Q is the sum of the potential and the kinetic energies.

At R, the ground or lowest level, the energy is all kinetic energy. There is no potential energy because the height is zero.

The total energy of the body at each of the points P, Q and R is equal to a constant (mgh) although the energy is all potential at P, potential and kinetic at Q and all kinetic at R.

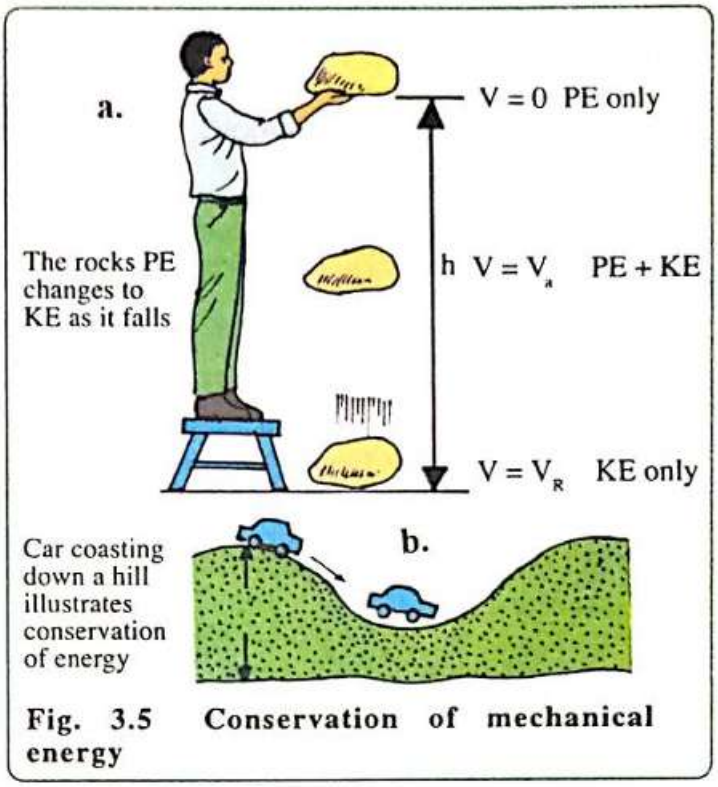


Fig. 3.5 Conservation of mechanical energy

(ii) Potential and kinetic energy of a simple pendulum

A simple pendulum consists of a bob attached to a string the other end of which is suspended from a firm support. When the pendulum swings from end to end, the energy of the system changes from potential to kinetic and vice-versa, but at each stage of the swing, the total energy remains constant.

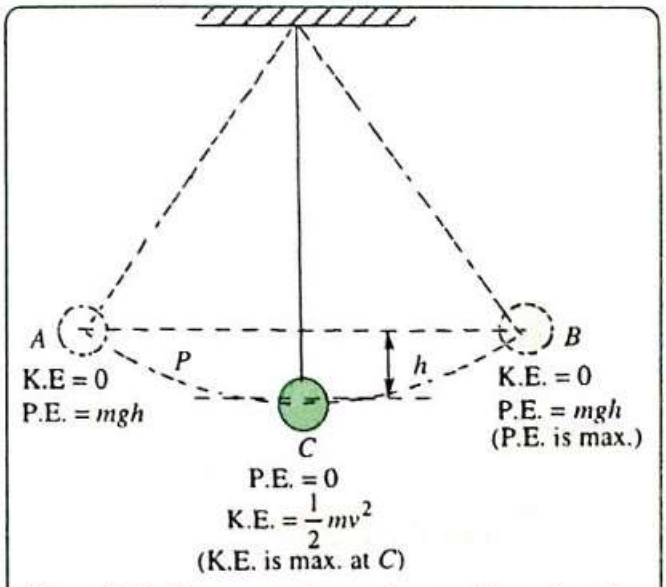


Fig. 3.6 Energy transformation in the motion of a simple pendulum

The pendulum swings from the highest point A through the centre of the swing C, to the other highest point B. At C, the bob is at the lowest position. Therefore the potential energy of the

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system is zero. Also at C, the speed of the pendulum is maximum and so the K.E. is maximum at this point. As the bob moves from C to B, the K.E. at C is gradually transformed to P.E. with the P.E. becoming maximum at B which is at a height h above C. At the point B, the total energy is potential energy and is equal to mgh. This is also the potential energy at A which is at the same height above C as B.

We can find the maximum velocity of the swinging bob, which occurs as the bob passes through C. Let this velocity be v_{max}. Since total energy is always conserved, we have:

$$\frac{1}{2}mv_{max}^2 = mgh$$

(K.E. at C) (P.E. at A or B)

Hence $v_{max}^2 = 2gh$

$$v_{max} = \sqrt{2gh} \qquad 3.5$$

At A and B, the energy is all P.E. At C, energy is all K.E. At any intermediate point e.g. P, the energy is partly potential and partly kinetic. At each point however, the total energy (P.E. + K.E.) is constant and is equal to mgh.

3.5 World Energy Resources

There are many resources currently available to the world. These can be classified into two broad classes -
(a) Renewable energy resources and
(b) Non-Renewable energy resources.

1. Renewable Energy Resources are those that can be replaced as they are used. They are continually replenished. Examples are:

(a) Solar Energy is the energy available from the sun. The sun is the ultimate energy resource available to man. It's energy does not deplete, it cannot be used up. Solar energy is used by green plants in the process of photosynthesis.
Solar energy can be converted to electrical energy using photocells. Solar radiation can be convert to chemical energy by photochemical process. Solar energy can be used for heating homes and boiling water.

(b) Wind energy driven by thermal energy of the sun is used by sailing ships and for turning windmills.

(c) Water energy provides the source of energy in modern dams, hydroelectric plants etc. Ocean tides are powered by water energy.

(d) Biomass: Energy from cornstalks, garbage, sugarcane, seaweed, etc.

2. Non-Renewable Energy Resources
Such energy resources are depleting, they cannot be replenished as they are used.

(a) Nuclear Energy from radioactive materials – this produces enormous heat to operate turbines and drive ships and aircrafts. Fission and fusion energy are examples of nuclear energy. They are further discussed in Chapter 10 of Book 3.

(b) Petroleum and Natural Gas – these often occur together and they currently constitute the world major energy resources. Such fuels known as **fossil** fuels are a depleting energy resources. Other examples of fossil fuels are coal and wood.

Wood is still a major sources of energy for the common man in Nigeria. Its use is bringing about the depletion of forests – deforestation

Summary

1 **Work** is said to be done when a force F moves a body a distance, S in the direction of the force. Work is force × displacement in the direction of the force. $W = F \times S$. It is measured in joules.
2 Work done in lifting a body of mass 'm' through a height 'h' in the gravitational field of the earth is equal to mgh joules.
3 **Energy** is the capacity to do work and is measured in joules.
4 The two types of mechanical energy are potential and kinetic energy.
5 **Potential energy** is energy by virtue of position or state. **Kinetic energy** is energy due to motion.
6 The total energy in a closed or isolated system is constant. Thus in such a system, energy can never be created or destroyed but can be transformed from one form to another. This is the **law of the conservation of energy**
7 **Power** is the time rate of doing work or the rate at which energy is expended. It is measured in Watts.

Exercise 3

(Take $g = 10\ ms^{-2}$)

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1 A man of mass 50 kg ascends a flight of stairs 5 m high in 5 seconds. If acceleration due to gravity is 10 ms^{-2}, the power expended is
A 100 W B 200 W C 250 W
D 400 W E 500 W
(JAMB)

2 What do you understand by the term *work*? What is its unit?

3 A man holds a child on his shoulders while watching a football match. How much work has he done if the game lasts for 2 hours?

4 A pendulum swings back and forth. Discuss the energy possessed by the pendulum at each stage of its swing.

5 For a freely falling body
a. the ratio of kinetic energy to potential energy is constant
b. the sum of kinetic and potential energy is constant
c. the total energy is entirely kinetic;
which of the above statements is correct?

6 A mass of 10 kg moves with a velocity of 4 ms^{-1}. Find its kinetic energy.

7 Under which of the following conditions is work done
a. A man supports a heavy load above his head with his hands
b. A woman holds a pot of water
c A boy climbs onto a table
d. A man pushes against a stationary petrol tanker
e. A bag of cocoa stands on a platform.
(SC/GCE)

8 A ball of mass 8 kg falls from rest from a height of 100 m. Neglecting air resistance, calculate its total energy after falling a distance of 40 m.

9 At what height above the ground must a body of mass 10 kg be situated in order to have potential energy equal in value to the kinetic energy possessed by another body of mass 10 kg moving with a velocity of 10 ms^{-1}?
a. 5 m b. 10 m c. 50 m
d. 100 m e. 500 m.
(SC/GCE)

10 A catapult is used to project a stone. Which of the following energy conversions takes place as the stone is released?
a. The kinetic energy of the stone is converted into gravitational potential energy.
b. The gravitational potential energy is converted into kinetic energy of the stone.
c. The elastic potential energy of catapult is converted into the kinetic energy of the stone.
d. The gravitational potential energy is converted into elastic potential energy.
e. The elastic potential energy of catapult is converted into gravitational potential energy of the stone.
(SSCE)

11 How much work is needed to stop a 20 g bullet moving with a speed of 150 ms^{-1}.

12 A boy weighing 350 N runs up a flight of stairs consisting of 20 steps each 10 cm high in 10 seconds. What is his power?

13 Distinguish between work, energy and power. If a body of weight 10 N is lifted from the floor to a table 2 m above the floor, calculate the potential energy gained. What happens to the energy if the body falls off the table and hits the floor?

14 Define the terms *potential energy* and *kinetic energy*. Give two examples of each.
State the *law of the conservation of mechanical energy*. How is the law illustrated by
a. the swing of a simple pendulum
b. the energy of a falling orange?

15 A body is displaced through a certain distance x by a force of 30 N. If the work done is 100 J and the displacement is in the direction of force, what is the value of x?

16 Find the electrical energy expended by an electric lamp labelled 100 W in 8 seconds.

17 What do you understand by *gravitational potential energy*? What is the increase in the gravitational potential energy of a 20 kg block which is lifted 10 m from the ground?

18 A man pushes a drum of oil 10 kg up an incline OA. If OA is 5 m and AB is 3 m, what is the potential energy of the drum at A?

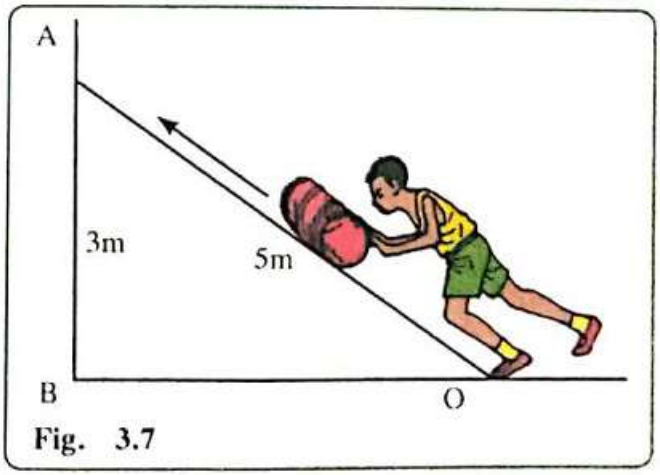


Fig. 3.7

What is the work done by the man? What work is done if the drum is just lifted from B to A?

19 A student eats a dinner containing 8.0 x $10^6 J$ of energy. He wishes to do an equivalent amount of work in a near by gym by lifting a 60Kg object. How many times must he raise the object to expand this much energy? Assume that he raises it a distance of 2.0m each time.

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4 HEAT ENERGY

In this chapter, we will introduce the concepts of *heat* and *temperature*. We will consider the effect of heat on various substances, and the various modes of heat transfer. The kinetic molecular theory will be introduced and used to explain temperature changes, expansivity and heat transfer.

At the end of the chapter, the student should be able to:

1 Explain temperature, expansion, change of state and vaporisation using the kinetic molecular theory.
2 Explain conduction, convection and radiation in terms of the kinetic molecular theory.
3 Solve simple problems involving linear, area and volume expansivity.
4 Know the difference between good and bad conductors of heat using metals, water and air as examples.

4.1 Concepts of Heat and Temperature

Heat is a form of energy called *thermal energy*. When we place a pot of cold water on a heating stove, after a few minutes the water feels hotter to the touch. We say that heat flowed from the heating stove to the cold water. It is our common experience that heat always flows from a hot to a cold object. Thus when placed in contact with a cold object, a hot object always warms the cold one. Never does the reverse occur. A cold object never warms a hot object. Heat energy is the energy that is transferred from a hot object to a cooler object as a result of their difference in temperature.

Temperature is the degree of hotness or coldness of an object.

It is a *scalar quantity*. It is a property of an object which decides which way heat will flow when it is placed in contact with another object. Heat flows from a body at higher temperature to one at lower temperature. The transfer of heat energy is as a result of the temperature difference between two objects. A hot body is at a higher temperature than a cold body. Temperature rises or falls as a body gets hotter or cooler respectively.

Thermometers are used to measure temperature. The commonest thermometer available in school laboratories is the Celsius-scale thermometer in which the temperature is measured in degree Celsius (°C).

4.2 Effects of Heat

Heat can have several effects on an object. These effects are summarised as follows:

1 **Change in temperature of the body** Addition of heat will cause the temperature of a body to rise (except during a change of state).
2 **Change of state of the body** Addition of heat can bring about a change of state of the body. For example, when heat is added to a body in the form of a solid, e.g. ice, it melts and changes to a liquid e.g. water. When enough heat is added to a liquid, it may vapourise or change from liquid to vapour state, or it may boil.
3 **Expansion of the body** Addition of heat will usually cause the expansion of a body. During expansion, the dimensions of the body increase.
4 **Change in the physical properties of a body** Addition of heat to a body may cause changes in the electrical resistance, magnetic properties, conductivity, elasticity, density and colour of a body.
5 **Thermionic emission** Addition of heat to a metal may result in the emission of electrons from the surface of the metal in a process known as *thermionic emission*.
6 **Chemical change** When heat is added to a body, it may bring about changes in the chemical properties of a body.
7 **Changes in pressure** When heat is added to a gas, the pressure of the gas increases, so also is its volume.

4.3 Kinetic Molecular Theory

The molecular theory of matter assumes that matter is made up of atoms that aggregate in molecules.

The **atom** is the smallest particle of an element that can have a separate existence and still retain the chemical properties of that element.

The **molecule** is a group of atoms of the same or different elements joined together in a simple proportion.

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The size of the molecule is of the order of 10^{-2} m.

In solids, the molecules are fixed in definite positions and vibrate about a mean or fixed position. Because the positions of the molecules of a solid are fixed due to intermolecular forces of attractions and repulsion, solids have a fixed shape and volume.

In liquids the molecules are more free to move about within the liquids than the molecules of a solid. Molecules of liquids and solids are held together by intermolecular forces.

In gases the molecules are in constant motion having overcome the intermolecular force. Gases therefore take up the shape and volume of their container. The molecules of a gas are also much farther apart than those of solids and liquids.

The kinetic molecular theory assumes that

(i) Every substance is made up of tiny particles called molecules.

(ii) The molecules are in a constant state of random motion, colliding elastically with one another and changing their direction as a result.

(iii) There is always an attractive force between the molecules.

(iv) The volume of the gas molecules is negligible compared with the volume of the gas container.

Kinetic theory explanation of temperature

Since the molecules of a substance are in constant motion, they possess kinetic energy.

Temperature of a body is a measure of the average kinetic energy of its molecules.

Addition of heat to a substance causes an increase in the speed of motion of the molecules and hence an increase in the average kinetic energy of the molecules of the substance and thus an increase in its temperature. Removal of heat from a body leads to a decrease in the motion of the molecules and hence reduction in their average kinetic energy. This results in a decrease in temperature.

Thus it follows that the temperature of a body is related to the average kinetic energy of its molecules.

4.4 Thermal Expansivity

When heated, most solids and liquids expand. They also contract when cooled. Expansion means an increase in the size of an object.

According to the kinetic molecular theory, when an object is heated the molecules acquire more kinetic energy which enables them to overcome their intermolecular forces. Therefore the vibrations of the molecules increase and their displacements about their mean positions increase. As a result of this, the average distance between the molecules of the substance becomes larger leading to an increase in the size of the substance. This increase in the dimension of the heated object depends on the strength of the intermolecular forces. If these forces are strong, the expansion will be small and vice versa.

The intermolecular forces are stronger in solids than in liquids and weakest in gases. Hence when heat is applied, gases expand more than liquids and liquids expand more than solids.

Each particular substance has intermolecular forces peculiar to it. Therefore the addition of heat causes different expansions in different substances.

Expansion in Solids

Solids expand when heated and contract when cooled. We can demonstrate this by the 'ball and ring experiment'.

Activity 4.1 Ball and Ring Experiment

The apparatus is as shown in fig. 4.1. It consists of a metal ring through which a metal ball just fits.

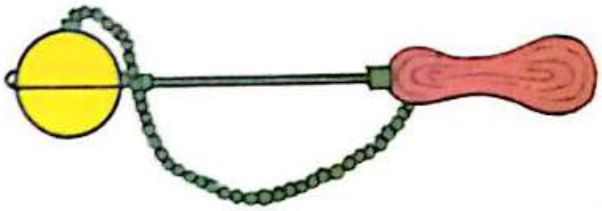


Fig. 4.1 Ball and ring experiment

1 Bring the metal ball out of the ring. Heat the ball strongly over a hot flame for several minutes.

2 Place the hot ball over the ring and observe that it no longer fits into the ring.

3 Allow the hot ball to cool sufficiently for several minutes and put it over the ring again. Notice that the ball now passes through the ring.

The metal ball had increased in size when heated and decreased in size when cooled.

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Activity 4.2 Bar and Gauge Experiment

In fig. 4.2, the bar just fits into the gauge at ordinary room temperature.

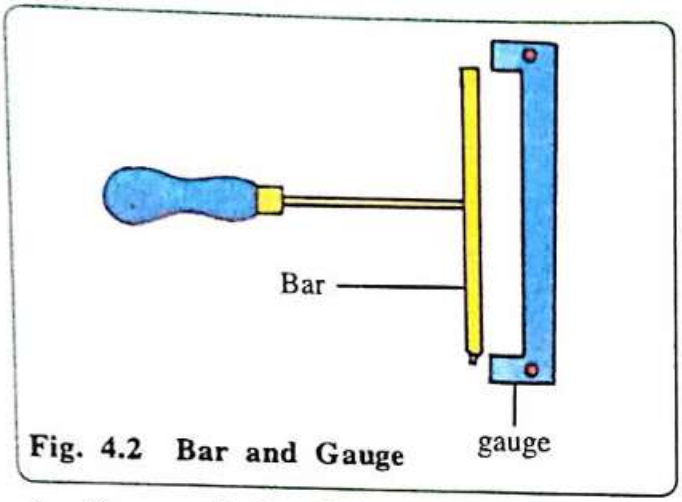


Fig. 4.2 Bar and Gauge

1 Remove the bar from the gauge and heat it strongly over a flame. Let the entire length of the bar be heated over the flame for several minutes.

2 Try to fit the hot bar into the gauge and observe what happens.

3 Cool the hot bar down to room temperature and attempt to fit it once again over the gauge. Observe the new situation.

The heated bar expands in length when hot and can no longer fit into the gauge. When cooled to room temperature the bar contracts and fits once more into the gauge.

Linear expansion

Different solids expand by different amounts when heated over the same temperature range. Copper for example expands more than steel when both are heated through the same rise in temperature. This is because they have different *coefficient of linear expansion* or *linear expansivity*.

The linear expansivity α of a substance is defined as the increase in length per unit length per degree rise in temperature.

In symbols, this is equivalent to:

$$\alpha = \frac{l_2 - l_1}{l_1(\theta_2 - \theta_1)} = \frac{e}{l_1\theta} \qquad 4.1$$

where α = linear expansivity

l_1 = length of metal at temperature θ_1

l_2 = length of metal at temperature θ_2

θ = temperature rise which is given by $\theta_2 - \theta_1$

$e = l_2 - l_1$ = expansion or increase in length

$$\alpha = \frac{\text{increase in length}}{\text{original length} \times \text{temperature rise}}$$

The unit of α is per °C or per K (K^{-1}).

The statement that *the linear expansivity of copper is 0.000017 K^{-1} or 0.000017/°C* means that a unit length of copper expands by 0.000017 units when it is heated through 1 K (or 1°C) rise in temperature.

From equation 4.1 we have that:

$$l_1\alpha\theta = l_2 - l_1$$

or $$l_2 = l_1 + l_1\alpha\theta$$

$$l_2 = l_1(1 + \alpha\theta) \qquad 4.2$$

Increase in length, $e = l_2 - l_1 = l_1\alpha\theta$.

The value of linear expansivity (α) differs from substance to substance. It is highest for metals (Table 4.1).

Table 4.1 Linear expansivity

Substance	Linear expansivity K^{-1}
Platinum	0.000009
Iron	0.000012
Copper	0.000017
Brass	0.000018
Aluminium	0.000023
Lead	0.000029
Zinc	0.000030
Invar (alloy)	0.000001
Glass	0.0000085
Silica	0.0000004

Example 4.1

A copper rod whose length at 30°C is 10.0 m is heated to 50°C. Find its new length. Take α for copper as 0.000017 K^{-1}.

Solution

Length at 30°C = l_1 = 10.0 m.

Let length at 50°C = l_2

then $l_2 = l_1[1 + \alpha(\theta_2 - \theta_1)]$

$= 10[1 + 0.000017 \times (50 - 30)]$

$= 10.0034$ cm

Example 4.2

A metal of length 15.01 m is heated until its temperature rises by 60°C. If its new length is 15.05 cm, calculate its linear expansivity.

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Solution

$$\alpha = \frac{l_2 - l_1}{l_1 \theta}$$

$l_2 = 15.05$ m, $l_1 = 15.01$ m

temperature rise, $\theta = 60°C$

$$\alpha = \frac{15.05 - 15.01}{15.01 \times 60}$$

$$= 0.0000444 \text{ K}^{-1}$$

Experiment: To determine the linear expansivity of a metal rod

The apparatus for this experiment is as shown in fig. 4.3

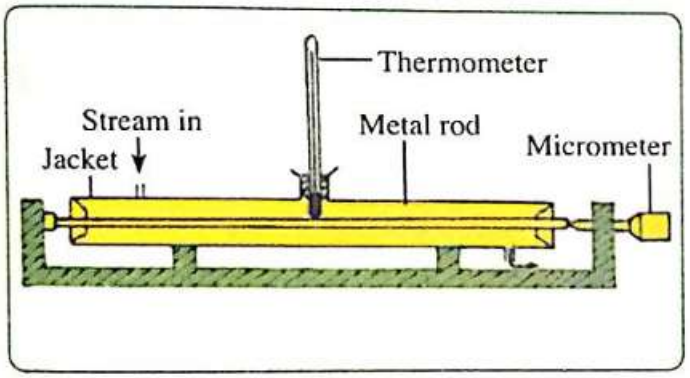


Fig. 4.3 Measurement of linear expansivity

The length of the metal rod is first measured with a metre rule. The rod is then inserted into the steam jacket. One end of the metal rod is fixed firmly at A, the other end B lies just outside the steam jacket and is free to move. Cold water is run through the jacket. The initial temperature of the cold rod is read with the thermometer.

The micrometer screw gauge is screwed up until it just touches the free end of the rod, and the reading of the screw gauge is taken. The micrometer screw gauge is unscrewed from the end B of the rod to leave enough space for the rod to expand.

Steam is now passed through the steam jacket for several minutes and the rod expands as a result of this heating. The micrometer screw gauge is screwed to make contact with the rod again and the new reading of the gauge is taken. The process is repeated until the reading of the screw gauge is constant. The final steady temperature of the rod is read using the thermometer inserted into the steam jacket.

The expansion of the rod is the difference between the initial and final micrometer readings.

The linear expansivity is calculated as follows.

Let the length of the rod be l (m)

Let the initial temperature of the rod be θ_1

Let the final temperature of the rod be θ_2

Let the initial micrometer reading be x_1 (m)

Let the final steady micrometer reading be x_2 (m)

$$\text{then } \alpha = \frac{\text{expansion (or change in length)}}{\text{original length} \times \text{temperature change}}$$

$$= \frac{x_2 - x_1}{l(\theta_2 - \theta_1)}$$

Area and volume expansivity

When a solid is heated, it expands in all directions – in length, in breath and in height. Hence there is an increase in the area as well as in the volume of the solid. The increase in area when a body is heated is known as *area or superficial expansion*.

The **area** or **superficial expansivity**, β, of a solid is the increase in area per unit area per degree Kelvin increase in temperature *or* the fractional increase in area per Kelvin rise in temperature.

Similarly an increase in volume when a body is heated is known as cubic or volume expansion and we define *cubic expansivity* as follows:

The volume or cubic expansivity, γ, is the increase in volume of a substance per unit volume per Kelvin rise in temperature *or* the fractional increase in volume per Kelvin rise in temperature.

Area expansivity β

$$= \frac{\text{change in area}}{\text{original area} \times \text{temp. rise}}$$

$$\beta = \frac{A_2 - A_1}{A_1 \times \theta} \qquad 4.3$$

where A_2 = area at temperature θ_2

A_1 = area at temperature θ_1

$\theta = \theta_2 - \theta_1$

From equation 4.3 we have

$$A_2 = A_1(1 + \beta\theta) \qquad 4.4$$

Increase in area = $A_2 - A_1 = A_1\beta\theta$

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Cubic expansivity (γ)

$$= \frac{\text{change in volume}}{\text{original volume} \times \text{temperature rise}}$$

$$\gamma = \frac{V_2 - V_1}{V_1\theta} \qquad 4.5$$

where V_2 = volume at temperature θ_2

V_1 = volume at temperature θ_1

$\theta = \theta_2 - \theta_1$

From equation 4.5 we have

$$V_2 = V_1(1 + \gamma\theta) \qquad 4.6$$

Increase in volume $= V_2 - V_1 = V_1\gamma\theta$

We now want to show the relationship between linear expansivity (α), area expansivity (β) and cubic expansivity (γ).

Consider a sheet of a metal of length l_1 and breath b_1. Let the length and breath be l_2 and b_2 after the sheet has been heated through a rise in temperature θ.

Initial area of sheet $= A_1 = l_1 b_1$

Final area of sheet $= A_2 = l_2 b_2$

From equation 4.2

$l_2 = l_1(1 + \alpha\theta)$

$b_2 = b_1(1 + \alpha\theta)$

Hence $A_2 = l_2 b_2 = l_1(1 + \alpha\theta) \times b_1(1 + \alpha\theta)$

$= l_1 b_1(1 + \alpha\theta)^2 = l_1 b_1(1 + 2\alpha\theta + \alpha^2\theta^2)$

Since α is a very small quantity (e.g. = 0.000026 for aluminium), we have that α^2 is extremely small and can be neglected. Hence we have

$$A_2 = l_1 b_1(1 + 2\alpha\theta) \qquad 4.7$$

But from equation 4.4,

$A_2 = A_1(1 + \beta\theta)$

Hence comparing 4.7 and 4.4 we have that:

$$\beta = 2\alpha \qquad 4.8$$

Hence the area or superficial expansivity, β, is twice the linear expansivity α.

Similarly we can show, using the same kind of calculation that the cubic or volume expansivity (γ) and linear expansivity (α) are related by the equation

$$\gamma = 3\alpha \qquad 4.9$$

Example 4.3

The linear expansivity of the material of a cube is $12 \times 10^{-6}\ \text{K}^{-1}$. If the length of each side of the cube is 10 cm, find the area of one face of the cube and the volume of the cube when its temperature is raised by 30 K.

Solution

Initial area of a face of cube $A_1 = 10\ \text{cm} \times 10\ \text{cm}$

$= 100\ \text{cm}^2$

Initial volume of cube V_1

$= 10\ \text{cm} \times 10\ \text{cm} \times 10\ \text{cm}$

$= 1000\ \text{cm}^3 = 10^3\ \text{cm}^3$

$\alpha = 12 \times 10^{-6}\ \text{K}^{-1}$

$\beta = 2\alpha = 24 \times 10^{-6}\ \text{K}^{-1}$

$\gamma = 3\alpha = 36 \times 10^{-6}\ \text{K}^{-1}$

$A_2 = A_1(1 + \beta\theta)$

$= 100(1 + 24 \times 10^{-6} \times 30)\ \text{cm}^2$

$= \underline{\underline{100.072\ \text{cm}^2}}$

$V_2 = V_1(1 + \gamma\theta)$

$= 10^3(1 + 36 \times 10^{-6} \times 30)\ \text{cm}^3$

$= \underline{\underline{1001.08\ \text{cm}^3}}$

Consequences and Applications of expansion

1 Expansion in buildings and steel bridges

We often hear creaking noises made by the galvanised iron sheets used in the roof of buildings as the roof is being heated by the sun. This is due to the expansion of the metal sheet as it gets heated.

Bridges made of steel girders expand during the day when they are heated. If both ends of the girders are fixed, the bridge will bend because of thermal expansion. To allow for this expansion, one end is fixed and the other rests on rollers in an expansion gap.

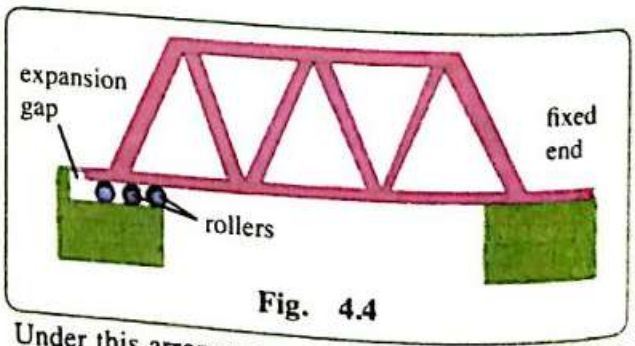


Fig. 4.4

Under this arrangement, the metal structure is free to expand or contract at various temperatures.

2 Railway Lines

Gaps are left between rails in railway lines to allow for free expansion and contraction of the rails. Without the gaps, the rail joints will swell up on hot days, the railway line will buckle and trains would be derailed.

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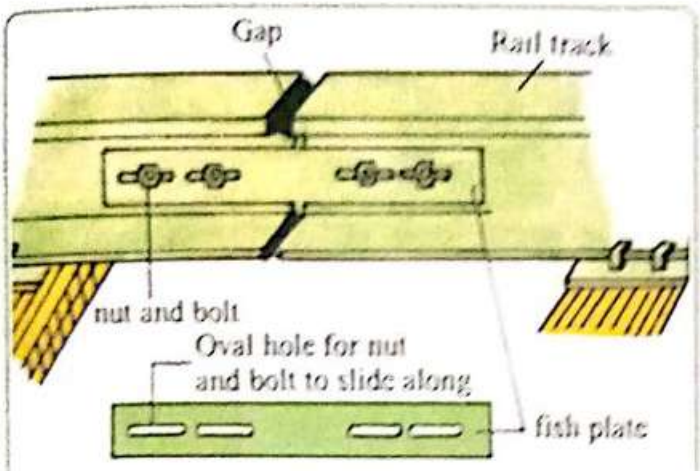


Fig. 4.5 Railway track

3 Sagging of telegraph wires

The metals used for making telegraph wires expand and sag during the hot season owing to the high temperatures. They also contract during the cold season. While fixing overhead telegraph wires, they are usually given a certain amount of sag so that they can contract during cold period without snapping.

4 The Bimetallic strip and its applications

The bimetallic strip consists of a strip of different metals such as iron and brass attached rigidly side by side.

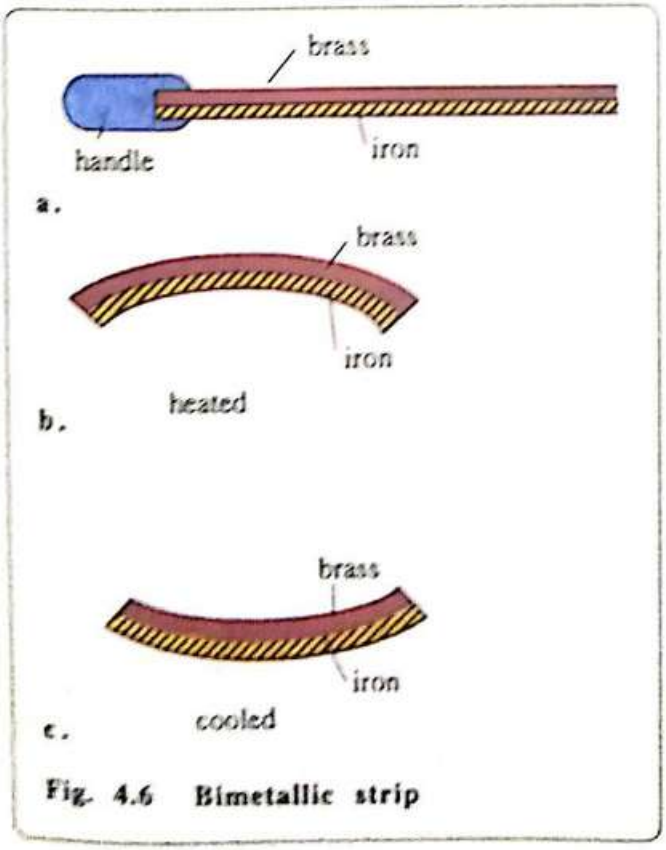


Fig. 4.6 Bimetallic strip

Because the two metals have different expansivity, they expand by different amounts when the strip is heated. The strip therefore bends when heated because of unequal expansion with the brass of greater linear expansivity being on the outside of the curve, and the iron in the inside.

The bimetallic strip has several useful applications.

a. The balance wheel of clocks and watches

The balance wheel of clocks and watches is made of a bimetallic strip usually of brass and steel. The strip bends inwards on expansion and compensates both for the outward expansion of the spokes of the wheel and the reduced elasticity of the hairspring which occur when the temperature of the clock rises in hot weather. Without this bimetallic device an increase of temperature increases the diameter of the balance wheel and weakens the elasticity of the hairspring, thus causing the watch to lose time.

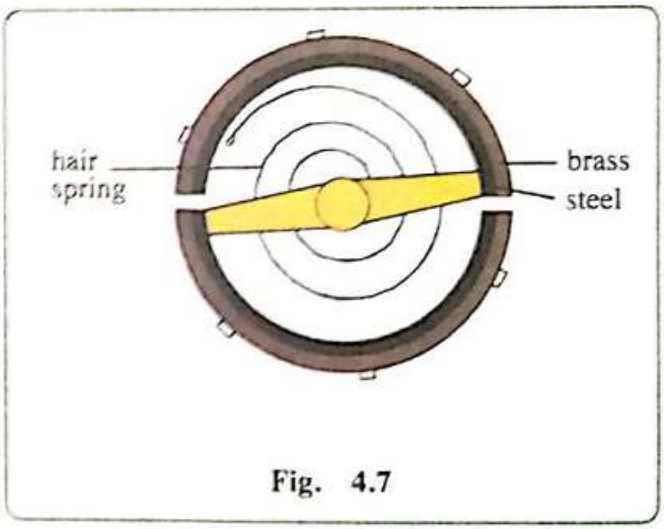


Fig. 4.7

b. The bimetallic strip thermometer

The thermometer consists of spiral form of bimetallic strip made of invar (which hardly expands), and brass. The brass is on the outside of the strip and the invar is on the inside. One end of the spiral strip is fixed and the other is attached to the spindle of a pointer.

Uneven expansion of the metals of the strip due to rise in temperature causes it to curve in a clockwise direction. This curving movement makes the pointer to move over a scale and record the increase in temperature.

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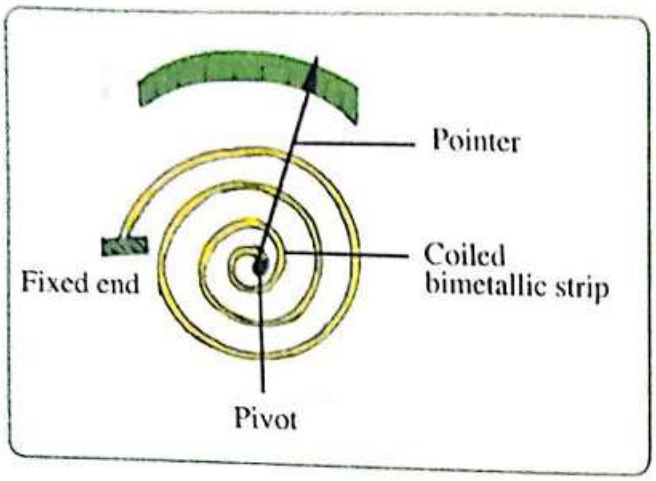


Fig. 4.8 A bimetallic strip used in a thermometer

c. Thermostat

The bimetallic strip is used in the electric thermostat, gas oven thermostat, and the electric fire alarm.

The bimetallic strip of an electric thermostat is used to control the temperature of an electric laundry iron.

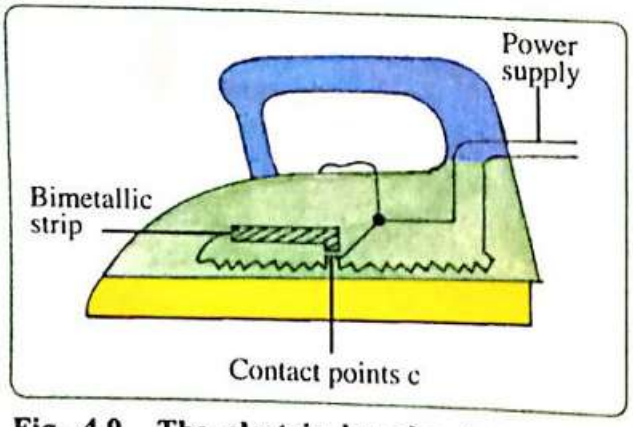


Fig. 4.9 The electric laundry iron

In fig. 4.9, the thermostat knob is set to the desired temperature. When the current is switched on, current flows through the circuit and produces heat in the heater coils. As the coil heats up, the bimetallic strip bends away from the contact point *C*, until at the required temperature, the contact is completely broken thereby switching off the flow of current. As the laundry iron cools, the strip straightens up again and remakes contact at *C*, thus switching on the electric current once more. This make-and-break device regulates the temperature of the electric iron.

d. Electric fire alarm

The circuit of an electric fire alarm is as shown in fig. 4.10.

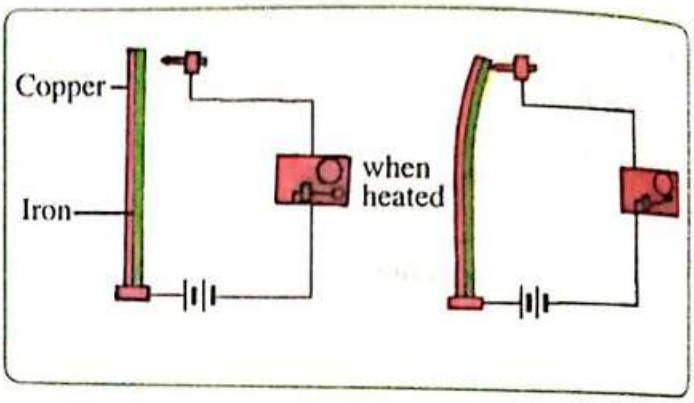


Fig. 4.10

When a fire breaks out in a building, the resulting heat causes the strip to bend towards the contact, thus completing the circuit and causing the bell to ring out a fire alarm.

5 Expansion of glass

A thick glass tumbler cracks when hot water is poured into it. The inside of the tumbler expands more rapidly than the outside and causes a strain in the glass. A type of glass known as *pyrex* is therefore used for making laboratory beakers and flasks to avoid the above effects. Pyrex has a low thermal expansivity.

We can remove a tight glass stopper of a glass bottle without cracking either the bottle or the stopper by either standing the bottle in hot water making sure the stopper is not in the water or by warming the neck of the bottle with a flame. As the bottle expands, the stopper becomes lose.

The use of bimetallic strip in fire alarms, electric pressing irons, etc., and the use of heat in removing glass stoppers are all useful applications of expansivity. Other such useful applications of expansion are in the fixing of metal tyres on to the metal wheels in locomotives and in *rivetting*. A force of contraction when a hot metal cools is utilized in rivetting together of steel plates and girders used in ship-building, boilers and large containers.

Also because platinum and glass have approximately the same expansivity, platinum wires are used when wires are to be fused through the walls of glass vessels.

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Expansion in liquids

Like solids, liquids expand on heating and contract on cooling. Because liquids are always held in a container, the expansion of liquids is often complicated by the expansion of the containers.

We can demonstrate liquid expansion by the following experiment

Activity 4.3

1. Colour some water by putting into it a few crystals of copper sulphate.
2. Fill a flask with this coloured water and cover the flask with a rubber bung with a hole. Push a narrow tube through the hole of the rubber bung until some liquid rises in the tube. Ensure there are no air bubbles under the bung and that the flask is completely filled with the coloured water.

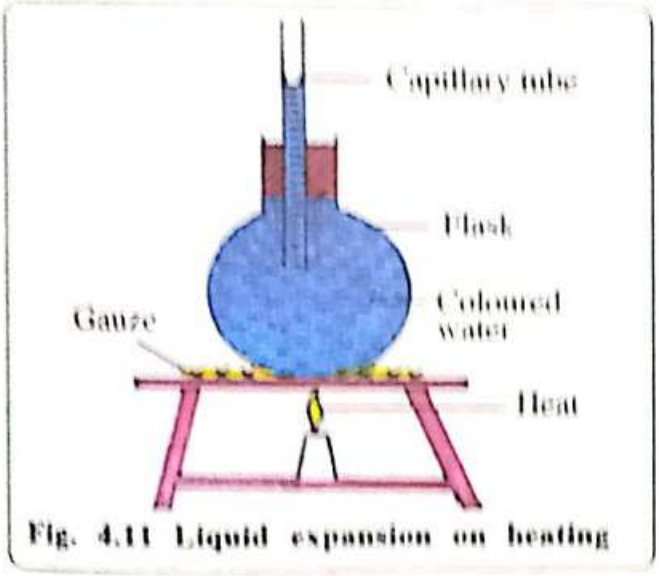


Fig. 4.11 Liquid expansion on heating

3. Warm the liquid gently with a bunsen burner and observe the level of the coloured liquid in the narrow tube.
4. Remove the heat and observe the movement of the liquid level in the tube.

You will observe that the liquid level initially falls slightly and then rises slowly and steadily as more heat is applied. The reason for this is because on the application of heat, the glass first expands faster than the liquid, hence the fall of the liquid level. As more heat is applied, the expansion of the liquid increases more than that of the glass and the liquid rises in the narrow tube. On removal of heat the liquid contracts and the level of liquid in the tube is seen to fall.

We can also show that equal volumes of different liquids expand by different amounts when heated through the same temperature range.

Activity 4.4

1. Fill three similar glass flasks with different liquids e.g. water, alcohol and benzene. Push narrow tubes into the rubber bungs of the flasks.
2. Stand the flasks in a trough of water and heat the trough, stir the heated water in the trough to ensure uniform temperature.
3. Observe the levels of rise of the liquids in the narrow tubes of the three flasks.

It will be observed that in all three flasks, the levels of liquids in the tube first falls and then rises steadily. The level of the alcohol is the highest and the level of the benzene is higher than that of the water.

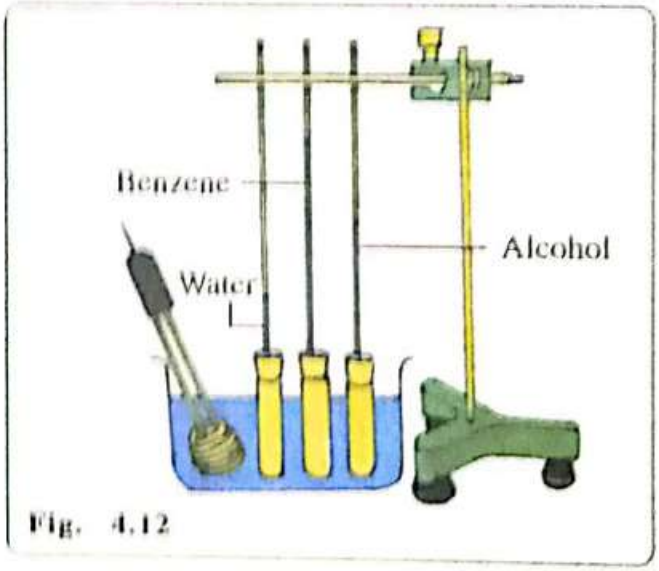


Fig. 4.12

Real and Apparent Expansivity

Because the expansion of liquids is complicated by the expansion of their containers, it is necessary to distinguish between the *real* and the *apparent* cubic expansivity of a liquid.

Real expansion = Apparent expansion + expansion of the container.

The real cubic expansivity of a liquid can be derived from the above relationship.

The real (or absolute) cubic expansivity (γ_r) of a liquid is the increase in volume per unit volume per degree rise in temperature.

The apparent cubic expansivity (γ_a) of a liquid is the increase in volume per unit volume per degree rise in temperature when the liquid is heated in an expansible vessel.

Because the apparent expansivity depends also on the cubic expansivity of the material of the

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vessel, the real expansivity of a liquid (γ_r) is always more than its apparent expansivity (γ_a).

It can be shown that the difference between the real and the apparent expansivity of a liquid is the cubic expansivity of the vessel.

Hence

$$\gamma_r = \gamma_a + \gamma \qquad 4.10$$

Activity 4.5 To determine the apparent cubic expansivity (γ_a) of a liquid

1 Clean and dry a relative density bottle and weigh it by suspending it from a spring balance and record the mass.

2 Fill the bottle with the liquid (e.g. paraffin) whose apparent cubic expansivity is required. Replace the stopper and wipe off any spilling liquid and wipe the bottle dry. Weigh the bottle again and record the new mass.

3 Suspend the bottle from the clamp of a retort stand with the bottle completely immersed in a beaker of water as shown. Record the temperature of the water.

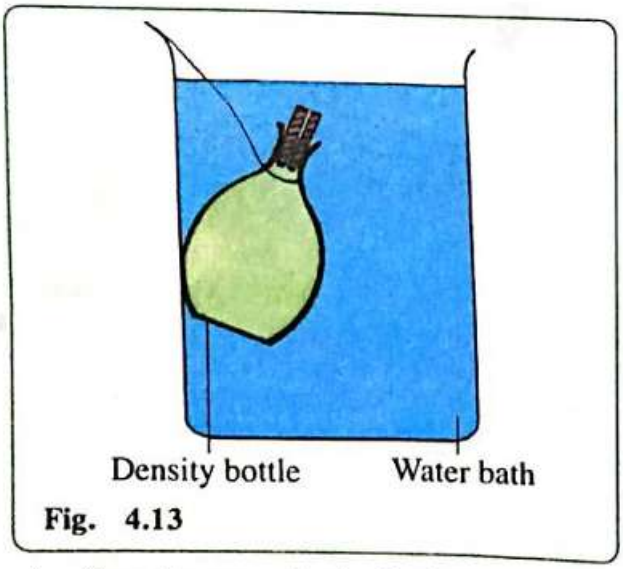


Fig. 4.13

4 Heat the water in the beaker gently with a bunsen burner until it boils. Notice that some liquid is expelled through the orifice of the bottle stopper.

5 Continue to heat and stir the water in the beaker until no more liquid is expelled from the relative density bottle. Read and record the temperature of the boiling water with the thermometer.

6 Remove the heat and then the bottle. Wipe dry the bottle and allow it to cool. Reweigh the bottle and record its new mass.

The apparent cubic expansivity of the liquid is calculated as follows:

Let mass of empty relative density bottle be = M_1
Let mass of bottle + liquid = M_2
Let final mass of liquid and bottle be = M_3
Let initial temperature be = θ_1
Let final temperature be = θ_2
Mass of expelled liquid = $M_2 - M_3$
Mass of remaining liquid = $M_3 - M_1$
Temperature rise $\theta = \theta_2 - \theta_1$
Apparent cubic expansivity γ_a is given by

$$\gamma_a = \frac{\text{Volume of liquid expelled}}{\text{Volume of liquid remaining} \times \text{temp. rise}}$$

Since the masses are proportional to the volumes of the liquid we have:

$$\gamma_a = \frac{\text{Mass of liquid expelled}}{\text{Mass of liquid remaining} \times \text{temp. rise}}$$

$$= \frac{M_2 - M_3}{(M_3 - M_1) \times (\theta_2 - \theta_1)} \qquad 4.11$$

Example 4.4

A density bottle holds 250 g of liquid at 30°C and only 248.5 g at 60°C. Find

a. the apparent and

b. the real cubic expansivity of the liquid, if the linear expansivity of the material of the bottle is $\alpha_g = 0.000006\ \text{K}^{-1}$

Solution

$$\gamma_a = \frac{\text{Mass expelled}}{\text{Mass remaining} \times \text{temp. rise}}$$

$$= \frac{250.0 - 248.5}{248.5 \times (60 - 30)}$$

$$= \frac{1.5}{248.5 \times 30} = \underline{0.0002012\ \text{K}^{-1}}$$

$$\gamma_r = (\gamma_a + \gamma_g) = (\gamma_a + 3\alpha_g),$$

where $\alpha_g = 0.000006\ \text{K}^{-1}$

$$= 0.0002012 + (3 \times 0.000006)$$

$$= \underline{0.0002192\ \text{K}^{-1}}$$

Variation of density with temperature

From equation 4.6 we have that $V_2 = V_1(1 + \gamma\theta)$ where V_1 is the initial volume of an object, V_2 its final volume, θ is the temperature change and γ is the cubic expansivity. Since the mass (M), volume (V) and density (d) of an object are related by

$$d = \frac{M}{V} \qquad 4.12$$

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We observe that density decreases with volume. Since volume increases with temperature, we see that density will be decreasing with temperature since mass of a substance, M is constant

$$V_1 = \frac{M}{d_1} \qquad 4.13$$

$$V_2 = \frac{M}{d_2} \qquad 4.14$$

Dividing equation 4.14 by equation 4.13, we have:

$\frac{V_2}{V_1} = \frac{d_1}{d_2}$, since M is a constant.

But $V_2 = V_1(1 + \gamma\theta)$ and hence

$$\frac{V_2}{V_1} = 1 + \gamma\theta$$

$$\therefore \quad \frac{d_1}{d_2} = 1 + \gamma\theta$$

$$d_1 = d_2(1 + \gamma\theta) \qquad 4.15$$

where d_1 is the density at a lower temperature and d_2 is the density at a higher temperature.

The anomalous expansion of water

Most liquids expand when heated and contract when cooled. Water is an exception to this rule as it behaves in an anomalous way when cooled. It contracts on cooling from any temperature until at 4°C when it starts to expand if cooled below this temperature. Thus water expands (rather than contracts) when cooled from 4°C to 0°C. Hence a given mass of water has its least volume (and hence its highest density) at 4°C.

This anomalous expansion of water is of utmost importance in nature. It makes ponds, lakes or rivers to freeze from the top surface rather than from the bottom. Thus Marine lives can survive during winter since ice forms at the surface of the water while the bottom of the lake remains at 4°C, a temperature warm enough for aquatic life.

We can illustrate the anomalous behaviour of water by the following experiment known as Hope's experiment.

Use is made of an apparatus known as Hope's apparatus. This consists of a metal cylinder filled with water and surrounded by a circular tray containing a freezing mixture of ice and salt about half-way up its height (fig. 4.14).

One thermometer A measures the water temperature near the surface and another thermometer B measures the temperature near the bottom. The readings of these thermometers are recorded at regular intervals. When the two sets of temperatures are plotted against time, we obtain graphs as shown in fig. 4.14b.

The initial temperatures indicated by A and B are the same. Thermometer B soon shows a rapid fall in temperature while the upper thermometer A shows a constant temperature. When the readings of B falls to 4°C, it remains constant there while the upper thermometer A starts to show a fall in temperature until at 0°C at which the water at the surface of the cylinder changes into ice.

We can explain the shape of the graph in this way: Initially, the water in the middle (near the freezing mixture) cools, decreases in volume, hence becomes denser than the water around it and so sinks to the bottom. The water below rises to replace it. The process continues until all the water from the middle to the bottom of the cylinder reaches the temperature of 4°C. At this temperature water has its least volume and therefore its

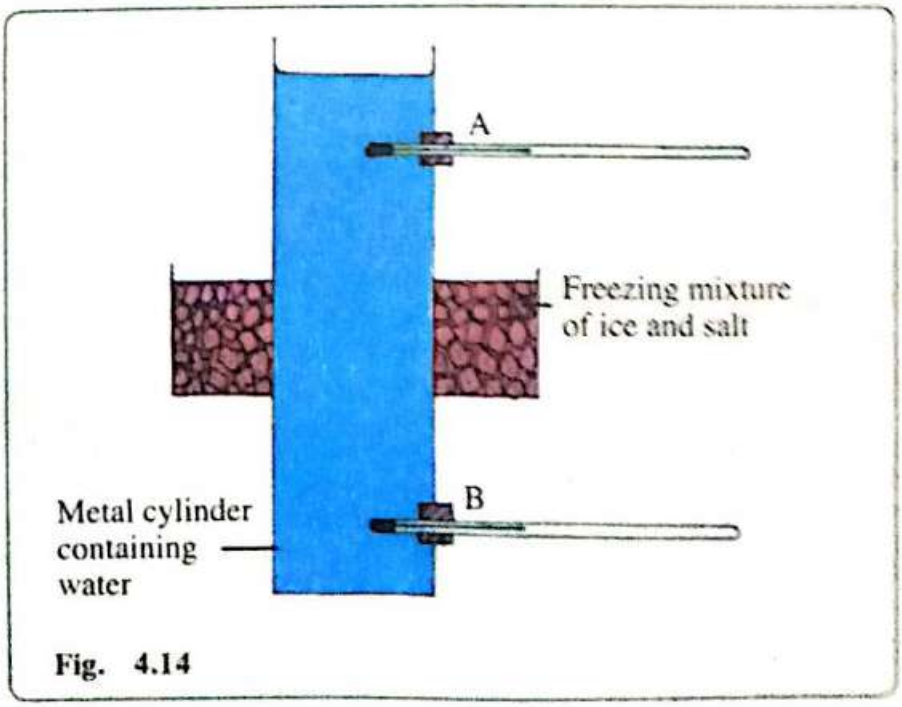


Fig. 4.14

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maximum density. On cooling below 4°C, the water begins to expand, becoming less dense and so moves up and is replaced by water from above the freezing mixture. The process continues with the upper layer getting colder and colder until it eventually turns to ice.

4.5 Change of State

Matter exists in three states – solid, liquid, and gas. When a solid is heated sufficiently it melts into a liquid form or state. Heating the liquid further causes it to change into the gaseous state. Hence the state in which a substance exists depends on its temperature. Using water as an example, we see ice as the solid state when the temperature is below zero degree Celsius (0°C). Between 0°C to 100°C it is in liquid state. Above 100°C it is in gaseous state. Hence apart from expansion, the addition of heat to a substance may lead to a change of state.

The processes involved in the change of state are *fusion* (or melting), *freezing*, *vaporization*, and *condensation*.

Fusion (or melting) is the process whereby a substance changes from a solid state to a liquid state when heat is applied to it.

The temperature at which the change occurs is known as *fusion or melting point*. Addition of heat causes a solid to change to a liquid. Conversely, the removal of heat may cause a substance to change from the liquid state back to the solid state, in a process known as *freezing or solidification*. The temperature at which this occurs is known as the *freezing temperature* or *freezing point*.

Some liquids such as water expand on freezing. Hence if a glass bottle is filled with water and cooled below 0°C, the water will expand in forming into ice and will crack the bottle. Most other liquids contract on freezing.

The melting point of a solid (e.g. ice) is found to be the same as the freezing point of its liquid. Some solids such as candle wax and glass have no definite melting or freezing point.

Vaporization is the process by which a substance changes from a liquid to a gaseous state when heat is applied to it.

Vaporization occurs at a definite temperature for a given substance. This temperature is known as the *boiling point* of that liquid.

In another similar process, some liquids turn into vapour before the boiling point.

The process where a liquid turns into vapour below its boiling point is known as **evaporation**.

Evaporation takes place at all temperatures. Kinetic molecular theory explanation of evaporation and melting as well as further details of the subject of *change of state* will be discussed in Book 2 of this volume.

4.6 Transfer of Heat

In section 4.1 we stated that heat flows from a body at a higher temperature to another at a lower temperature. This transfer of heat can happen in three ways: *conduction*, *convection*, and *radiation*.

Conduction

It is our everyday experience that when we pour hot water into an aluminium cup, the handle of the cup soon feels hot to the touch. If you dip a silver spoon into a hot tea, the handle of the spoon very quickly feels hotter than it was before. The heat from the hot water or hot tea has been transferred along the metal handle to the other end of the cup or spoon by the process of conduction.

Conduction of heat is the process by which heat energy is transferred through a material, the average position of the particles of the material remaining the same.

Heat energy is always transferred when different parts of a solid body are at different temperatures. The direction of heat transfer is always from the hotter to the cooler parts of the solid. For example, for the spoon dipped into a hot tea, the heat is transferred from the bowl of the spoon inside the hot tea to the handle of the spoon outside the tea.

Most metals (e.g. copper, aluminium, silver, iron) allow heat energy to pass through them very easily. Such materials are known as *good conductors of heat*. Most non-metals (e.g. water, air, wood, plastic, cloth, cotton wool, cork) are *poor conductors of heat* or *insulators*. Such materials do not allow heat to pass easily through them. The handle of a cooking utensil is made of wood or plastic which are insulators, but the cooking utensil itself is made of a good conductor (e.g. aluminium), so that heat can be conducted quickly through utensil but not through the handle.

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The ability of a metal to conduct heat is known as its thermal conductivity.

Though most metals are good conductors, their thermal conductivities differ from metal to metal.

Experiment to compare the thermal conductivities of materials

We can compare the thermal conductivities of different metals and wood with the set up as shown in fig. 4.15.

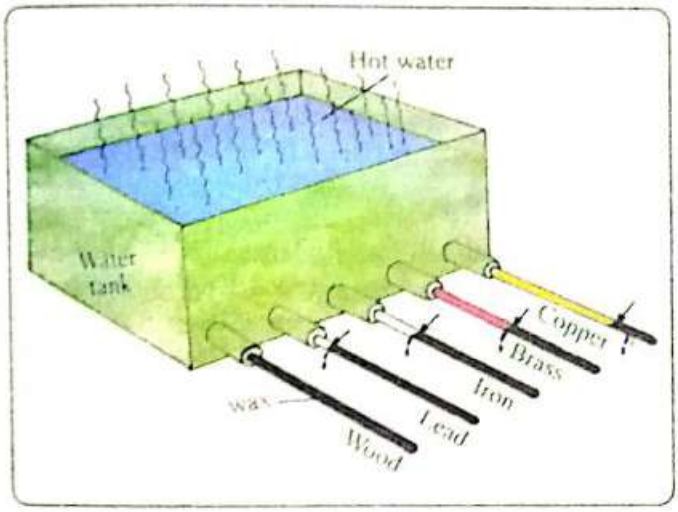


Fig. 4.15 Comparing the conductivities of different materials

Equal lengths of rods of different metals such as copper, iron, aluminium, brass and one rod of a non-metal such as wood, all of equal cross-sectional area are stuck lightly into a small water tank, so that a good length of the rods protrude outside the tank. The rods are coated with paraffin wax. Boiling water is then poured into the tank. After several minutes, it is noticed that different lengths of the wax have melted from the metals with the greatest length of copper and the least length of wood. More length of wax on the copper was melted than that of brass, iron and wood in that order. Thus copper has the greatest thermal conductivity and wood has the least.

Kinetic molecular theory explanation of conduction

When one end of a material is heated, the molecules there gain energy and vibrate with increasing speed. They vibrate faster and faster and farther out. They therefore bump against neighbouring molecules more than before and so transfer greater energy to them. This goes on all the way along the solid. All the molecules of the solid eventually vibrate more rapidly about their fixed positions. Therefore in this way thermal energy is transferred along the solid although the average positions of the molecules remain unchanged.

Thermal conductivity of liquids

Water and other liquids are poor conductors of heat. The only exception is mercury which is a metal in liquid form.

We can demonstrate that water is a poor conductor of heat by the following simple experiment.

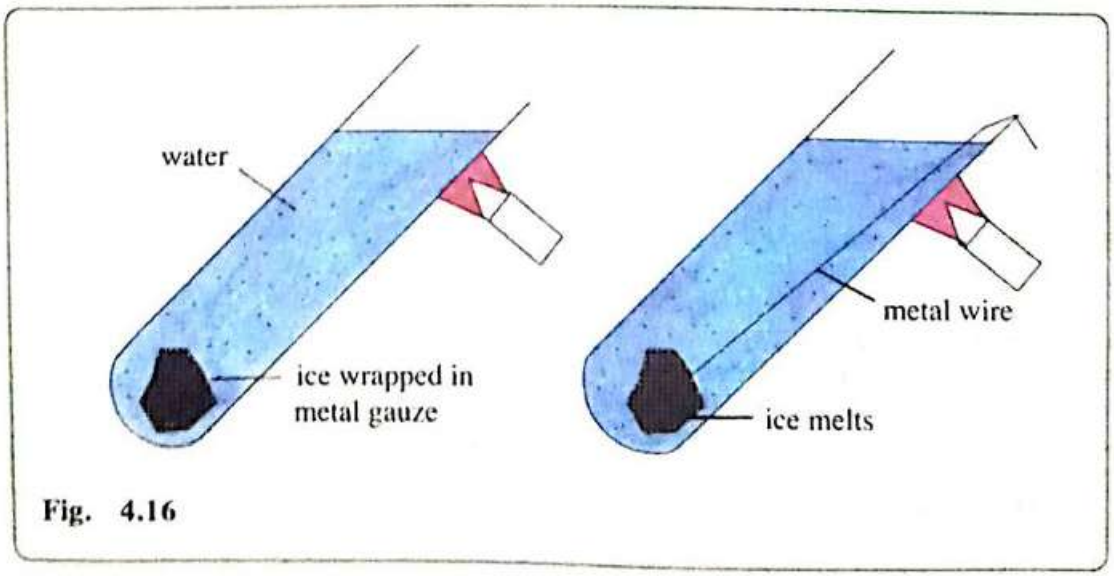


Fig. 4.16

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A piece of ice is wrapped in metal gauze and put into a test tube full of water. The tube is held in a tilted position and heat is applied to its top portion. It will be observed that even though the water near the top is boiling, the ice near the bottom of the tube does not melt for some time. This thus shows that water is a poor conductor of heat since little heat is conducted from the top to the bottom of the water.

Practical application of good and bad conductors

1 Cooking Utensils

Good conductors are used wherever it is necessary to conduct heat quickly. Thus cooking pots and frying pans are usually made of metals such as aluminium which is a good conductor. This ensures quick transfer of heat from the fire to the food being cooked, and a quick cooking of the food.

On the other hand, the handle of cooking utensils are made of insulators like wood or plastic materials, so that the utensils when hot, can be held comfortably by the handles.

2 Use of rugs on floors

A tile floor is much colder to the feet than a rug-covered floor at the same temperature. This is because tile is a better conductor of heat than rug. Heat transferred from one's foot to the rug is not conducted away rapidly, so the rug quickly heats up on its surface to the temperature of the foot; but the tile conducts the heat away rapidly from one's foot, and thus can take more heat from the foot, so its surface temperature drops and the foot feels cold.

3 Home Cooling in the Tropics

It is our common experience that thatched-roofed houses are cooler inside than galvanised iron roofed ones, especially in the absence of asbestos ceilings. The iron conducts the heat from the sun and heats up the inside of the house. To reduce this heat, asbestos ceilings are used. Asbestos is an insulator and prevents the heating up of the house. Thatched-roofed houses need no ceilings to maintain cool temperatures because their materials are insulators and do not conduct the heat from the sun into the room.

4 The use of cloth to keep warm

Clothes keep us warm by holding air between the cloth and the body. It is not the clothes that warm the body but the air (a poor conductor) that the cloth traps around the body. Hence clothing made from woollen and fur materials are worn in the cold climates to retain heat and keep the body warm.

Convection in liquids and gases

Liquids and gases are generally poor conductors. Heat is however effectively transferred in them by *convection*. Convection involves the transfer of heat by the actual movement of the heated molecules from the hot parts to the cooler parts of the fluid.

Convection is the process by which heat is transferred in a liquid or gas by the actual movement of the heated fluid from the hotter to the colder parts.

We can observe convection in liquids by the following two experiments.

Activity 4.6

1 Fill about two-thirds of a large round bottomed flask with water. By using a long glass tube, drop a tiny crystal of potassium permanganate through it to the bottom of the flask.

2 Gently heat the bottom of the flask just below the crystal and observe the movement of the coloured liquid.

You will observe the colour streaks of the potassium permanganate to rise upwards from the bottom of the flask where the heat was applied. At the top of the water, the streaks curve downwards as shown in fig. 4.17.

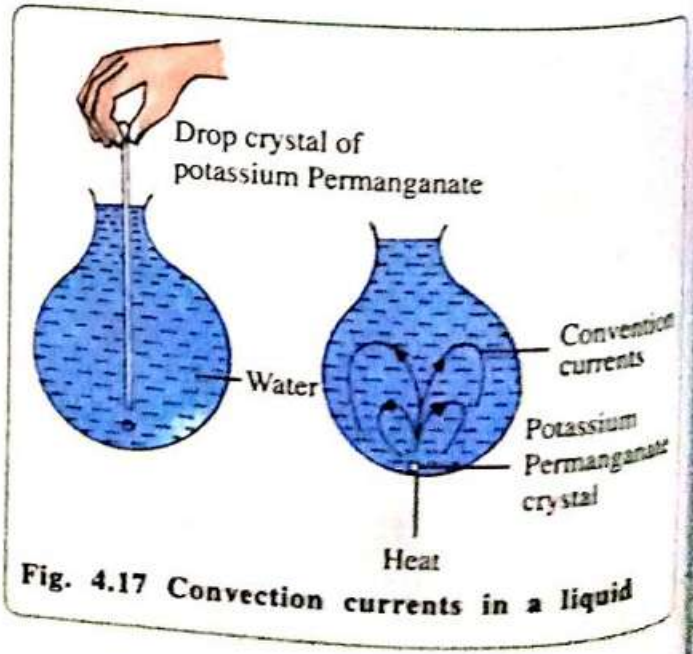


Fig. 4.17 Convection currents in a liquid

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The upward movement of the coloured streaks are due to *convection*, and the circulation of the liquid is called *convection current*.

When the liquid at the bottom of the flask is heated, its volume expands, and since the mass remains constant, the density of the liquid decreases. As the liquid becomes lighter than the cold surrounding liquid, it rises upwards and is replaced by the colder denser liquid from the top. This in turn gets heated and moves upwards. Thus a cyclic motion of liquid, a *convection current*, is set up.

You will also observe convection current in a kettle when you boil water in the kettle.

In terms of the kinetic molecular theory, we explain convection as follows: When the water is heated below, its molecules near the heat source gain more kinetic energy (in the form of heat energy). The heated molecules vibrate farther away from their fixed position and become less densely packed and lighter than the ones above. They therefore rise from the bottom to the top carrying thermal energy along with them. The denser molecules from the top sink to the place of those that have risen above. These molecules, in turn, get heated and rise, thus setting up a cyclic motion of liquid molecules, until the water boils.

The difference between conduction and convection, is that in conduction the average position of the molecules remain the same but in convection, heat is carried to the other parts of the liquid by the actual movement of the warm liquid itself.

Convection Current in Gases

Gases expand much more readily than liquids. Therefore convection currents are easily set up in gases when their temperature rises.

The following simple experiment demonstrates convection current in air.

Activity 4.7

1 Obtain a wooden box with a plane glass front. Fit two vertical chimneys, A and B on to the top of box.

2 Light a candle below chimney A and after a few minutes hold a smouldering paper over chimney B. Report your observation.

You will notice that the smoke from the smouldering paper instead of rising up as usual, flows down chimney B and rises up chimney A as shown in fig. 4.18.

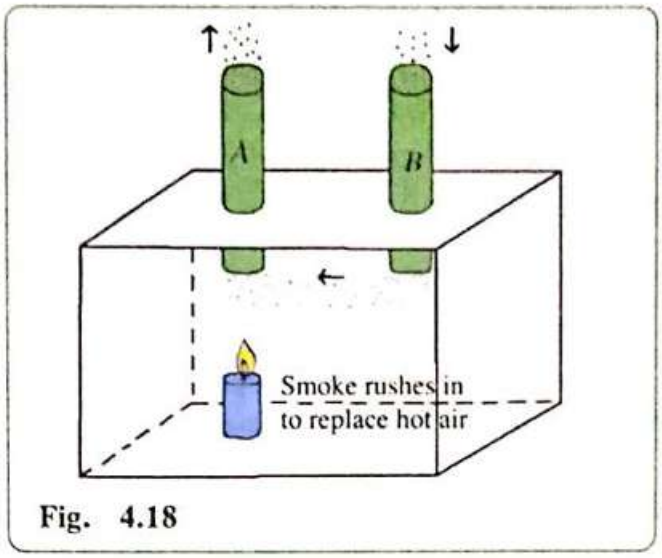


Fig. 4.18

This indicates the direction of the convection current. The smoke is being sucked in at B because of the convection current set up in the air in the box by the burning candle at A. The air heated at A by the candle, becomes light and rises upwards through chimney A, and cold air is drawn down at chimney B to take its place. As a result of both movements, a convection current is set up as indicated by the arrows in fig. 4.18.

The use of chimneys in bringing about the circulation of fresh air in a kitchen or a room is based on the above principle

Convection Current in Nature: Land and Sea Breezes

Land and sea breezes are the result of convection current in nature. In a hot day the sun warms the land more quickly than the sea, because the land is a better absorber of heat and also has a lower specific heat capacity (see p. 208). As a result, the air near the land warms up and rises. Cooler air from the sea moves in to replace the risen air. The convection current is completed by the air in the upper atmosphere moving down in the opposite direction.

The cool breeze that blows inland from the sea is known as a *sea breeze*.

At night, the reverse process takes place. The sea retains much of its heat in the night, while the land temperature drops considerably. As a result the sea is warmer than the land. Convection current is set up in the opposite direction to the daytime. The breeze moving from the land to the sea is known as *land breeze*.

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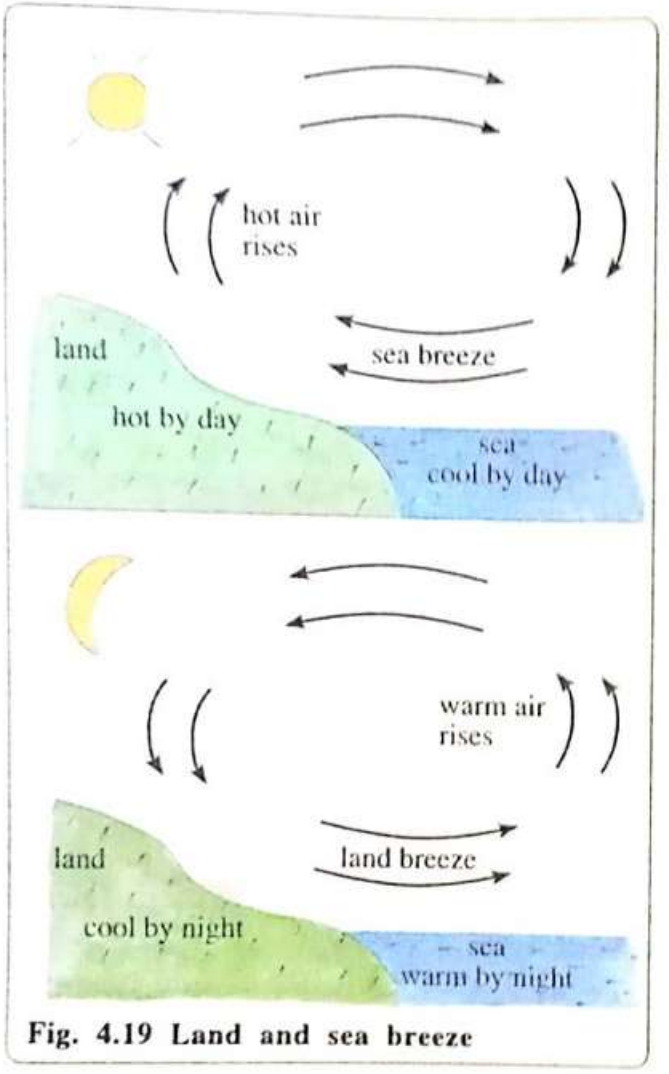


Fig. 4.19 Land and sea breeze

Practical application of convection currents in cooling devices

1 Ventilation

Good ventilation in houses relies on the continuous circulation of convection air currents. Air heated by respiration and fires, rises towards the ceiling and escapes through the ventilators placed near the ceiling. This is replaced by fresh, cool air from outdoor which enters the rooms through the windows and other openings. In this way the room is ventilated.

2 Cooling of motor car engine

The motor car engine requires to be cooled to prevent overheating. Continuous convection currents are utilized in the cooling process of the car engine. Water circulates round the engine by convection currents. The heat generated by the engine is conducted by the metal to the water in the jacket (fig. 4.20). The water itself is cooled by the draught of air created round the radiator by the motion of the car and the movement of the fan.

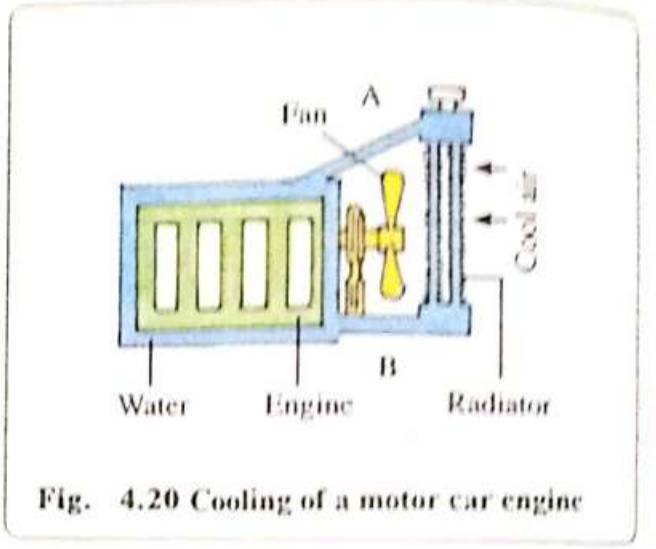


Fig. 4.20 Cooling of a motor car engine

*3 **The Domestic hot water system*** also works on the convection current principle.

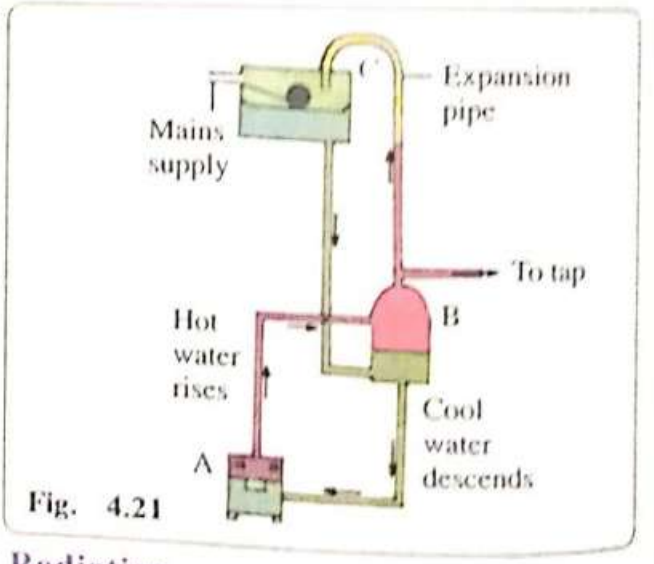


Fig. 4.21

Radiation

A third mode of heat transfer is by *radiation*.

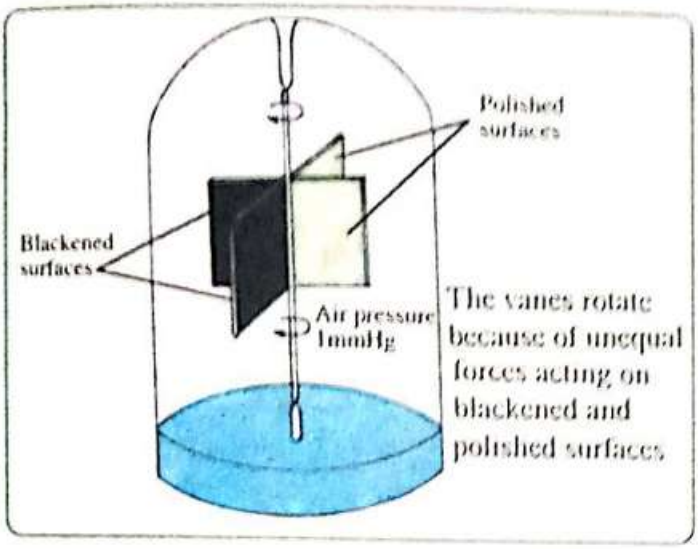
Radiation is the process by which heat is transferred from a hotter to a cooler place without heating of the intervening medium.

Thus no material medium is required in radiation, but in conduction and convection, material medium is necessary. Heat from the sun reaches the earth by radiation. Also when you warm yourself by the fire side, the heat from the fire reaches you by radiation. The heat energy given out by radiation is known as *radiant energy* or *radiant heat*. Such a radiation is an *electromagnetic radiation* (see p. 356) of the type called Infra-Red. It travels with the velocity of light (3×10^8 ms^{-1}).

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Detection of Radiation

Radiation can be detected by a *radiometer* (fig. 4.22) and a thermopile. A thermopile can both detect and measure radiant energy.



A thermopile detects and measures radiant energy by the deflection of a galvanometer, a sensitive current measuring instrument. The radiation is detected by the conversion of the radiant energy into electrical energy. This electricity causes the deflection of the galvanometer. The greater the amount of radiation, the more the deflection of the galvanometer of the thermopile.

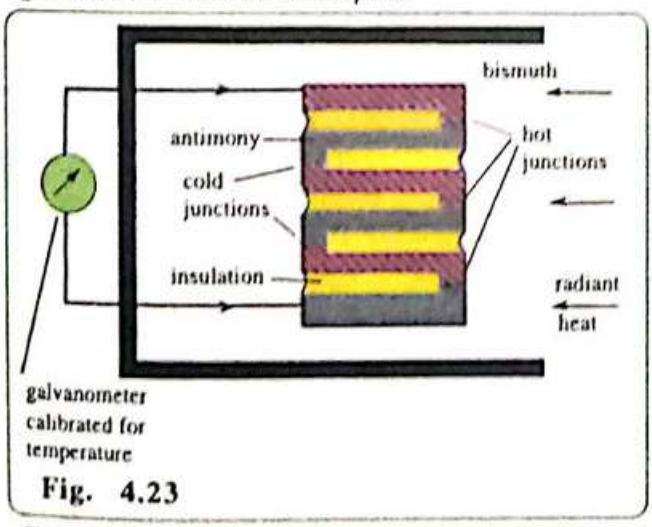


Fig. 4.23

The details of the construction and principle of operation of the thermopile will be discussed in Book 2 of this volume.

Emission and absorption of radiation by different surfaces

Different types of surfaces at the same temperature emit or radiate heat at different rates which is dependent on the nature of the surface. It is found that, *for a given temperature, a dull black surface radiates most and a highly polished surface radiates least.* We can demonstrate this by using a *Leslie cube* and a *thermopile*.

The Leslie cube is a cubical hollow tin box with one of its vertical faces painted dull-black, another face highly polished, and the third face painted with a grey colour. Hot water is poured into the cube and each of the surfaces is then exposed in turn to the thermopile.

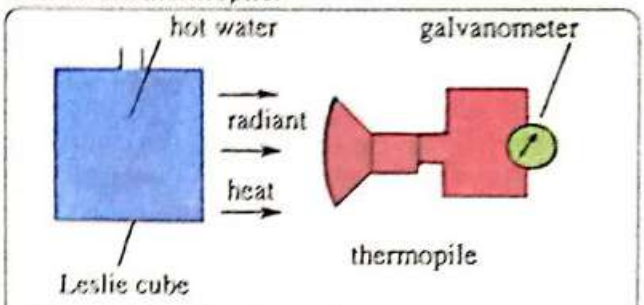


Fig. 4.24 Leslie cube

It will be noticed from the deflection of the pointer of the thermopile, that the dull-black surface radiates the most heat followed by the grey surface. The polished surface radiates the least heat. The experiment is repeated with water at a different temperature. The result will show that the hotter the water the greater the deflection of the pointer of the thermopile.

A black surface besides being a better radiator of heat than a polished surface, is also a better absorber of heat. We can demonstrate this by the following simple experiment.

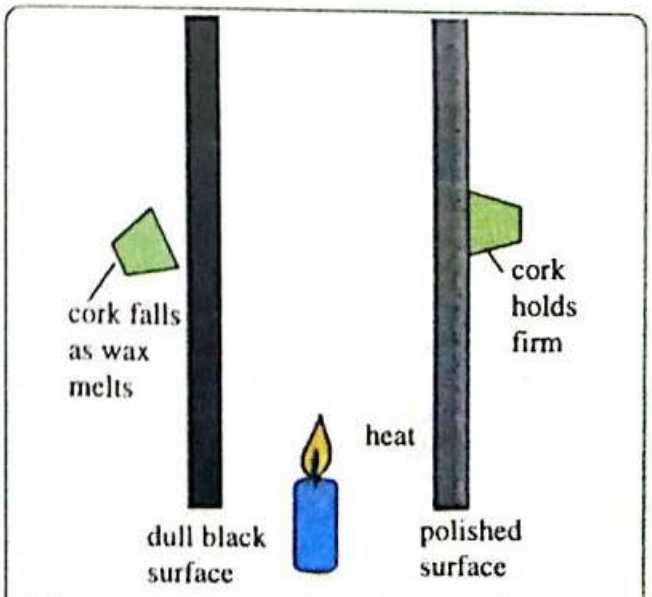


Fig. 4.25 Comparison of heat absorption by two surfaces

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Two sheets of tinplate are used, one painted black, the other highly polished. Cork is attached to the reverse side of each plate using a candle wax. A hot object, e.g. a burning candle is placed half way between the two surfaces fixed vertically facing each other and a few centimetres apart. After a few minutes, it will be noticed that the wax behind the dull black surface begins to slide off the plate. The wax behind the polished surface does not melt and the cork remains fixed.

This shows that the dull black surface absorbs more radiation than the polished surface.

Applications

1 On the basis of the above it is easy to see why it is not advisable to wear a dark coloured jacket or shirt in a hot afternoon. The dark cloth will absorb the radiant heat of the sun and cause the wearer to feel hot and uncomfortable. It is preferable to wear a white cloth which will not absorb the heat but will rather reflect it away from the body.

2 The outer surface of a teapot is brightly polished. Such a surface radiates less heat than an unpolished one and will therefore retain heat inside the teapot for a longer period.

3 A brightly painted car is preferred to a black-painted car in Nigeria and other hot tropical countries. The black-painted car will absorb and retain heat from the sun and inside the car will be very hot.

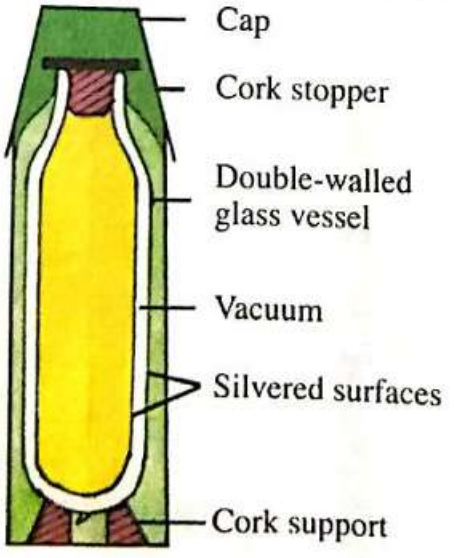


Fig. 4.26 Thermos flask

4 An electric pressing iron has a silvered surface at the base. The bright surface reduces the heat lost from the iron by radiation.

5 Factory roofs are brightly painted with aluminium to keep the interior cool, since the bright surface will not absorb much heat from the sun.

6 ***The Thermos flask (or vacuum flask)***
In the Thermos flask the three modes of heat transfer are brought into good application. The principal features of the flask are shown in fig. 4.26.

The flask has a double-walled glass vessel with a vacuum between the walls. The vacuum is to reduce the loss or gain of heat by conduction or convection. The double walls of the glass vessel are silvered on the vacuum side. This minimizes the loss or gain of radiant heat to the outside. The bottom of the flask rests on cork supports and the mouth of the flask is also closed by a cork stopper or a hollow plastic stopper with air inside. These materials, cork, plastic, and air are all poor conductors and hence no heat is lost or gained by conduction. The cork stopper also prevents heat loss by evaporation and convection if the flask contains hot liquid. The glass vessel is placed in an outer container made of plastic.

Because of all these features, heat lost or gained by the flask from the surroundings is very small. Hence the flask keeps cold liquids cold and hot liquids hot, for a long time.

Summary

1 Heat is a form of energy called thermal energy. It is the energy communicated from one body at higher temperature to another at lower temperature.

2 **Temperature** is the degree of hotness or coldness of a body. It is a scalar quantity.

3 Application of heat to a body may have the following effects: change in temperature; change of state; expansion; thermionic emission; change in physical and chemical properties such as electrical resistance, magnetic properties, conductivity, elasticity, density and colour, pressure change.

4 According to the kinetic molecular theory, the temperature of a body depends on the average kinetic energy of its molecules.

5 **The linear expansivity α** of a substance is the fractional increase in its length per degree change in temperature.

6 **Area or superficial expansivity β** is the fractional increase in area per degree change in temperature.

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7 **The volume or cubic expansivity** γ is the fractional increase in volume per degree change in temperature.

8 $\beta = 2\alpha$; $\gamma = 3\alpha$.

9 For a liquid, the real expansion is the apparent expansion + the expansion of the container. $\gamma_r = \gamma_a + \gamma$.

10 Advantages of expansion are utilized in the fitting of wheels on metal rims, fire alarms, thermostat, bimetallic thermometers, compensated pendulum.

The disadvantages of expansion are in the sagging of overhead cables, the buckling of railway lines, in the gaining or losing of time as a result of expansion of balance wheels of watches or clocks.

11 Water contracts when its temperature is raised from 0°C to 4°C; above 4°C it expands with temperature like other liquids. Hence, a given mass of water has its minimum volume and its maximum density at 4°C.

12 **Fusion (or melting)** is the process whereby a substance changes from a solid to a liquid state when heat is applied to it. The temperature at which this occurs is the fusion or melting point or temperature.

13 **Freezing temperature** is the temperature at which a substance changes from a liquid to a solid state.

14 **Vaporization** is the process in which a substance changes from a liquid to a gaseous state when heat is applied to it. Vaporization occurs at the boiling point of the liquid. This same process is known as **evaporation** when it takes place below the boiling point.

15 The three modes of heat transfer are *conduction*, *convection* and *radiation*. Conduction may occur in solids, liquids and gases; in it the average position of the molecule of the substance remains the same. In convection, heat is transferred by the actual movement of the molecules. It takes place in liquids or gases but not in solids. In heat transfer by radiation, no material medium is required. Heat from the sun reaches the earth by the radiation process.

16 Dull-black surfaces are good absorbers and good emitters of radiation. Highly polished or shiny surfaces are poor absorbers and poor emitters of radiation. Generally good absorbers are also good emitters of radiant energy.

17 The Thermos flask keeps hot liquids hot, and cold liquids cold for a long time. Heat is prevented from getting into or out of the flask by the following features: a double-walled silvered surface, a vacuum between the walls, an insulating support and an insulating stopper.

Exercise 4

1 The increase in the volume of 10 cm^3 of mercury when the temperature rises by 100°C is 0.182 cm^3. What is the cubic expansivity of mercury?

A $0.000182\ K^{-1}$ B $0.0000182\ K^{-1}$
C $0.000178\ K^{-1}$ D $0.000187\ K^{-1}$
E $0.00182\ K^{-1}$.

2 A brass rod is 2 m long at a certain temperature. What is its length for a temperature rise of 100 K, if the expansivity of brass is $18 \times 10^{-6}\ K^{-1}$?

A 2.0036 m B 2.0018 m
C 2.1800 m D 2.0360 m.

(JAMB)

3 A room is heated by means of a charcoal fire. A man standing away from the fire is warmed by

A conduction B reflection
C radiation D convection.

4 On which of the following is the design of the thermostat of an electric iron based?

A apparent expansion of a body
B thermoelectric effect
C expansion of metals when heated
D change in the resistance of a body on heating
E change in the density of a body when heated.

5 Which of the following surfaces will absorb radiant heat energy best?

A White B Red C Yellow
D Black E Blue.

6 On a cold morning, the metal blade of a cutlass feels colder to touch than the wooden handle because

A The blade is at a lower temperature than the handle.
B The hand is at a lower temperature than both blade and handle.
C The blade is a better conductor of heat than the handle.
D The handle is a better conductor of heat than the blade.
E The handle contains some heat which is absent in the blade. (SC/GCE)

7 It is advisable to wear white dresses in the tropics because white can

A absorb sweat readily

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B absorb all radiant heat
C conduct heat away more readily from the body
D reflect radiant heat
E absorb the rays from the sun.

(SC/GCE)

8 In which of the following is the expansion of solids a disadvantage?
A The fitting of wheels in rims
B Fire alarms
C The thermostat
D The bimetallic strip
E The balance wheel of a watch.

9 Which of the following is a reason why a concrete floor feels colder to the bare feet than a mat on the same floor?
A Mat is a better conductor of heat than the feet.
B Mat loses heat to the bare feet at a faster rate than the concrete floor.
C Mat loses heat to the bare feet while the concrete floor extracts heat from them.
D Concrete floor is a better conductor of heat than mat.
E Mat is a better conductor of heat than the floor.

(SSCE)

10 Define *linear expansivity*.
An iron rod is 1.58 m long at 0°C. What must be the length of a brass rod at 0°C if the difference between the lengths of the two rods is to remain the same at all temperatures.
Linear expansivity of iron = 1.2×10^{-5} K^{-1}
Linear expansivity of brass = 1.9×10^{-5} K^{-1}.

11 Describe an experiment each to show
a. Convection current in water
b. Copper is a better conductor of heat than wood
c. Water is a poor conductor of heat.

12 Draw a labelled diagram of a vacuum (thermos) flask. Explain how its construction minimizes heat exchange with the surroundings.

(SSCE)

13 a. What is meant by the statement: *the linear expansivity of a solid is* 1.0×10^{-5} K^{-1}.
b. (i) Describe an experiment to determine the linear expansivity of a steel rod.
(ii) Steel bars each of length 3 m at 29°C are to be used for constructing a rail line. If the linear expansivity of steel is 1.0×10^{-5} K^{-1}, calculate the safety gap that must be left between successive bars if the highest temperature expected is 41°C.
c. State *three* advantages and *two* disadvantages of thermal expansion of solids.

(SSCE)

14 Explain the three modes of heat transfer. State the major differences between them. By which of the modes does heat from the sun reach the earth?

15 Describe some application of conduction, convection and radiation.

16 State the assumptions of the kinetic molecular theory. How does the theory explain conduction, and convection.

17 Describe an experiment to show that blackened surfaces are better heat absorbers than shiny surfaces.

18 Explain why a tile floor feels colder to the feet than does a rug in the same room.

19 What is a convection current? Describe a phenomenon of convection current in nature.

20 Distinguish between *real* and *apparent* cubic expansivities of a liquid.
A glass bottle full of mercury has mass 500 g. On being heated through 35°C, 2.43 g of mercury are expelled. Calculate the mass of mercury remaining in the bottle (cubic expansivity of mercury is 1.8×10^{-4} per K, linear expansivity of glass is 8.0×10^{-6} per K.

(SC/GCE)

21 The density of mercury is 1.36×10^{4} kgm^{-3} at 0°C. Calculate its value at 100°C and at 22°C. Take cubic expansivity of mercury as equal to 180×10^{-6} K^{-1}.

22 Find the values of *A*, *B*, *C*, *D*, *E* in the following table.

	Original length, area, or volume	Linear expansivity 10^{-6} K^{-1}	Rise in temperature in K	Increase in length, area or volume
1	0.5 m	10	15	*A*
2	2.0 m	25	*B*	3.0 mm
3	0.6 m	*C*	80	1.0 mm
4	*D*	16.7	100	25 mm^2
5	500 mm^2	18	*E*	0.36 mm^2

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23 a. Define *linear expansivity* of a solid.

b. Describe a method of measuring the linear expansivity of a copper rod.

c A compound strip of iron and brass is straight at room temperature. Draw a labelled diagram to show its appearance when it is heated. Linear expansivity of iron = $1.2 \times 10^{-5}\ K^{-1}$. Linear expansivity of brass = $1.8 \times 10^{-5}\ K^{-1}$

d. A brass measuring tape is correct at 20°C. The value obtained when the length of a field is measured with the rule at 50°C appears to be 70.5 m. What is the true length of the field? Linear expansivity of brass = $1.8 \times 10^{-5}\ K^{-1}$.

24 A solid metal cube of side 10cm is heated from 10°C to 60°C. If the linear expansionity of the metal is $1.2 \times 10^{-5} K^{-1}$, calculate the increase in its volume.

(a) $0.6cm^3$ (b) $1.2cm^3$ (c) $1.8cm^3$
(d) $3.63cm^3$ (e) $6.0cm^3$ (SSCE)

25 (a) Define upper fixed point and lower fixed point as used in thermometry.

(b) The electric resistance of the element in a platinum resistance thermometer at 100°C, 0°C and room temperature are 75.000, 63.000 and 64.992 ohms respectively. Use these data to determine the room temperature. (SSCE)

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5 ELECTRIC CHARGES

In this chapter, we shall introduce Electrostatics or static electricity, a type of electricity that does not move from one point to another in the substance in which it is produced. The study of electrostatics is the study of charges at rest. The chapter will discuss the different ways of producing electric charges, the types of charges, their distribution and storage.

At the end of the chapter, the student should be able to

1 Charge a body by friction, induction and contact.
2 Identify bodies charged either similarly or oppositely.
3 Construct simple electroscope.
4 Show how an electroscope can be used to detect the presence and magnitude of electric charges.
5 Distinguish between electric force and other types of force.
6 Explain the action of lightning conductors.

5.1 Production of Charges

The following simple experiments will introduce the student to some phenomenon that are due to static electricity.

Activity 5.1

1 Ensure your hair is dry, then rub the barrel of a plastic biro or ball pen vigorously on it. Hold the rubbed pen near some small pieces of paper and observe the effect of the pen on the paper.
2 Rub a plastic ruler vigorously against the sleeve of your shirt or blouse. Then hold the rubbed ruler over small pieces of paper and observe the effect on the paper.
3 Comb your hair on a dry day with a plastic comb. Observe that tufts of hair will be attracted to various parts of the comb.

Repeat this activity in front of a mirror in a dark room and report your observations.

The rubbed plastic biro or ruler attracts small pieces of paper. The comb in the above experiment attracts some hair tufts and in a dark room we can observe tiny sparks between the comb and hair especially during the dry harmattan season. Also during a dry day you can feel some crackling sensation as you comb your dry hair with a plastic comb.

The plastic passing through the dry hair as well as the hair itself acquire some electric charge by friction between the two and is said to be electrified or charged or to have a charge. Similarly the plastic biro or ruler rubbed vigorously on a shirt acquires electric charge by friction between the rulers and the shirt. These electrostatically charged materials attract the small pieces of paper through a *force field*, an electric force, since they do not actually come in contact with the piece of paper, before attracting it. The piece of paper is also charged before being attracted.

There are other observations of static electricity in our daily experience. For example, we often hear some crackling noise near the ear when taking off a dry sweater or a nylon garment especially during the dry harmattan weather. Also when we attempt to take off nylon or terylene clothes in a dry weather, the clothes tend to stick to the body because they are strongly electrified. The hair on our body has a strong pull on these types of clothes.

5.2 Types of Charges

There are two types of charges – positive and negative charges.

One type of electric charge could be produced by rubbing a glass rod with silk. This is the *positive charge*. A different type of charge could be obtained on an ebonite rod by rubbing it with fur. This type is known as the *negative charge*.

Activity 5.2

Rub a glass rod vigorously with silk and suspend it by a thread with a paper stirrup. Rub another glass rod with silk and bring it near the first rod. Notice the repulsion between the two as the suspended glass rod moves away from the second glass rod.

In a similar fashion, rub an ebonite rod with fur and bring it near the suspended glass rod. Observe the attraction between the two rods.

Now, replace the suspended glass rod with an ebonite rod which has been charged by rubbing with fur. Repeat the previous experiment by bringing a charged ebonite rod and a charged glass rod in turn towards the suspended ebonite rod; observe that attraction and repulsion occur when the charged glass and ebonite rods are brought near the suspended charged ebonite rod.

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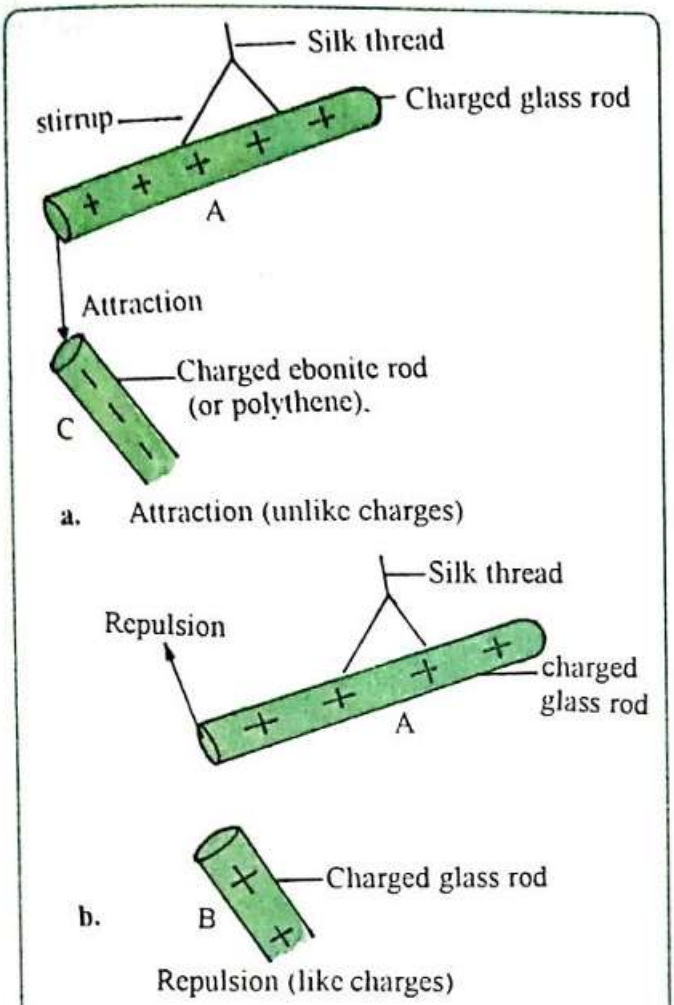


Fig. 5.1 Attraction and Repulsion by charges

The above experiments prove that the charge on the ebonite rod is different from that on the glass rod. An ebonite rod rubbed with fur has a negative charge, but a glass rod rubbed with silk has a positive charge. Nowadays we can also obtain a positive charge by rubbing cellulose acetate strip with silk and a negative charge by rubbing polythene rod with fur. The cellulose acetate strip and the polythene rod are less affected by moist conditions in the laboratory.

The results of our experiments in *Activity 5.2* show that a positively charged glass rod attracts a negatively charged ebonite rod but repels another positively charged glass rod. Also a negatively charged ebonite rod attracts a positively charged glass rod but repels another negatively charged ebonite rod. These results lead to the conclusion that

Like or similar charges repel each other; unlike or opposite charges attract each other

as a fundamental law of static electricity.

Atomic structure and static electricity

A basic knowledge of the structure of the atom will help us to understand more clearly the phenomenon of static electricity. Only a short overview of the atomic structure will be given here. Greater details about atomic structure will be discussed in Book 3.

Atomic theory assumes that matter is made up of molecules and the molecule is made up of atoms. The atom itself, is made of a central part known as the nucleus, made up of protons and neutrons. The proton is a positively charged heavy particle and cannot be transferred easily. The neutron has no charge. The other part of the atom is made of electrons which circle in orbits round the central nucleus. The electron is a light particle carrying a negative charge. The total number of positive charges of the proton is equal to the total number of negative charges of the electrons. This makes the atom in its normal state to be electrically neutral.

When the neutral atom gains an electron, it becomes negatively charged, because it has more negative charges than positive charges. When the neutral atom loses an electron, it has a net positive charge. It is therefore positively charged.

When two different materials are made to rub against each other, e.g. silk on glass, friction brings about a net transfer of surface electrons from glass to the silk. Therefore the glass becomes positively charged and the silk becomes negatively charged. Likewise a negative charge gained by an ebonite rod when rubbed with fur is due to a net transfer of electrons from the atoms of fur to the ebonite rod. The fur itself therefore becomes positively charged. It is the gain or loss of electrons that determines whether a body is negatively or positively charged. The positively charged protons deep in the nucleus are not free to be transferred.

Conductors and insulators

In static electricity, substances are classified according to the degree of resistance they offer to the passage of electrons through them. There are two such groups – Conductors and Insulators.

Conductors are materials that allow electrons to pass through them easily.

Examples of conductors are almost all metals, damp air, graphite, acids, salt solutions, the earth, and the human body.

Insulators are materials that do not allow electrons to pass through them easily.

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Examples of insulators are plastic, polythene, bakelite, ebonite, paper, dry hair, silk, oils, glass, sulphur and wood.

As a general rule, good conductors of heat are also good conductors of electricity.

Metals are usually supported with insulating materials before they are charged. If they are supported with the hand, the charges produced in them will immediately flow through the human body to the earth.

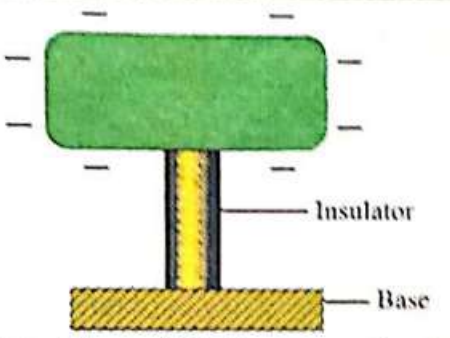


Fig. 5.2 Charged body on an insulating stand

The Gold Leaf Electroscope

An electroscope is an instrument used for the detection and testing of small electric charges. Its essential features are as shown in fig. 5.3.

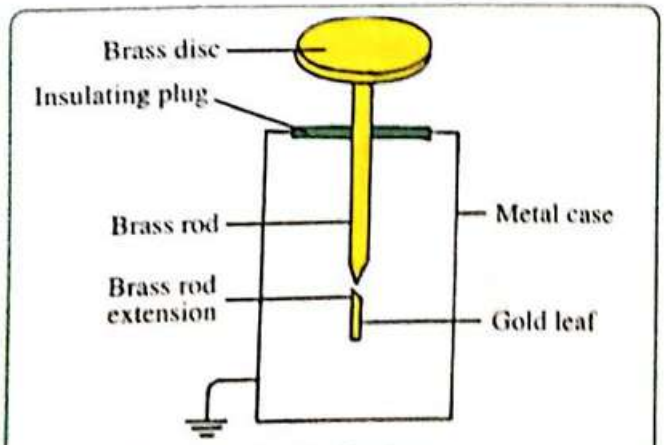


Fig. 5.3 Gold-leaf electroscope

It consists of a flat brass disc or cap and a brass stem or rod with a gold leaf (or leaves) attached to the end of the rod. The brass rod is insulated from the metal case which surrounds the rod and leaf. The metal case is made draught-proof with two plane glass sides such that the leaf can be clearly visible. The metal case is normally connected to the earth (or earthed) to prevent the accumulation of charges due to external influence. The flat brass disc is one terminal of the instrument and the earthed metal case is the other terminal. The leaf is the sensitive part of the instrument, its position shows the presence of charges in the disc and rod.

Charging the Electroscope

a. By contact

The electroscope can be charged by sliding a charged body on the brass disc of the instrument. Charge passes from the charged body to the disc and stem of the instrument down to the bottom through this *contact*. The leaf is observed to open when this happens. The leaf is repelled from the metal as both carry similar charges. The degree of divergence of the leaf indicates the amount of charge acquired by the instrument. The leaf remains in this divergence position even after the charging body has been removed showing that a charge has been acquired by the electroscope.

The charge acquired by the instrument is the same as that of the charging body, i.e. if a negatively charged ebonite rod is brought in *contact* with the brass disc or cap, the electroscope acquires a negative charge. If a positively charged glass rod is brought in contact with the electroscope, the instrument acquires a positive charge.

b. By electrostatic induction

If a charged rod is brought near the cap of the electroscope *without touching it*, the leaf is seen to diverge, indicating that the instrument has acquired some charges. When this happens, we say that charge has been induced in the electroscope through the phenomenon of *electrostatic induction*.

By electrostatic induction, we mean the act of charging a neutral body by placing a charged body near it without any contact between the two.

To charge an electroscope by induction we require four steps

(i) A negatively charged rod is brought near the cap of the electroscope. Some free electrons in the metal cap are repelled to the leaf which now diverges leaving some positive charge on the cap.

(ii) The cap is touched with a finger and electrons are repelled towards the earth through the hand. The leaf closes as a result.

(iii) The finger is removed, and a net positive charge is left on the cap.

(iv) Finally the rod is removed, leaving a net positive charge on the cap. This positive charge is now distributed between the cap and the leaf causing the leaf to diverge due to repulsion.

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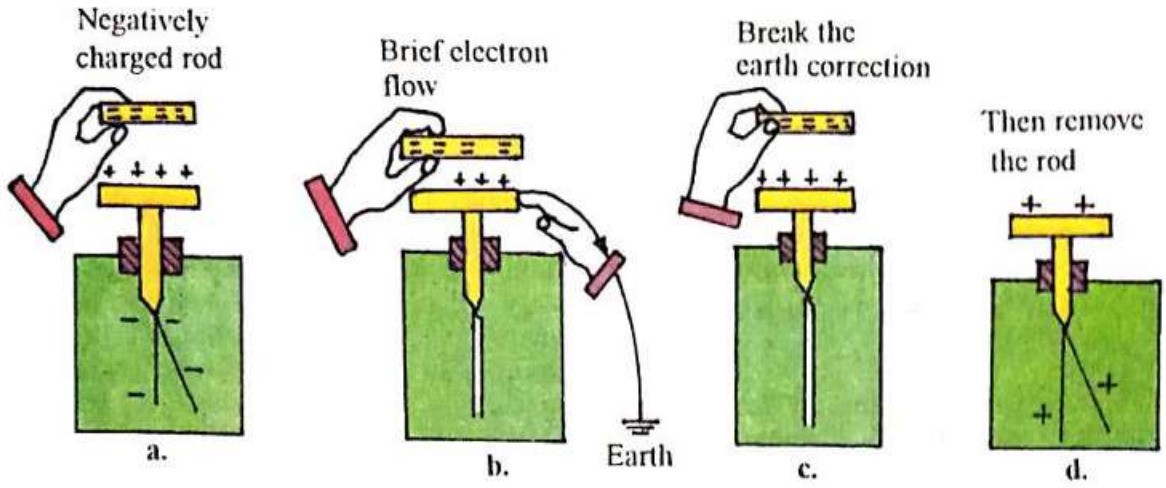
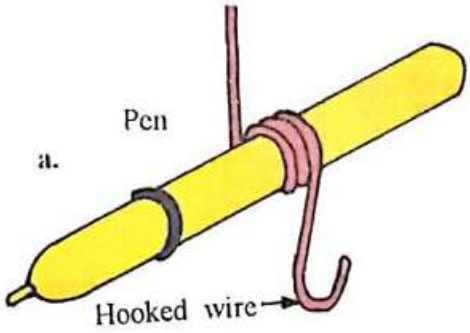
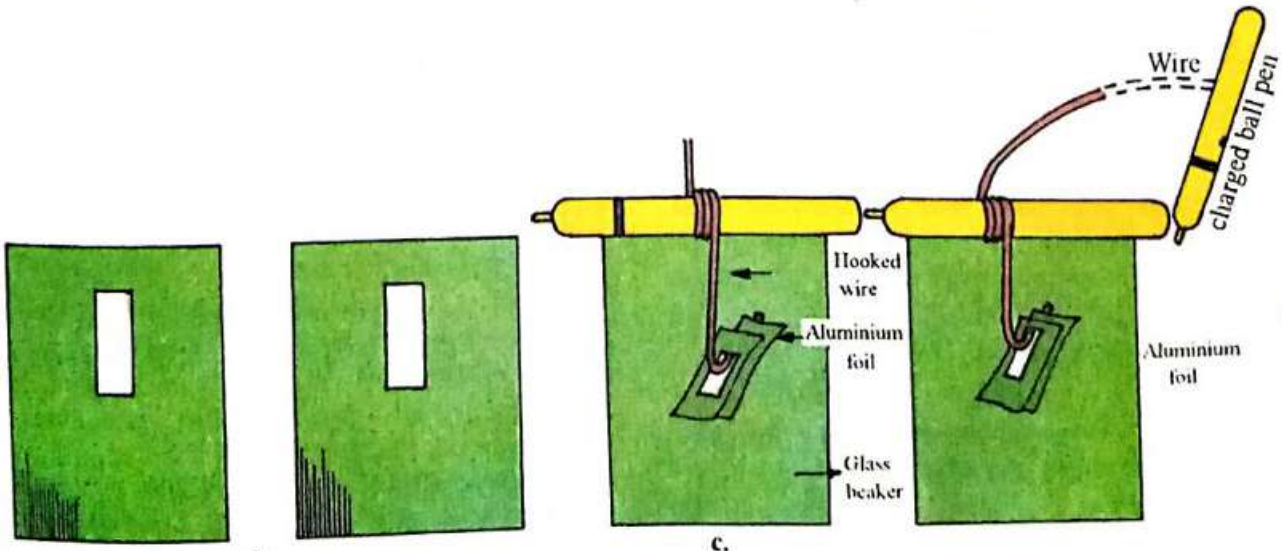


Fig. 5.4 Charging an electroscope by induction



An electrosope made from a ball pen and copper wire



Slits in aluminium foils

A simple electroscope

Electroscope being charged by a wire

Fig. 5.5 Construction of a simple electroscope

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Note that if the rod is removed before the finger is removed, no induced charge is obtained, because the electrons return to the electroscope through the body.

If we start with a positive rod instead of a negative one, we shall end up with a negatively charged electroscope.

To construct a simple electroscope

The student can construct a simple electroscope by the following activity.

Activity 5.3

1 Wind a bare copper wire around the barrel of a biro pen. Bend one end of the wire into a hook as shown in fig. 5.5a.

2 Obtain an aluminium foil from a cigarette packet and from it cut out two small rectangular sheets to act as the leaf of your electroscope. Cut two slits in the aluminium sheet as shown in fig. 5.5b.

3 Suspend the sheets freely from the hook. Cover the open top of a glass beaker with the biro pen barrel so that the hook and the aluminium leaf hang freely in the beaker (fig. 5.5c). This set-up is your simple electroscope.

4 Now test your electroscope through the following activity.

   a. First rub the plastic barrel of another biro pen on your dry hair or shirt sleeve and bring it near some small pieces of paper. What happens? What does it show about the biro pen barrel?

   b. Rub the barrel of the pen again on your dry hair and bring it to touch the top of the copper wire. Report your observation.

   c. Touch the copper wire with your finger. What happens? Finally remove the charged biro pen barrel and report your final observation.

5.3 Ways of Producing Charges

The results of our discussions so far show that there are three methods of producing charges (1) By friction (2) By contact and (3) By induction.

1 By friction

From the results of our *Activity 5.1*, we say that the charges have been acquired by friction, the friction between the comb and the hair, the plastic biro and the hair or shirt and between the plastic ruler and the shirt. Also when an ebonite rod is rubbed with fur, the ebonite rod acquires a charge by friction. Similarly a glass rod acquires a charge by friction when rubbed with silk.

Equal and opposite charges are produced by friction. Thus when an ebonite rod is rubbed with fur, negative charges are produced on the rod and an equal number of positive charges are left on the fur.

Moving vehicles become charged by passage through the air especially in dry weather. The friction due to the rubbing of air on the fast moving vehicle may cause electric charges to collect on the body of the vehicle. Hence people stepping out of cars and buses occasionally complain of a slight electric shock as soon as their feet touch the ground. Electrostatic charges acquired by friction when a petrol tanker moves through the air may cause a spark especially when inflammable vapour is present that can set the tanker ablaze. Hence to prevent this type of occurrence, a chain (or metal strip) is often left hanging from the rear of such petrol tankers.

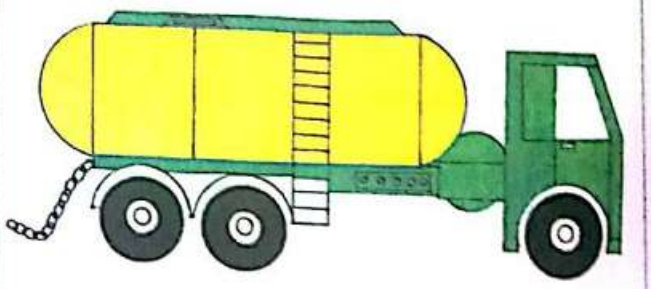


Fig. 5.6 A Petrol tanker with discharge chain

Static electric charges acquired by the tanker pass through the chain down to the earth where they are neutralized.

2 By contact

We have shown that an electroscope can be charged when a charged ebonite rod is brought in contact with the brass cap. As another example, we bring two uncharged conductors C and D in contact and on insulating stands. A positively charged glass rod is brought near to C as shown. If D is then separated from C while the rod is still near C, D can be shown to be positively charged while C is negatively charged. C is charged by induction but D receives its charge from C by contact. When the rod is eventually removed the negative charge spreads on C, and the positive charge spreads on D.

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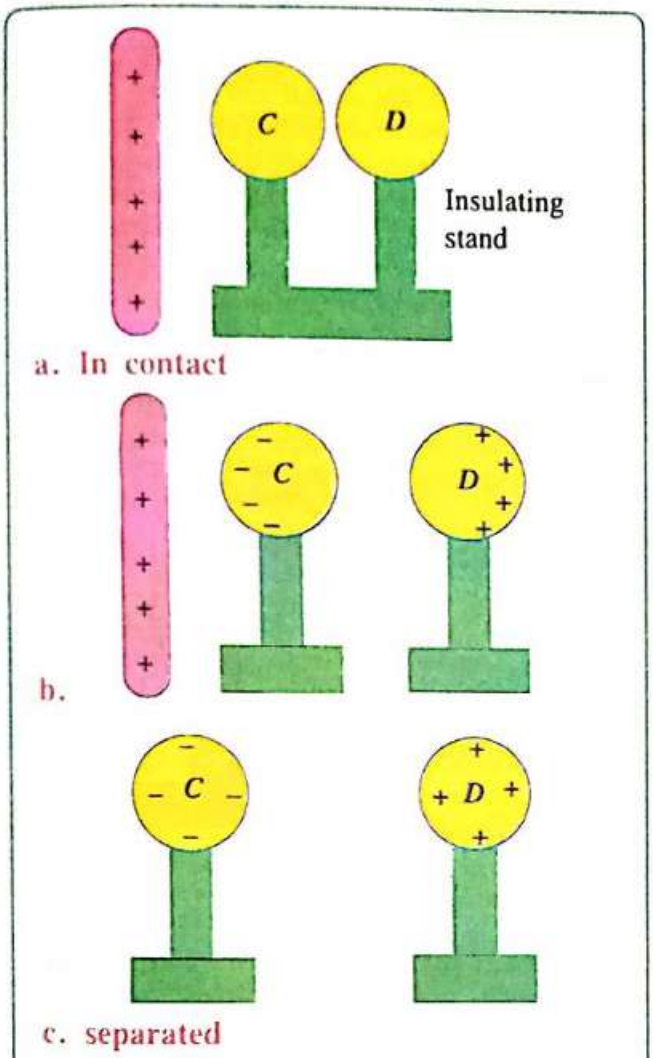


Fig. 5.7 Production of charges by contact

Charging by contact is not a very good method.

3 By electrostatic induction

A body can also receive charge by electrostatic induction. If an uncharged conducting body *CD* is mounted on an insulating stand and a positively charged glass rod is brought near one end, *C* as shown in fig. 5.8, it can be shown that the end *C* becomes negatively charged while the other extreme end *D* becomes positively charged. The charges produced are called *induced charges*, and the charge on the glass rod is called the *inducing charge*.

When the charging rod is removed the charges on *CD* will disappear. If we had a negatively charged ebonite rod near the end *C*, that end will be positively charged while end *D* will be negatively charged. The conducting body has been charged by electrostatic induction since the rod has made no contact with the conductor. How do we explain this? When for instance the negatively charged ebonite rod is brought near the end *C*, the electrons in the rod repel those at *C* and force them to the extreme *D*. Consequently *C*, having lost electrons becomes positively charged. The end *D* having received electrons becomes negatively charged.

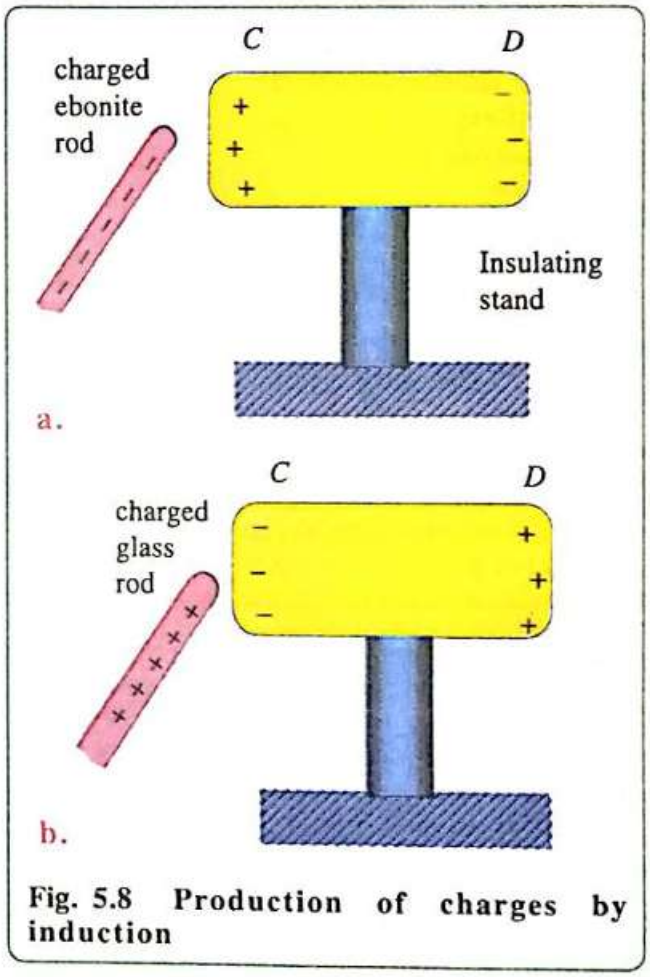
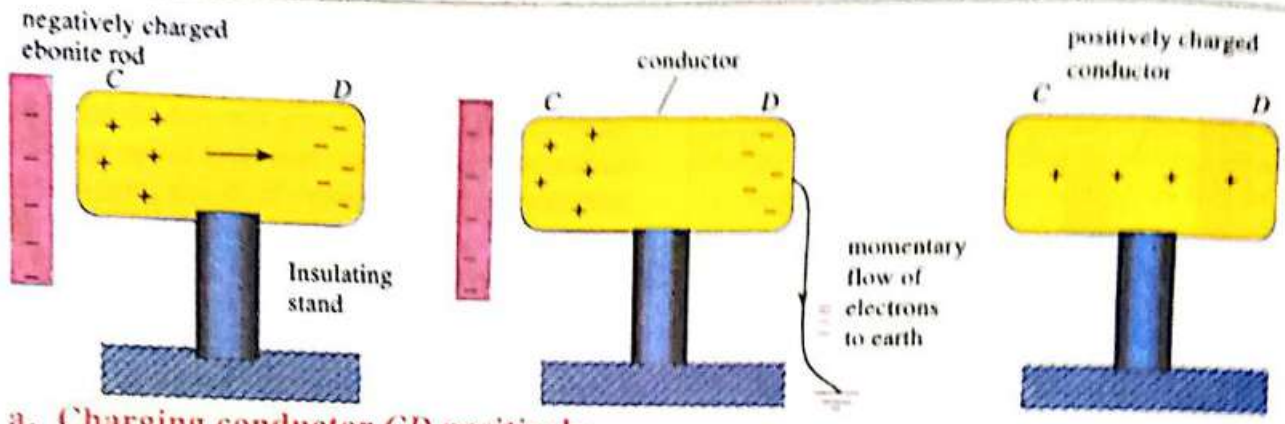


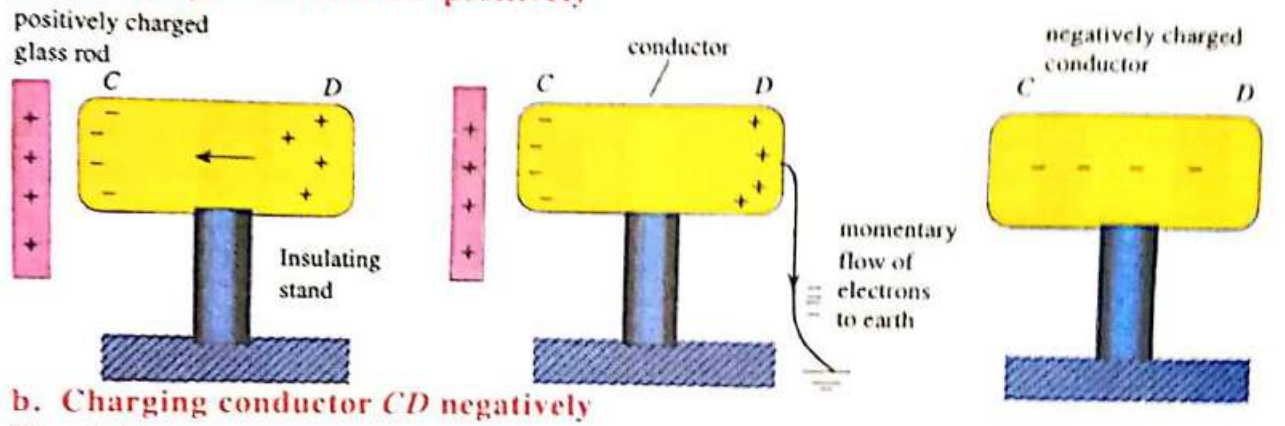
Fig. 5.8 Production of charges by induction

To make the charge on *CD* permanent, we bring the negatively charged ebonite rod near *C* and touch the end *D* with the fingers thus earthing the conductor. The negatively charged electrons leak away to the earth through the human body, leaving only the positive charges on the conductor.

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a. Charging conductor *CD* positively



b. Charging conductor *CD* negatively

Fig. 5.9

These positive charges remain on the conductor as long as the inducing charge remains. If the finger is now removed, and then the ebonite rod is removed, *CD* will be left with a net positive charge which then distributes itself throughout the conductor. These steps are illustrated in fig. 5.9.

To Distinguish Between Conductors and Insulators Using an Electroscope

If the brass cap of a charged electroscope is touched with the finger, the leaves of the electroscope collapse showing that the human hand has conducted away some of the charges. The human body is a fairly good conductor of electric charges.

Similarly when any other body is brought in contact with the metal cap of a charged electroscope, a rapid collapse of the leaves indicates that the body is a good conductor while a gradual collapse will indicate that the body is a poor conductor. When there is no change at all in the divergence of the leaf, then the body is an insulator.

In this way, we can distinguish between a good conductor, a poor conductor and an insulator.

To determine the type and magnitude of charge on a body using an electroscope

We can use a gold leaf electroscope to test the sign of the charge on an object. We summarize in the table below the effect on the gold leaf of a charged electroscope when a charged body and an uncharged body are brought near its metal cap.

Charge on electroscope	Charge brought near the cap	Effect on leaf
negative	negative body	divergence increases
negative	positive body	divergence decreases
positive	positive body	divergence increases
positive	negative body	divergence decreases
positive	uncharged body	divergence decreases
negative	uncharged body	divergence decreases

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We see from the table that an uncharged body also causes a decrease in the divergence of the gold leaf. Hence we cannot take a decrease in the divergence of the leaf of a charged electroscope as an indicator of the presence of a charged body or the type of charge possessed by the body. *Only an increase in divergence of the leaf can be taken as a sure sign test of the type of charge on a body.* An increase in divergence indicates that the charge on the electroscope and that of the test charge are the same.

The magnitude of the charge is also measured by the degree of divergence of the leaf. A greater degree of divergence indicates a greater amount of the test charge.

5.4 Distribution of Charges on Conductors

We can charge an insulated pear-shaped conductor and examine the distribution of charges over its surface. The conductor can be charged by electrostatic induction and examination of charge distribution can be done with an electroscope and a small metal disc D with an insulating handle H known as a *proof plane*.

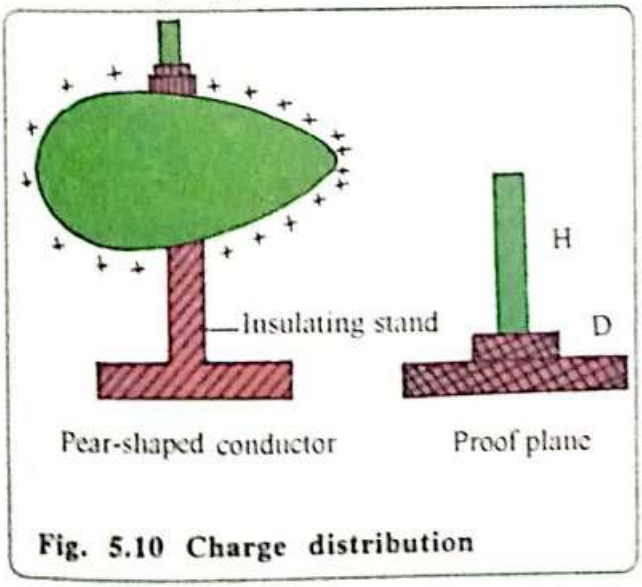


Fig. 5.10 Charge distribution

Such an examination shows that charges are usually concentrated at places where the surface is sharply curved. *The charge per unit area of surface, or charge density is highest at the sharpest point of the conductor.* The sharpest point has the smallest surface area, and the greatest concentration of charges.

Similar examination of a charged hollow conductor shows that the charge on a conductor collects only on its outside surface.

Action at a point: Lightning conductors

We noted in the last section, that charges are most highly concentrated at the sharp points of a conductor. As a result, when such a highly charged pointed conductor is placed in the air, it causes the air around the sharp point to be ionized, i.e. broken into positive and negative ions. Suppose the sharp point is positively charged, its high charge density will strongly attract the negative ions of the air and strongly repel the positive ions.

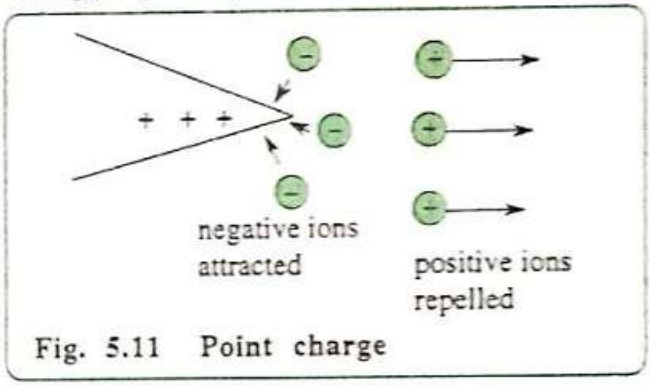


Fig. 5.11 Point charge

The fast moving ions will collide with surrounding air molecules stripping off electrons from them and causing further ionization and repulsion of positive ions. This process has a cummulative effect and within a short time an avalanche of such ions will cause the surrounding air molecules to be dragged along with it producing an electric wind.

The negative ions will move towards the positively charged sharp point and gradually neutralize its charges.

This whole process is known as *point action.*

Lightning and Lightning conductors

Dramatic displays of electrical discharges in nature are seen in lightning flashes which are common during the rainy season. When electrical charges formed in thunderclouds build up, the attraction between unlike charges within a cloud or between different clouds increases steadily until opposite charges suddenly jump from cloud to cloud or from cloud to the earth in the form of huge sparks of electrical discharges. This spark is seen as a very bright flash which we call *lightning*. A great amount of heat is released in this process and a building, a tree or any other tall structure can be set on fire if a safe path is not provided for this discharge.

A lightning conductor, by its design, is used to provide this safe path and to protect buildings from being damaged. It consists of sharp pointed metal rods or spikes attached to the highest point of a

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building or structure and joined by a thick strip of copper cable to a copper plate buried in the ground beside the building.

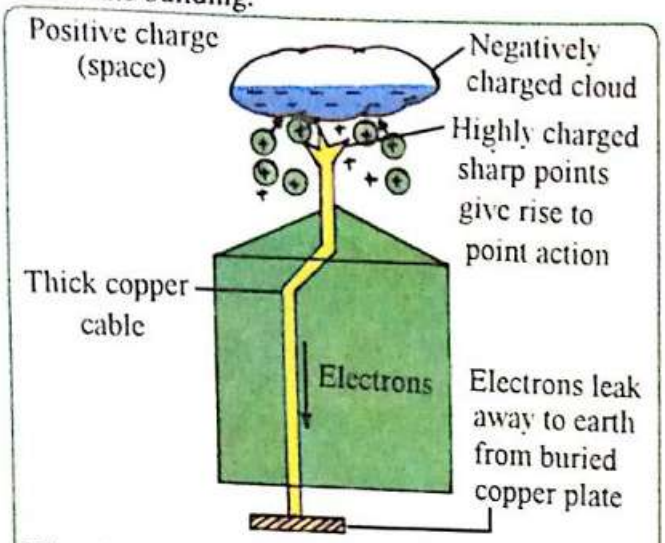


Fig. 5.12 Action of lightning conductor

The thick copper cable which is fixed to the building, is able to carry a large current without melting. The lightning conductor works on the principle of *point action* or *action at a point.*

When for instance, a negatively charged thundercloud passes over the area of a building, the sharp points of the spikes gain induced charges opposite to those of the cloud. The air around the spikes is further ionized due to point action, leading to a huge avalanche of positive ions which now move upwards and cause the formation of a positively charged space higher up in the atmosphere. This space charge slowly neutralizes part of the negative charges of the cloud and reduces the possibility of lightning.

If however lightning still strikes the building, the negatively charged electrons are attracted towards the spikes and are carried harmlessly down to the earth, through the conducting copper cable.

5.5 Storage of Charges – The Electrophorus

The electrophorus is a device for transferring and storing charges. It produces electric charges by electrostatic induction.

It consists simply of a circular metal disc or plate P fitted with an insulating handle H and another flat disc made of an insulating material such as ebonite or polythene tile.

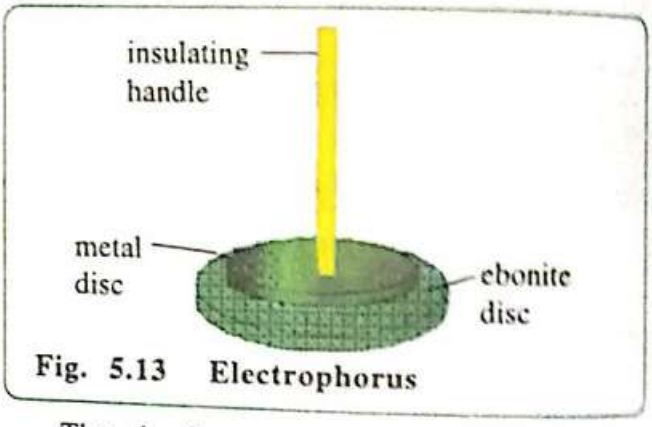


Fig. 5.13 Electrophorus

The ebonite or polythene disc is given a negative charge. The metal disc is then placed over the ebonite disc. There are only a few contact points between the two discs, hence no charges flow between them. Positive and negative charges are given to the metal disc by electrostatic induction. The metal disc is now touched with the finger, the negative charges flow away through the body to the earth, and the metal disc is left with only the positive charges. These charges are then stored in the disc when the handle is lifted from the ebonite. The amount of charge stored can be built up by repeating the above process.

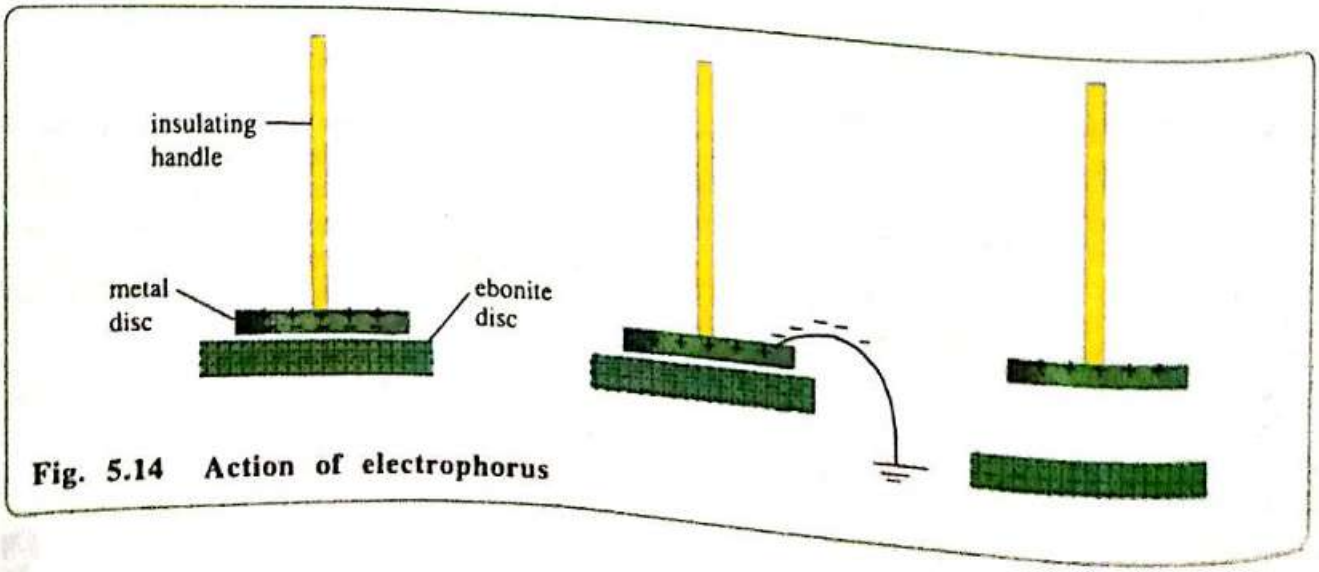


Fig. 5.14 Action of electrophorus

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Another device for the storage of electric charges is the *capacitor*. This will be discussed in Book 3.

Force between charges – Force field

In chapter 2 we had distinguished between two main classes of force – the contact force and the non-contact force or force field.

When we rub the barrel of a ball point pen vigorously on a dry hair or shirt, it attracts small pieces of paper with which it had no contact. Hence we can say that electrostatic force is a non-contact force or force field. We will learn later that other examples of such non-contact forces are gravitational fields, electric fields and magnetic fields.

Summary

1 An ebonite rod (or a polythene strip) rubbed with fur acquires a negative charge but a glass rod (or a cellulose acetate strip) rubbed with silk acquires a positive charge.
2 A fundamental law of static electricity is that like charges repel but unlike charges attract each other.
3 When a body loses some of its electrons, it becomes positively charged. When it gains electrons, it becomes negatively charged. When it has equal number of electrons and protons the body is said to be electrically neutral.
4 Electrostatic conductors allow electrons to pass through them easily but insulators do not allow electrons to pass through them easily.
5 Charges can be produced by friction, by contact and by electrostatic induction.
6 By electrostatic induction we mean the act of charging a neutral body by placing a charged body near it without any contact between the two.
7 If an unknown charge is brought near to a charged electroscope, and the leaf diverges more, the known charge is similar to the charge on the electroscope.
8 The concentration of charge on a conductor is greatest at the sharpest point of the conductor.
9 Lightning conductors work on the principle of point action.
10 An electrophorus is used to store charges.
11 The forces between two charges is a non-contact force or force field.

Exercise 5

1 Fill in the gaps in the following
   A When an ebonite rod is rubbed with fur, the rod acquires a ... charge.
   B When a glass rod is rubbed with silk, the rod acquires a ... charge.
   C When two similarly charged rods *A* and *B* are brought together they ... each other.
   D When an electron is transferred to a neutral rod, the rod becomes ... charged.
   E When a neutral rod loses an electron, it becomes ... charged.

2 When charging an insulated conductor by induction, the following steps are usually taken.
   I Bring a charged body near to the conductor
   II Remove the charged body
   III Touch the conductor with a finger
   IV Remove the finger.
   What is the correct sequence in this activity?
   A I, II, III, IV
   B I, III, IV, II
   C I, IV, III, II
   D I, III, II, IV
   E III, IV, I, II

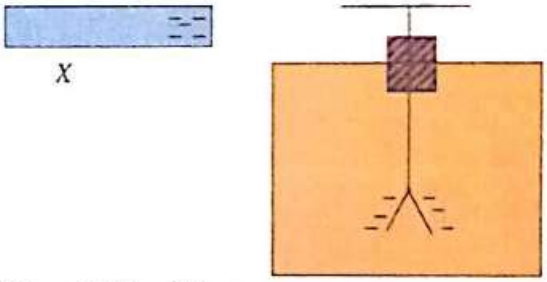
3 

Fig. 5.15 Electroscope

In the above figure, when the negatively charged rod X is brought near the cap of the electroscope.
   A The leaves will open further
   B The leaves will close up
   C The rod will become positively charged
   D The leaves will become positively charged
   E The cap will become negatively charged.

4 A glass rod rubbed with silk becomes positively charged because:
   A some positive charges are transferred from the silk to the glass
   B some electrons are transferred from silk to the glass
   C the silk becomes deficient in electrons
   D some electrons have been transferred from glass to silk
   E the glass and silk share their free electrons equally.

5 A building can be adequately protected from lightning by
   A using asbestos for the roof of the house
   B planting trees around the house

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C fixing a long copper strip from the ground along the outside wall to a sharp vertical spike

D fixing a long wooden pole with sharp spikes to the outside wall

E fixing a long ebonite rod with sharp spikes to the outside wall.

6 In fig. 5.16 below, the greatest charge density is at point

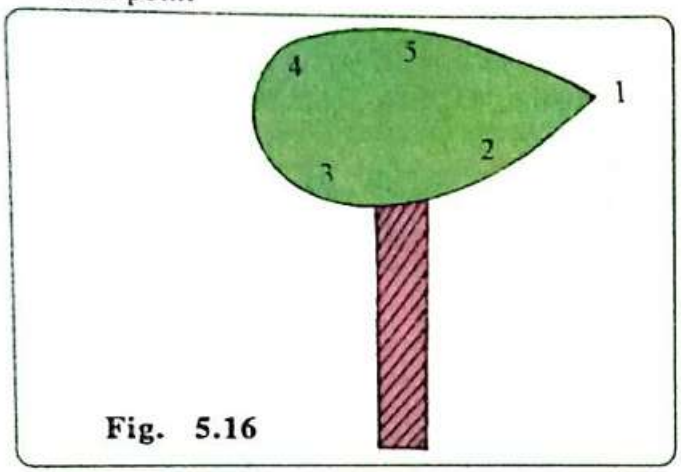


Fig. 5.16

A 1 B 2 C 3 D 4 E 5

7 A short chain is usually attached to the back of a petrol tanker trailing behind it to ensure that the

A petrol tanker is balanced on the road

B heat generated by friction in the engine can be conducted to the floor

C charges generated by friction in the tanker is conducted to the earth

D tanker moves slowly as the chain touches the road surface

E chain produces sound for the resonance of the tanker's engine.

(SSCE)

8 Which o the following instruments can be used to compare the relative magnitudes of charge on two given bodies?

A The electrophorus

B Ebonite rod

C Proof blanes

D Gold-leaf electroscope

E Capacitor

9 Describe how a lightning conductor protects a building from lightning.

10 What is meant by *electrostatic induction*? How would you charge a rod through this method?

11 Explain why

a. an ebonite rod rubbed with fur attracts small pieces of paper

b. nylon undergarments crackle in dry weather

c. the upper end of a lightning conductor is pointed.

12 Distinguish between *conductors* and *insulators*. Which of the following are good conductors: aluminium, graphite, rubber, copper, plastic, sulphur, ceramic materials.

13 Draw a labelled diagram of a gold-leaf electroscope. Given a plastic rod and fur, how would you charge the electroscope

a. positively b. negatively.

14 Explain why when a polythene rod which has been rubbed with dry cloth is held high above small pieces of paper on a table, the pieces of paper jump up and down repeatedly between the table and the rod.

15 List and describe the three ways of charging a conductor.

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DESCRIPTION AND PROPERTY OF FIELDS

6

The concept of fields and the properties of such fields are vast subjects which we will only be introducing in this chapter and the next. Detail discussion of fields will be taken up in Book 3.

At the end of this chapter, the student should be able to:

1 Explain the concept of force fields.
2 Identify force fields from a set of forces.
3 Identify the properties of fields – gravitational and magnetic.
4 Explain why two bodies of different masses when released from rest at the same point simultaneously, fall to the ground at the same time.
5 Use iron filings and a bar magnet to show field lines.
6 Use the compass needle to plot field lines.
7 Define magnetic lines of force.

6.1 Concept of Fields

We define a field as a region of space under the influence of some physical agency such as gravitation, magnetism and electricity.

There are many types of fields but for this chapter we shall be discussing gravitational and magnetic fields. In chapter 2, we had defined force fields as *forces whose sources do not require contact with the body to which they are applied.* We identified such force fields as gravitational force, electric force, magnetic force and electromagnetic force.

There are two classes of force fields – scalar fields and vector fields. A *scalar field* is one that has only magnitude but no direction, e.g. temperature, energy and density. A *vector field* is a field that has both magnitude and direction, e.g. gravitational, magnetic and electric fields.

6.2 Gravitational Field

If we throw up massive objects, it is our common observation that they move up to their highest points, stay stationary very briefly and eventually move downwards, falling faster and faster until they hit the ground level or the lowest level on their path. The up and down movements of objects on the earth's surface are subject to the influence of the Gravitational field of the earth.

Gravitational field is a convenient and informative method of describing the gravitational attraction of one body for another at a distance. *Gravitational field is a force field* and influences the motion of objects in the space where it operates, without coming in contact with the objects. Gravitational field acts over a distance. It surrounds every object that has mass and it permeates all of space.

The earth attracts every object existing in the earth's gravitational field. This attraction is called *gravitational attraction* and its effect is to change the velocity of objects under its influence, i.e. to accelerate such objects.

The acceleration of objects due to the earth's gravitational attraction is called the *acceleration due to gravity*. It is represented by the symbol g whose average value is about 9.81 ms^{-2}.

The acceleration due to gravity g is uniform at a given place and is the same for all bodies irrespective of their masses. It however varies from place to place. It is minimum at the equator (9.78 ms^{-2}) and increases with latitude reaching the maximum value of 9.83 ms^{-2} at the poles of the earth. Hence it is less in Lagos than in London.

It is because the acceleration due to gravity is the same for all bodies in the same locality that all objects whatever their masses, when released from rest at the same point above the ground fall to the ground simultaneously (at the same time).

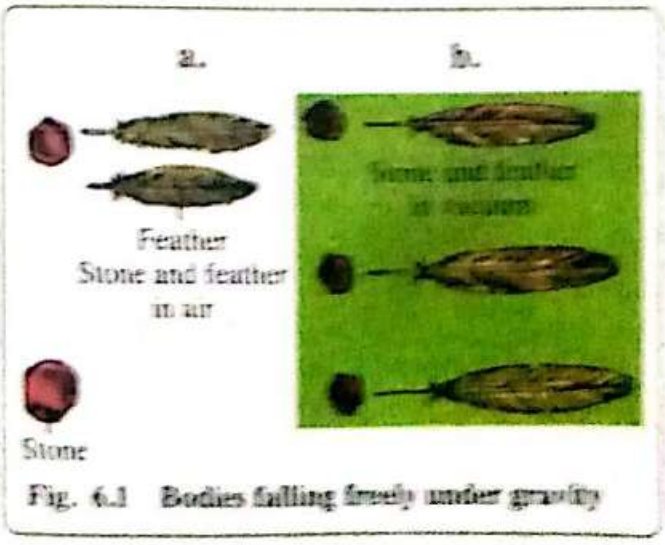


Fig. 6.1 Bodies falling freely under gravity

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The force of gravitational attraction is given by $F = mg$, where g is the acceleration due to gravity and m is the mass of the object. When $m = 1$, $F = g$. This is known as the Acceleration of free fall (g) due to gravity. This is the force of attraction on a unit mass.

All bodies at the same place in the earth's gravitational field are accelerated at the same rate, g, due to the earth's gravitational attraction.

We should therefore expect a stone and a feather or a sheet of paper, when dropped from the same height to reach the ground at the same time. This is not however the observation, as the stone is seen to fall faster and thus takes a shorter time to reach the ground than the feather.

This is due to air friction or resistance which retards the movement of the feather more than that of the heavier stone. Due to its large surface area, the feather falls more slowly than the stone because of the large amount of air that must be pushed aside to let it pass. However in the absence of air resistance (i.e. in a vacuum) both the heavier stone and the lighter feather will reach the ground at the same time (fig. 6.1b).

In the absence of air resistance or friction all bodies fall with the same acceleration irrespective of their masses.

Galileo demonstrated in his famous experiment at the tower of Pisa, that the two balls hit the ground together, showing that they fell with the same acceleration.

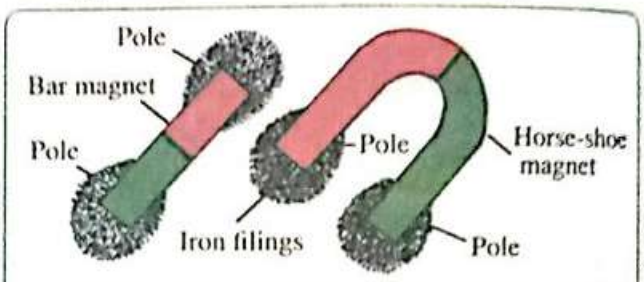


The strength of the gravitational field is smaller on the Moon than on the Earth. So an astronaut has a lower weight on the Moon than on the Earth. The force pulling him down is smaller, so he can move easily on the Moon.

6.3 Magnetism and Magnetic Field

6.3a Magnets

If we put a bar magnet in a plate of iron filings, on bringing out the bar magnet, it is noticed that the iron-filings cling to the magnet with most of them clinging in heavy bunches or clusters around the ends of the magnet (fig. 6.2). These ends of the bar magnet where the iron filings concentrate are called the *poles* of the magnet. The power of the magnet thus appears to be concentrated at these poles.



Iron filings cling in tufts at the ends of magnets

Fig. 6.2 Poles of magnets

The pole of a magnet is the portion of the magnet where its magnetic attraction appears to be strongest.

A magnet can attract pieces of iron, nickel and cobalt. Such *substances that can be attracted by a magnet are known as magnetic substances* and this *ability of a magnet to attract magnetic substances is called magnetism.*

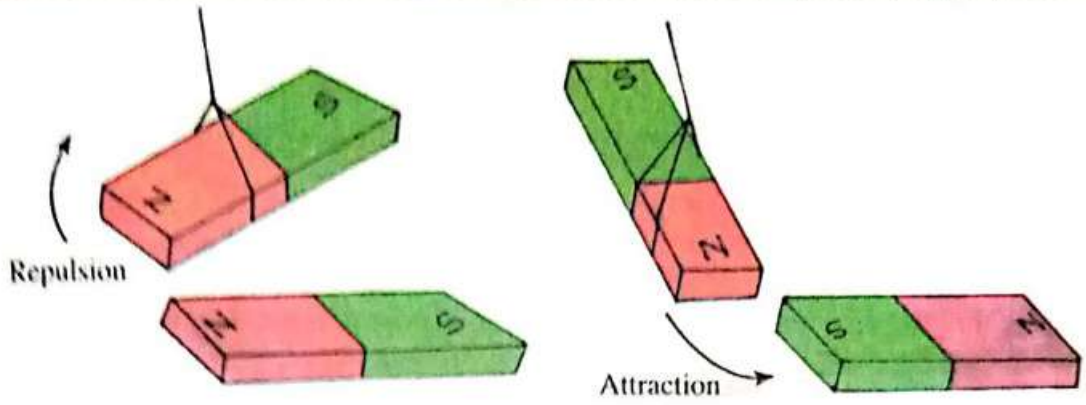
A bar magnet suspended about its centre in such a way as to swing freely, quickly comes to rest with its axis aligned approximately in the North-South direction.

The end of the magnet that points in the northward direction is the *North pole*. The other end that points approximately in the southward direction is the *South pole*. If we bring the north pole of another magnet near the north pole of the suspended magnet, we will notice that the suspended magnet moves away from the new magnet. Thus there appears to be a *force of repulsion* between the two north poles of these magnets. Similarly a south pole of a magnet brought near the south pole of the suspended magnet will repel it.

Now we can reverse this experiment and bring the north pole of a magnet near the south pole of the suspended magnet. We will notice that the two poles now attract each other. The same attraction is noticed when south pole is brought near the north pole of the suspended magnet. We therefore conclude that:

Like or similar poles of magnets repel one another but unlike or dissimilar poles attract one another.

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a. Like poles repel b. Unlike poles attract

Fig. 6.3 Attraction and repulsion between poles

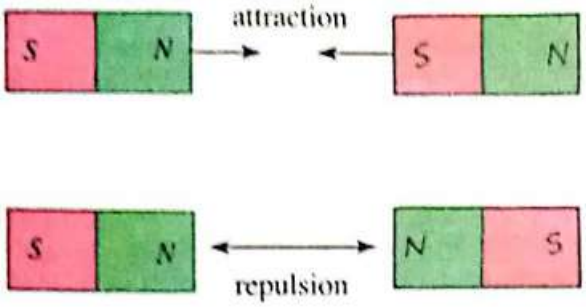


Fig. 6.4 Attraction and repulsion between poles

6.3b Magnetic Field

The region or space around a magnet in which the influence of the magnet can be felt or detected is called a magnetic field.

This space can be mapped out using a magnetic compass needle. The compass needle when placed in a magnetic field, swings around and settles in a definite direction. The swinging of the needle when in the vicinity of the magnet shows that it has been subjected to a force. This force is known as the *magnetic force*. Thus *magnetic field is a force field*. Magnetic force can be felt at a distance. It influences an object even when not in contact with it.

Magnetic field is also a *vector field*, i.e. it has both magnitude and direction.

We can demonstrate the pattern or nature of the magnetic field around a bar magnet by the use of (a) iron filings (b) magnetic compass needle.

Activity 6.1

a Field pattern with iron filings

Place a thin sheet of glass over a bar magnet and sprinkle some iron filings on the glass. Tap the glass gently and notice how the iron filings arrange themselves in a clear and definite pattern on the glass as shown in fig. 6.5.

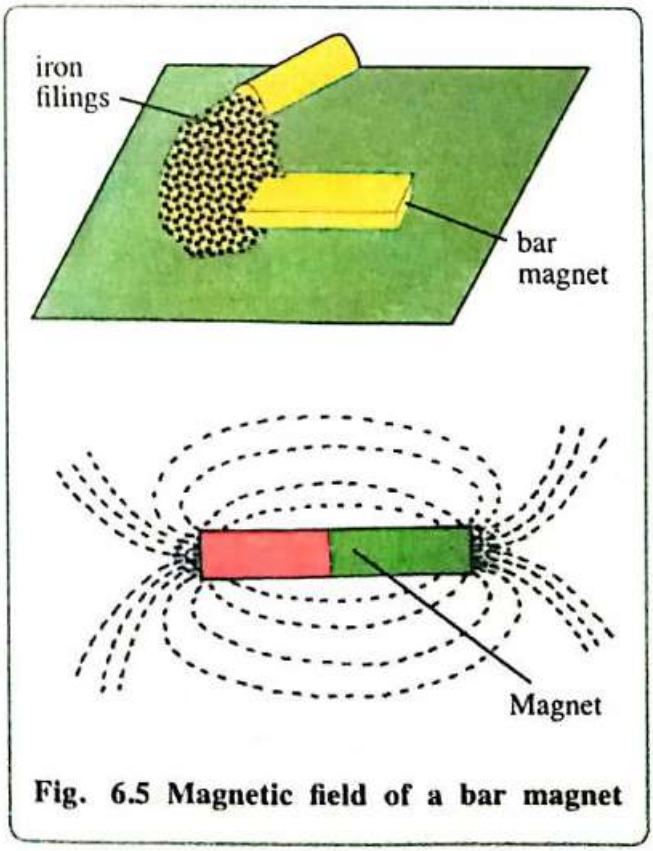


Fig. 6.5 Magnetic field of a bar magnet

The pattern of the iron filings show the form of the magnetic field. The filings are seen to arrange

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themselves in a series of closed loops around the magnet extending from pole to pole.

b. Field pattern using a compass needle

The magnetic field, being a *vector field*, has both magnitude and direction. The compass needle is a sensitive instrument for finding the direction of the magnetic field. We can therefore demonstrate the pattern and direction of the magnetic field of a bar magnet using a compass needle.

Activity 6.2

Fix a piece of drawing paper firmly on top of a drawing board. Place a bar magnet centrally on the sheet. Place a compass needle near the North pole of the magnet. Notice that the needle swings and settles in a particular direction. Mark two dots n_1, and s_1 on the paper to correspond with the North and South poles of the compass needle (fig. 6.6). Move the compass until the *S*-pole of its needle is at n_1 and mark another dot, n_2 to correspond with the new position of the *N*-pole of the needle. Repeat the procedure by moving the compass in this way until the other pole is reached. Join the dots together with a smooth curve. This curved line is a magnetic line of force.

Other lines of force are similarly obtained by placing the compass of other locations around the magnet and repeating the above procedures.

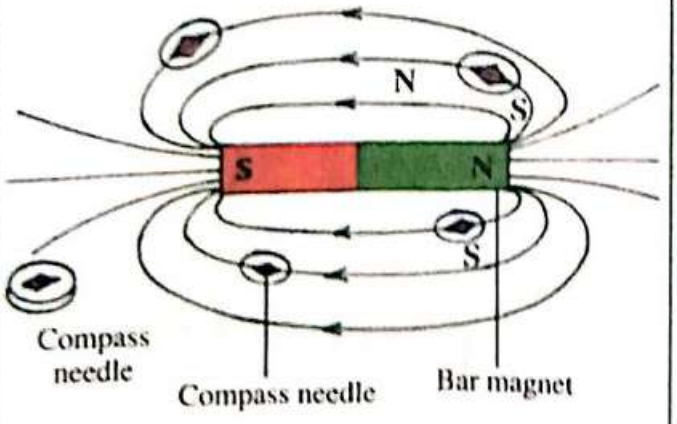


Fig. 6.6 Plotting the magnetic field using a compass needle

These emergent lines constitute the *magnetic lines of force* of the magnetic field.

The magnetic line of force of a magnet is defined as the line along which a free *N*-pole would tend to move if placed in the field or a line such that the tangent to it at any point gives the direction of the field at that point.

Thus *the direction of a line of force at any point is taken as the direction in which a free N-pole placed at that point would tend to move.*

From the form of the field pattern depicted in fig. 6.6, we note that *lines of force do not cross each other*, otherwise the magnetic field would have two possible directions at the point of intersection.

Summary

1 A **field** is a region of space under the influence of some physical agency. Examples of fields are (1) gravitational field (2) magnetic field (3) electric field.

2 **Force fields** are forces whose sources are not in contact with the bodies they influence.

3 **A scalar field** is a field that has only magnitude but no direction, e.g. temperature, energy, density.

4 **A vector field** is a field that has both magnitude and direction, e.g. gravity, magnetism, electricity.

5 **A gravitational field** is a force field that affects every object that has mass. The earth's gravitational field attracts every object in the earth and gives it an acceleration known as acceleration due to gravity.

6 The acceleration due to gravity is uniform at a given place and is the same for all bodies irrespective of their masses. It varies from place to place, being minimum at the equator and maximum at the poles.

7 In the absence of air resistance or friction, all bodies fall with the same acceleration irrespective of their masses.

8 Every object in the earth's surface attracts every other body around it.

9 There is a magnetic field around a magnet. It is a force field.

10 **The pole** of a magnet is the portion of the magnet where its magnetic attraction appears to be strongest.

11 Like or similar poles of magnets repel one another, unlike or dissimilar poles attract one another.

12 **A magnetic field** is a region or space around a magnet in which a magnetic force can

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be detected. A magnetic field can also exist around a current carrying conductor.

13 A magnetic field is a vector field. The compass needle is a sensitive instrument for detecting the direction of the magnetic field.

14 **The magnetic line of force** of a magnetic field is the line along which a free N-pole would tend to move in the field.

15 Lines of magnetic force do not cross each other.

Exercise 6

1 a. Explain what is meant by a *field* as used in physics

b. Mention three types of field known to you.

2 Distinguish between scalar fields and vector fields. Give two examples of each.

3 Define (a) *gravitational field*, (b) *acceleration due to gravity*. How does the acceleration due to gravity vary with latitude?

4 Explain why two solid bodies of different masses released from rest at the same point simultaneously fall to the ground at the same time. Is this observed when a feather and a coin are released in air? If not, why not?

5 What do you understand by the *poles* of a magnet? Using a magnet of known poles, how can you differentiate between the two poles of another magnet?

6 Explain what you understand by

(i) the magnetic field,

(ii) the magnetic lines of force.

How can you plot the lines of force in the magnetic field of a bar magnet? Draw a diagram to show the arrangement of such lines of force.

7 The lines of force of a magnetic field do not cross. Why is this so?

8 Identify the force fields from the following:
Density field, temperature field, gravitational field, magnetic field, electrostatic field, electric field.

9 Which of the following will experience a force when placed in a magnetic field?
Glass, plastic, iron, steel pins, paper pieces, aluminium.

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7 ELECTRIC FIELD

In chapter 6, we discussed gravitational and magnetic fields. In the present chapter we will discuss another type of field known as the *electric field*. This is another example of a force field.

The electric field is created by charged bodies and these bodies are in turn affected by the electric field.

At the end of this chapter the student should be able to:

1 Plot electric lines of force around
   a. isolated positive charge
   b. isolated negative charge
   c. two like charges placed near each other
   d. two unlike charges placed near each other.
2 Generate a continuous flow of charge.
3 Explain electric current.
4 Set up a simple electric circuit.
5 Distinguish between conductors and insulators.
6 Understand the meaning of resistance as opposition to the flow of charges (electrons).
7 Calculate the electrical work done in a given circuit.
8 Convert electric energy into mechanical energy, heat energy and light energy.
9 Convert solar energy to electric energy and to heat energy.

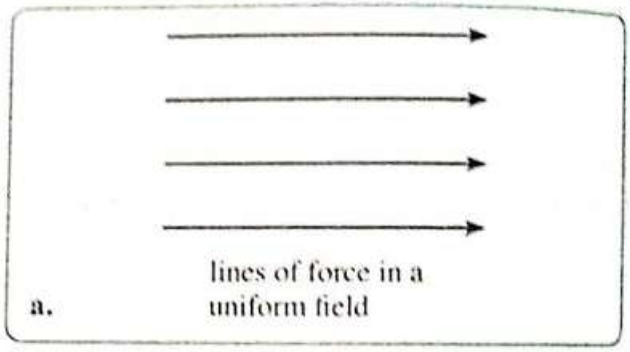
7.1 Electric Lines of Force

An electric field is a region of space where a charged body experiences an electric force.

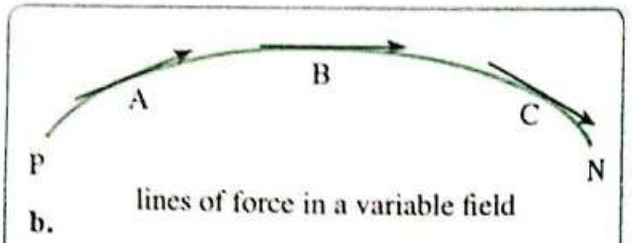
The electric field is just a way of describing the action at a distance of one charge upon another. Such a field is another example of a force field and is represented by *electric lines of force*. Since force is a vector quantity, an electric force field has both magnitude and direction.

The direction of an electric field at any point is given by direction of the force acting on a small positive charge placed at that point.

If, for example, the curved line such as PN represents an electric line of force, then the tangents to this line at points such as A, B, or C (fig. 7.1b) indicate the directions of the electric field at these points. The arrows indicate the direction of motion of a small positive charge placed at each such point and hence the direction of the field line at such points.



a. lines of force in a uniform field



b. lines of force in a variable field

Fig. 7.1 Electric lines of force

There are two types of fields –

a. *uniform field*, in which the field lines are straight,
b. *variable field*, in which the field lines are curved.

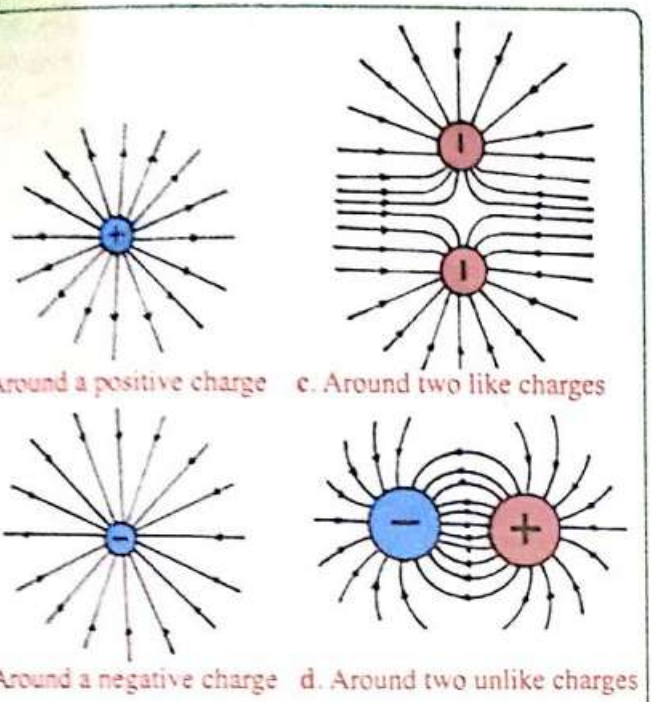
Electric line of force is an imaginary line drawn in an electric field in such a way that the direction at any point (or the direction of the tangent) gives the direction of the electric field at such a point. *OR*, the path which an isolated small positive charge would follow if placed in the field.

Patterns of electric field

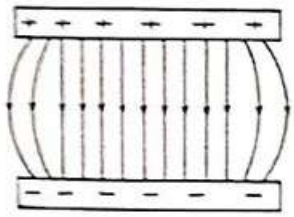
The field about an isolated positive charge is radially outward (fig. 7.2a) because a positive charge placed at any point around it is repelled outward along a line through the two charges.

Similarly the field about a negative charge is radially inwards as shown in fig. 7.2b. The fields for other charges or a combination of charges are as shown in fig. 7.2 (c & d).

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a. Around a positive charge
c. Around two like charges
b. Around a negative charge
d. Around two unlike charges



Between a pair of parallel conducting plates

Fig. 7.2 Electric fields

Properties of field lines

We summarize the properties of electric lines of force as follows:

1 Lines of force start only on positive charges and end only on negative charges. The number of field lines starting or ending is proportional to the magnitude of the charge.

2 Lines of force do not cross each other. If they did, it would mean that the electric field would point in different directions at the same point which does not make any sense.

3 In a uniform field, the lines of force are straight, parallel and uniformly spaced.

4 Lines of force indicate the direction of the electric field. The field points in the direction tangential to the lines of force at any point.

5 The lines of force are continuous in any region with free charges.

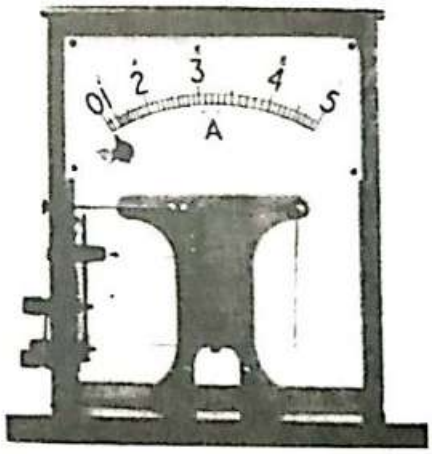
6 The lines are drawn such that the electric field is proportional to the number of lines crossing unit area perpendicular to the lines. The closer the lines are together, the stronger the electric field in that region.

7.2 Potential Difference and Electric Current

An electric charge can be at rest or in motion. We speak of *static electricity* when the charge is at rest, but when the charge is in motion, it is referred to as *current electricity*.

Electric current, I, is defined as the rate of flow of electric charge along a conductor.

A stream of moving charges (or electrons) constitute an electric current. We can describe the flow of electric charge along a conductor, e.g. a metallic wire, by expressing it in terms of the quantity of electric charge Q which moves past each section of the conductor in a given time t. The quantity of charge Q is measured in *coulombs* and the time t in seconds. Hence, current I is given mathematically by the expression:



Picture of an ammeter



Picture of a galvanometer

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$$I = \frac{Q}{t} \qquad 7.1$$

$$\text{Current} = \frac{\text{quantity of charge}}{\text{time}}$$

If Q is in coulombs, and t is in seconds, the current I is in *amperes* (A). Hence *1 ampere = 1 coulomb per second* and thus we can *define the coulomb as the quantity of electricity passing a section of a conductor in one second when the current is one ampere*. The commonly used submultiples of the ampere are the milliampere (mA) and the microampere (μA)

$1\ \text{mA} = 10^{-3}\ \text{A}$

$1\ \mu\text{A} = 10^{-6}\ \text{A}$

Such low currents are common in transistor radios and electric calculators.

The *ammeter* is an instrument for measuring current. Milliameters measure smaller currents. Very small currents are detected by sensitive instruments called galvanometers.

An ammeter is said to be sensitive if it can detect or measure very small currents. It is said to be accurate if the current it measures is close to the true value of the current flowing the instrument.

The Electric Circuit

An electric circuit is the path provided for the flow of electric current. The circuit consists of the source of electric energy (e.g. a battery) connected through a conductor (e.g. a wire) to a load (e.g. an electric bulb), and a key or a switch. The switch serves to complete (close) or break (open) the circuit.

A *closed circuit* is the circuit in which there is no gap (key close) along the conducting path. In such a circuit the current flows through an external resistor (or load) and the bulb in fig. 7.3a lights up.

An *open circuit* is a circuit with a gap or opening (key open) in the conducting path. In such a circuit, the battery maintains no current in an external resistor (or load) and the bulb in fig. 7.3b does not light up.

A *short circuit* is a closed circuit which has no load on it.

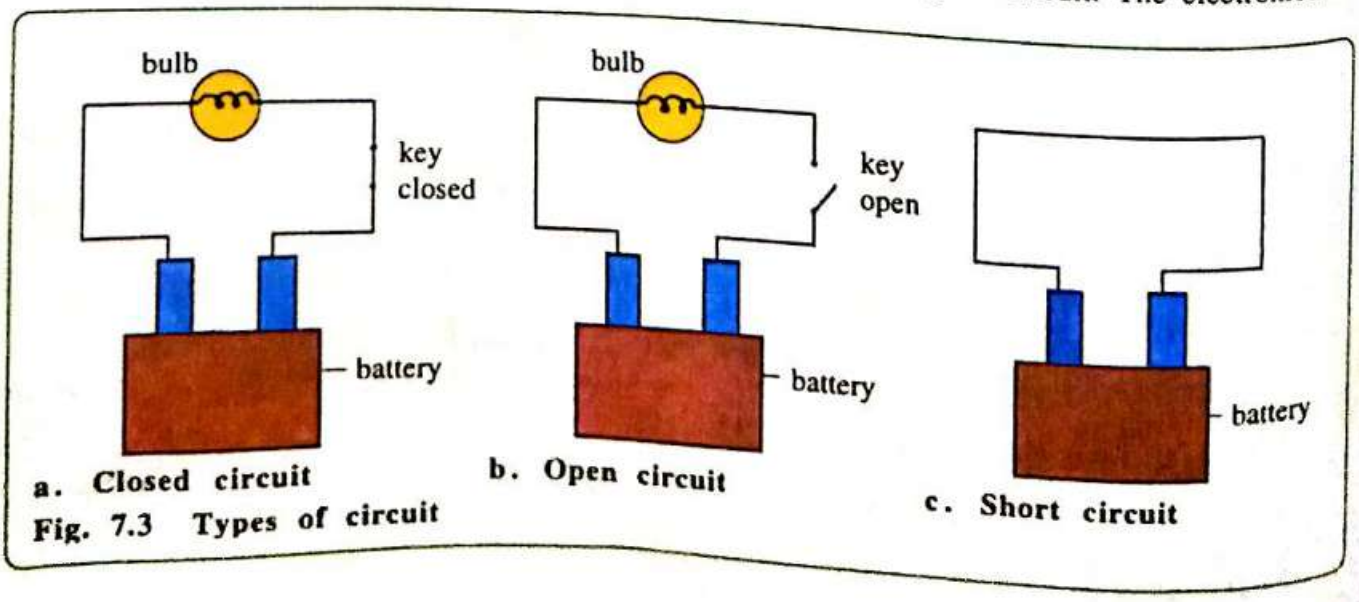
Potential Difference

The potential difference between any two points in an electric field is defined as the work done in moving a positive charge of 1 coulomb from one point in the electric field to another.

Potential difference is an essential condition for the flow of current in an electric circuit. We can get a feel of its effect on current flow by considering the flow of water along a pipe. For the water to flow along a pipe there must be a pressure difference between one end of the pipe and the other. Similarly for electric current to flow along a conductor, there must be an electrical pressure difference between the ends of the conductor.

We can think of potential difference between any two points as the *electrical pressure difference between those points*. Potential difference is measured in *volts* (V).

In order to maintain a flow of current round a circuit or along a conductor there must be a source of electricity that keeps the charge (electron) flowing through the conductor. Some energy is expended in maintaining the flow of current round the circuit. Such energy can be supplied by an agency, e.g. a cell (or battery) or by a generator. The battery serves as the source of potential difference that makes it possible for current to flow. The battery is said to have an electromotive *force* (e.m.f.) when it is able to maintain a current of electricity through a circuit. The electromotive



a. Closed circuit b. Open circuit c. Short circuit

Fig. 7.3 Types of circuit

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force is measured by the energy expended in causing a quantity of electricity of 1 coulomb to flow round the circuit. The unit of the e.m.f is the *volt*. We can thus define **the electromotive force as the total work done in driving one coulomb of electricity round a circuit or the total energy per coulomb obtained from a cell or battery.**

$$\text{Emf } (E) = \frac{\text{work}}{\text{charge}} = \frac{W}{Q} \qquad 7.2$$

When W is measured in joules, Q in coulombs, the e.m.f. (E) is in volts. Hence

$$V = \frac{W}{Q} \qquad 7.3$$

or $W = QV$

The e.m.f. of a cell is also defined as the p.d. between the terminals of a cell when it is not delivering any current in an external circuit (or when it is in an open circuit).

Since the e.m.f. of the cell provides the potential difference that causes a current to flow in a circuit, we can define potential difference in terms of work done by the e.m.f. of a cell. Hence the definition of potential difference (p.d.) between two points as the work done in moving one coulomb of electricity from one point to the other. The unit of p.d. is also the volt. Thus *the volt is the p.d. between two points on a conductor when one joule of work is done in moving one coulomb of charge from one point to the other.*

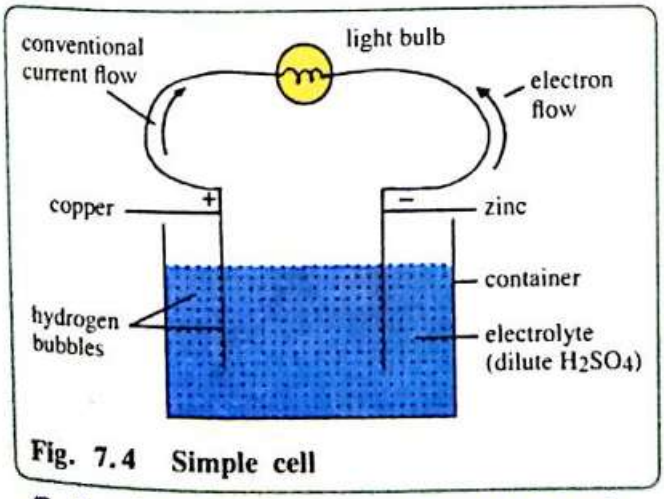


Fig. 7.4 Simple cell

7.3 Production of Electric Current

Electric current or a continuous flow of charge can be generated from (a) chemical energy (b) heat energy (c) mechanical energy (d) solar energy.

a . Electricity from chemical energy

Electricity is produced from chemical energy through the use of electric cells. *A cell* is a device for converting chemical energy into electrical energy.

A cell consists of two dissimilar (or unlike) metals separated by solutions of various acids or salts. The metals are known as the *electrodes* and the solutions are the *electrolyte*. The positive electrode is known as the *anode*, the negative electrode is the *cathode*.

A *simple cell* consists of a copper rod and a zinc plate immersed in a container filled with dilute sulphuric acid (tetraoxosulphate (VI) acid).

When the copper rod and zinc plate are joined by a conductor (e.g. a metallic wire), the zinc slowly dissolves in the acid and bubbles of hydrogen gas are formed on the copper rod. At the same time as a result of chemical reaction, electrons flow through the connecting wire from zinc to copper. Copper is the anode (+ve electrode). Zinc is the cathode (−ve electrode). By convention, current is taken to flow from copper to zinc even though the electrons move in opposite direction from zinc to copper. A bulb connected between the copper and zinc terminals will light up, showing that current is flowing through the external circuit.

Defects of simple cells

Simple cells supply current only for a short time. This is because of chemical processes known as (i) *polarization* and (ii) *local action*.

(i) **Polarization** is due to the formation of hydrogen gas bubbles around the copper plate of the simple cell. The hydrogen is produced by the reaction of zinc and the dilute sulphuric acid. An electromotive force (e.m.f.) is set up between the zinc and the hydrogen bubbles. This e.m.f. opposes the e.m.f. of the cell generated between the copper and zinc. This opposing e.m.f. known as *back e.m.f.* tends to reduce the current in the external circuit and will eventually stop the chemical action in the cell when enough hydrogen layer is well-established around the copper rod. The cell is then said to be *polarized*. This defect of the cell is known as *polarization*.

Polarization of cells is reduced or prevented by the addition of a suitable chemical known as a *depolarizer*; e.g. manganese dioxide, and potassium dichromate. The depolarizer removes the hydrogen by oxidizing it to water.

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(ii) **Local action** is due to impurities in the zinc plate. These impurities (e.g. iron and carbon) set up tiny cells around the zinc surface, producing bubbles of hydrogen. This activity leads to the dissolution of the zinc plate into solution even when no current is being supplied to the external circuit. The zinc is thus continuously wasted.

. Local action can be prevented by the process of *amalgamation*, that is the rubbing of mercury over the surface of the zinc plate. The mercury on the zinc plate prevents the impurities from coming into contact with the acid. With this local action cannot occur.

b. Electricity from heat energy (The thermoelectric effect)

We can show that heat energy can be converted into electric currents by joining two different metallic wires (e.g. copper and iron) at one end and connecting the free ends to the terminals of a sensitive current detecting device, e.g. a milli-galvanometer. When the junction of the metals is put in hot water, a current is observed to flow along the wires as indicated by the milli-galvanometer. The junction placed in hot water is known as the hot junction, while the ends connected to the instrument constitute the *cold junction*.

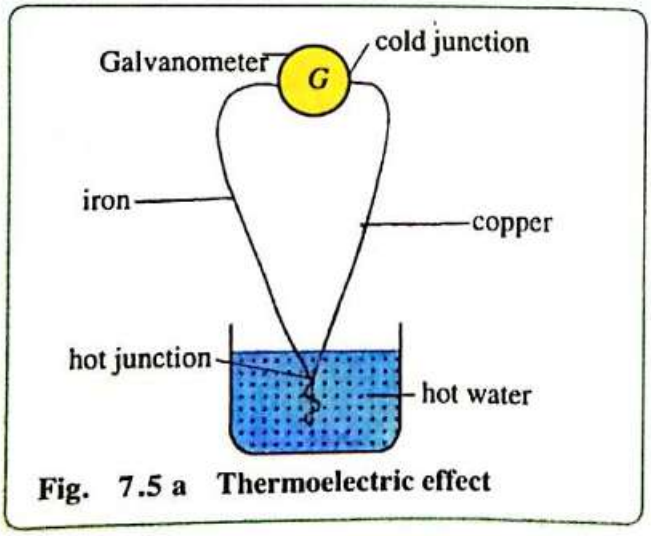


Fig. 7.5 a Thermoelectric effect

The greater the difference in temperature between the hot and cold junctions, the greater the current flow.

The two metals joined in this way constitute a device known as a *thermocouple* and the electricity (known as thermoelectricity) is produced by a process known as the *thermoelectric effect*. The current produced by the thermoelectric effect is small and can only be detected by very sensitive galvanometers.

c. Electricity from mechanical energy

A great majority of the world's electricity is produced from the conversion of mechanical energy into electric energy. When coils of insulated wire move across the magnetic field between two powerful magnets, current is induced in the coils because the coils cut across the magnetic lines of force. The current is tapped using split-ring commutators and carbon brushes.

The electric generator or dynamo produces electricity in this way. A common example of such device is the bicycle dynamo used to supply light to the headlamp of a bicycle.

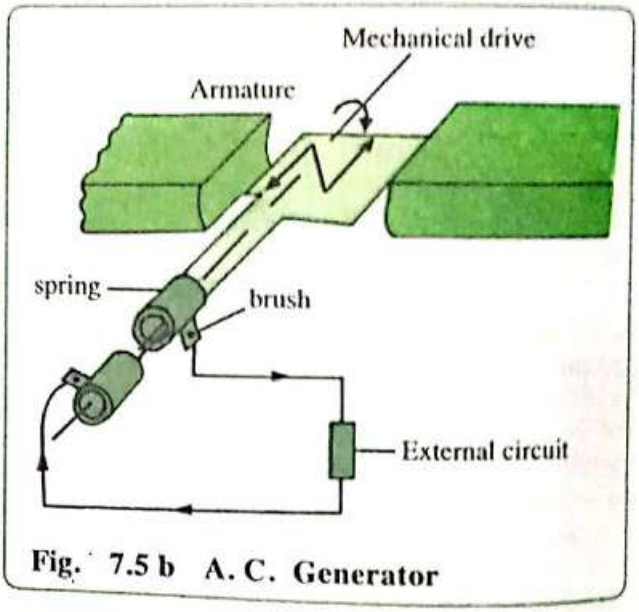


Fig. 7.5 b A. C. Generator

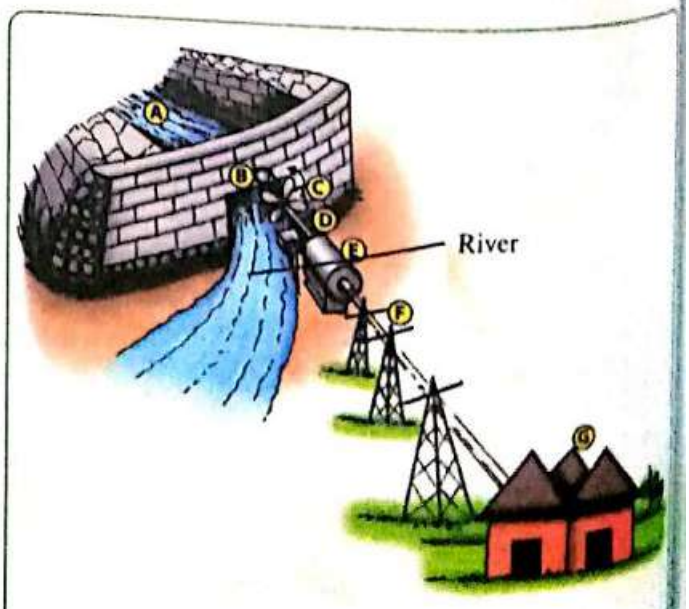


Fig. 7.6 a Working principle of hydroelectric power generation from dams

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d. Dams and Energy Production

Hydroelectric power generation is one of the best methods of electric power generation. This is because it is renewable and does not contaminate the environment. It is also an example of electricity production from mechanical energy.

A dam is a barrier that is built across a river in order to stop the water from flowing, used especially to make reservoirs (i.e. a lake for storing water) or to generate electricity. There are many dams in Nigeria, especially in the Northern parts of the country. The dams for producing electricity are located at Kainji, Jebba, Shiroro. Thus, we have Kainji dam power station, Jebba power station and Shiroro power station.

Electricity produced by the transformation of the mechanical energy of water from such dams to electric energy is known as Hydroelectricity.

When the huge amount of water in the dam is let out through some gates, large mechanical kinetic energy of the moving water is made to turn a water wheel or a turbine, and this in turn will work a dynamo and generate electricity.

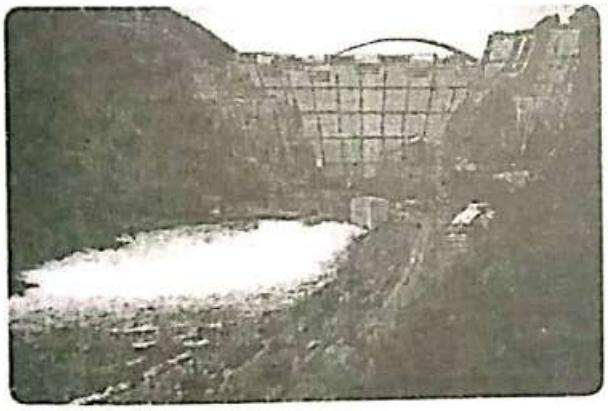


Fig. 7.6 b Hydroelectric power generation

7.4 Principle of production of electricity from a Dam

(i) When an object falls from a height, it does work. In this case, its potential energy before it falls is transferred to kinetic energy during its fall. This kinetic energy can be converted to useful work. This can be seen in hydroelectric power generation. This is illustrated in figure 7.6 (a &b).

Using figure 7.6a, the mechanism of generating electrical energy using water fall is summarized below:

i. A: Large volume of water is held back by the dam.
ii. B: Water forces itself out through a small opening in the dam.
iii. C: The falling water does work by falling on a turbine; thereby rotating it.
iv. D: Through fan belt or gear system, the dynamo, E, rotates at the same time.
v. E: The dynamo generates electricity.
vi. F: Electric cables are used to transmit the electrical power to residential areas.

Dams are walls built such that they can hold large volume of water in the river (A).

Through small opening, water falls (B). This falling water does the work of turning the turbine (C) as it hits it; and the turbine rotates as a result of this. Through a gear system or fan belt (D), a dynamo (or electrical generator) (E) at the same time rotates. This generates electricity which can be transmitted through electric cables (F) to residential areas.

e. Electricity from solar energy

When sunlight falls on a photosensitive surface (e.g. the surface of potassium, an alkaline metal), electrons are produced whose movements constitute a current. A *photoelectric cell* or *photocell* consists of a photosensitive surface as a cathode and a wire ring as the anode. If visible light falls on this surface, electrons are emitted by the *photoelectric effect* and the flow of these electrons can be detected as current by a micrometer.

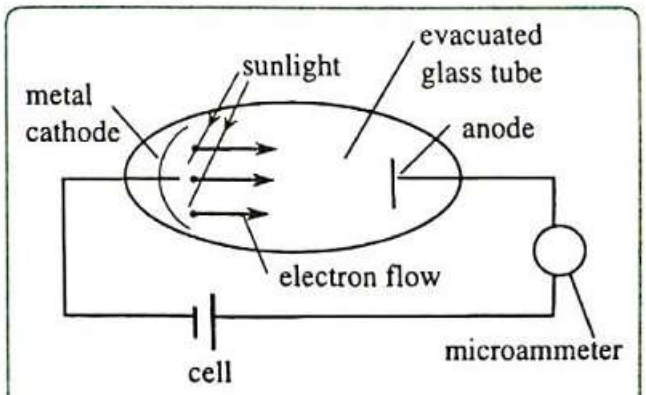


Fig. 7.7 Photocell

The electrons are usually accelerated from the cathode to the anode which is at a higher potential with respect to the cathode.

Solar electricity can be used to light up a whole village.

Conductors and insulators

All materials offer a certain degree of opposition or resistance to the flow of current through them. Some materials allow electric currents to flow through them more easily than others. *Materials through which electricity can flow easily are*

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known as conductors. Examples are (i) almost all metals (ii) impure water (iii) inorganic acids solutions and (iv) damp air.

Insulators are those materials that do not allow electric currents to flow easily through them. Examples are (i) ebonite, (ii) glass, (iii) dry wood, (iv) dry air, (v) silk, (vi) plastics, (vii) organic acid solutions.

The best metallic conductor is silver followed by copper. Most connecting wires and electric cables used in electricity are however made of copper instead of silver because copper is less expensive than silver.

Since a good conductor allows current to flow through it very easily, such conductors have a low opposition to the flow of current and are said to have a *low electrical resistance*.

Resistance can be defined as the opposition to the flow of charges (electrons) or current. Its unit is the Ohm.

Insulators are bad conductors and have a high electric resistance. The insulating properties of materials may break down under certain conditions and they become temporal conductors. For example, during a thunderstorm, the insulating property of air may break down and air conducts electricity. This occurs in lightning flashes which reveal the conducting path of the air during thunderstorm.

7.5 The Electric Circuit

An electric circuit is the complete path provided for the flow of electric current. The circuit usually consists of the source of electric energy, *the battery* (or cell), *a switch or key* to start or stop the current flow, *an ammeter* to measure the current flow, a *voltmeter* to measure the potential difference, *a resistor* or *load* and *a rheostat* to adjust the flow of current. These components are connected as shown in fig. 7.8.

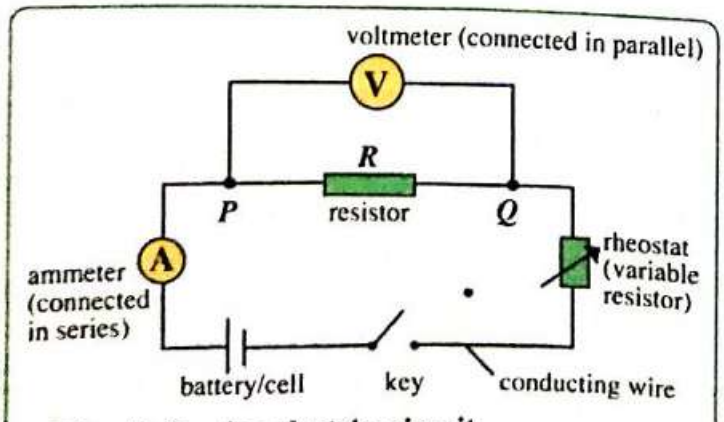


Fig. 7.8 An electric circuit

The ammeter must always be connected in a circuit in such a way that the current it measures flows directly through it. The ammeter is said to be connected *in series* in the circuit. In order not to alter the current it measures, the ammeter is constructed to have a low resistance.

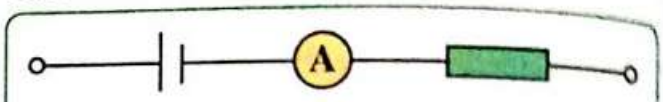


Fig. 7.9 Ammeter in series

The voltmeter which measures the potential difference between (or across) two points along a conductor must always be connected across or *in parallel* with these two points (e.g. the points P and Q in fig. 7.8). Voltmeters are constructed to have a large electrical resistance compared to the resistance across the points they are connected. They therefore take only a negligible amount of current.

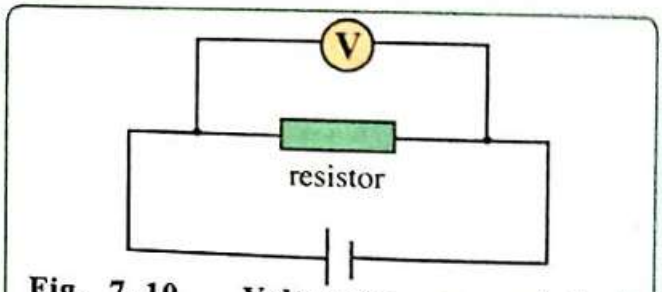


Fig. 7.10 Voltmeter connected in parallel across a resistor

Another component of the circuit is the *resistor:* A resistor is a component which is specially designed to provide a known amount of resistance in a circuit. *Standard resistors* are those that have a fixed resistance value. They are usually in the form of lengths of resistant wire or a piece of carbon. *Variable resistors* are those whose resistance can be varied. An example is the *rheostat*.

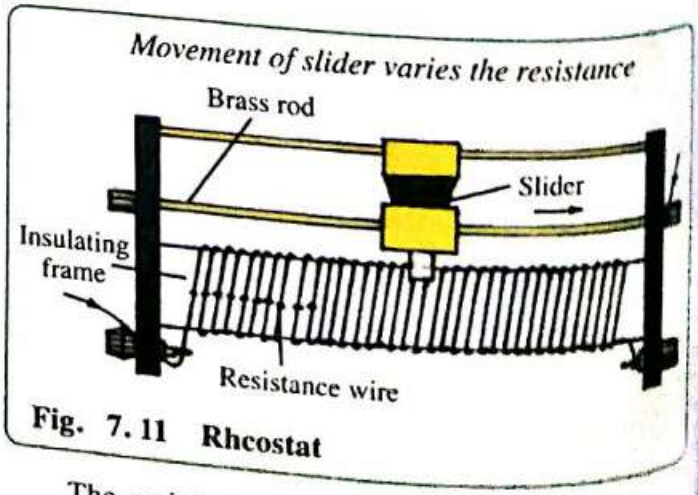


Fig. 7.11 Rheostat

The switch or key is the component of the circuit by which the circuit is completed or broken and the current be made to flow or to stop flowing.

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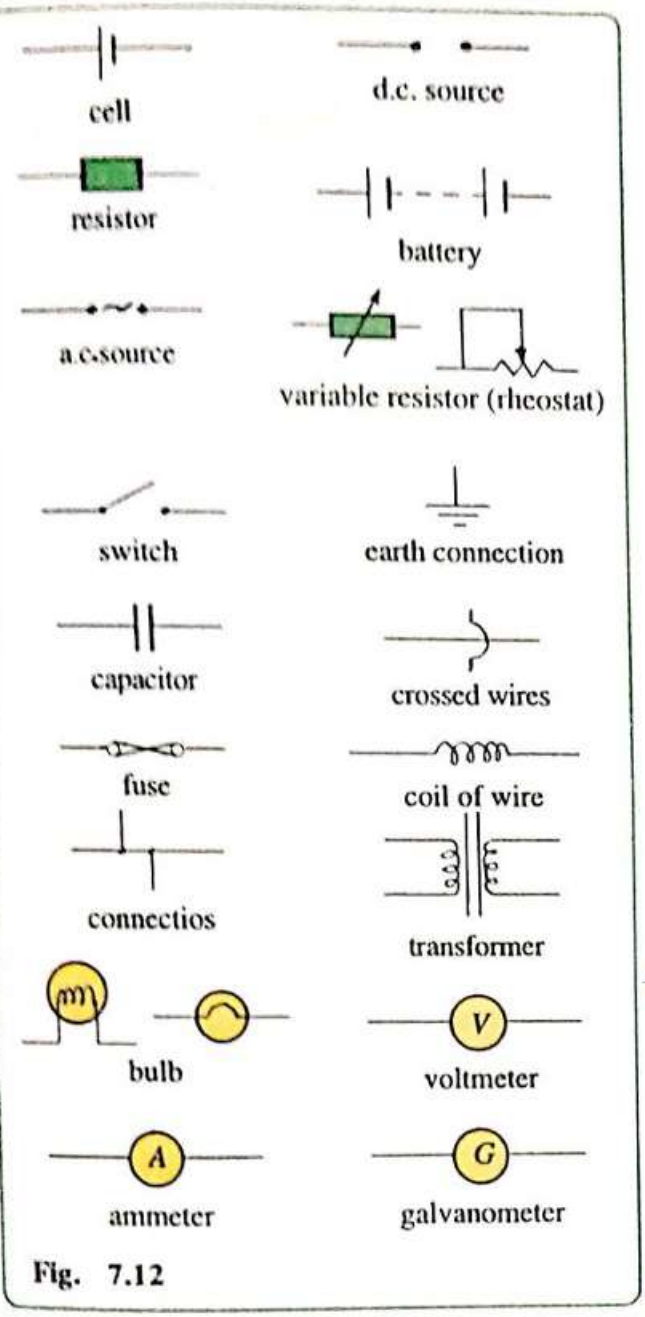


Fig. 7.12

Parallel and series connection of resistors in an electric circuit

There may be more than one resistor in an electric circuit. Such resistors may be connected in *parallel* or in *series*.

When the resistors are connected end to end as shown in fig. 7.13 they are said to be in series.

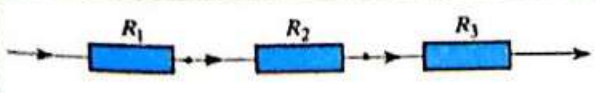


Fig. 7.13 Resistors in series

When they are so connected, the equivalent resistance of the combination is given by

$$R = R_1 + R_2 + R_3 \qquad 7.4$$

When resistors are arranged side by side such that their corresponding ends join together at two common junctions P and Q as shown in fig. 7.14, the arrangement is known as *parallel* connection of the resistors. In such an arrangement, the p.d. across the resistors is the same but the current through each resistor is different.

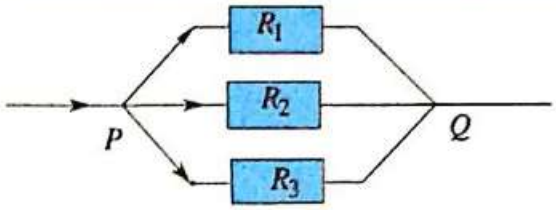


Fig. 7.14 Resistors in parallel

The combined or equivalent resistance (R) is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \qquad 7.5$$

Example 7.1

Three resistors of resistances 1 Ω, 2 Ω and 4 Ω are connected in series. Find the equivalent resistance of the combination.

Solution

$R = R_1 + R_2 + R_3$ for series connection

$\therefore \quad R = 1 + 2 + 4 = 7\ \Omega.$

Example 7.2

If three resistors 2 Ω, 3 Ω, and 5 Ω are connected in parallel in a circuit, calculate the equivalent resistance of the combination.

Solution

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \text{ for parallel connection}$$

$$= \frac{1}{2} + \frac{1}{3} + \frac{1}{5} = \frac{31}{30}$$

$$R = \frac{30}{31}\ \Omega = \underline{0.97\ \Omega}$$

7.6 Electric Conduction through Materials – Ohm's Law

As a result of many experiments conducted by George Simon Ohm, in which he sought to investigate the relationship between the current passing through a wire and the potential difference (p.d.) between the ends of the wire, this German scientist discovered a linear relationship between

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such a current and the potential difference. The relationship now known as Ohm's law is stated as follows:

Ohm's Law: The current flowing through a metallic conductor (e.g. a wire) is directly proportional to the potential difference across its ends, provided that temperature and other physical conditions of the conductor are kept constant.

In symbols, Ohm's law states that

$I \propto V$

where I is current and V is the potential difference (p.d.).

Removing the proportionality sign we have that

$\frac{V}{I}$ = Constant

The value of this constant is a measure of the opposition the conductor offers to the flow of current or the resistance of the conductor.

Hence

$$\frac{V}{I} = R \qquad 7.6a$$

or $$V = IR \qquad 7.6b$$

or $$I = \frac{V}{R} \qquad 7.6c$$

From equation 7.6c, we observe that when R is high I is small, and when R is low I is large.

From equation 7.6a, we note that the resistance of a conductor is the ratio of the potential difference (p.d.) across a conductor to the current flowing through it.

When I is in amperes and V is in volts, R is in ohms (Ω).

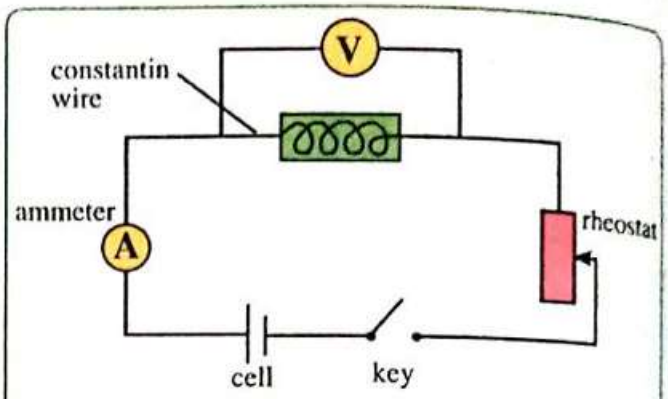
Thus the **ohm** is defined as the resistance of a conductor when a p.d. of 1 volt applied across its ends causes a current of 1 amp to flow through it.

Ohm's law is not obeyed by such conductors as radio valves (e.g. diodes), transistors, rectifiers, and gases.

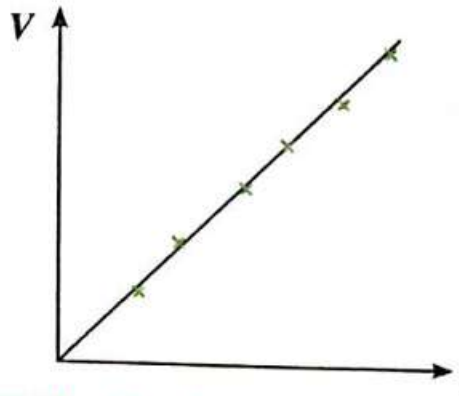
Experimental verification of Ohm's law

We can verify Ohm's law using the set of apparatus as shown in fig. 7.15.

The ammeter which measures the current flowing through the metallic conductor (a constantine wire) is connected in series with the wire and the cell. The voltmeter is connected in parallel with the conductor and measures the p.d. across it. The rheostat is used to vary the current flowing in the circuit.



a. Verification of Ohm's law



b. Graph of voltage against current (Ohm's law)

Fig. 7.15

We cause current to flow in the circuit by pressing down the switch. The values of current (I) and P.d. (V) recorded by both the ammeter and the voltmeter respectively are recorded. The rheostat is adjusted to a new setting, and the new values of I and V are read and recorded. The process is repeated four more times to obtain a total of six values of I and V, which are now set up in a table.

The graph of V against I is plotted and a straight line graph passing through the origin is obtained (fig 7.15b). The straight line graph shows that V varies directly with I, thus verifying Ohm's law.

The slope of the graph gives the resistance of the wire.

This experiment has thus been used to demonstrate Ohm's law and to obtain the resistance of the conductor.

Factors affecting electrical resistance

Four factors affect the resistance of a conductor. These are

1 length of conductor
2 area of conductor
3 temperature and

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4 type or substance of material.

1 Length The resistance of a conductor of uniform cross-sectional area is directly proportional to its length:

$$R \propto l$$

2 Area The resistance of a conducting wire is inversely proportional to its area:

$$R \propto \frac{1}{A}$$

That is the more the area (or the thicker the wire) the less the resistance.

3 Temperature The resistance of a conductor varies directly with its temperature. For certain conductors (e.g. metallic conductors) the higher the temperature, the higher the resistance i.e.

$$R \propto T$$

For non-metallic conductors, e.g. carbon, there is a decrease of resistance with temperature. Some other materials are little affected by temperature changes (e.g. constantine or manganin). Such materials are therefore used for making standard resistors.

Experiments show that the resistance R of a wire or a metallic conductor maintained at a constant temperature is related to its length (l), and its cross-sectional area (A) by the expression:

$$R = \frac{\rho l}{A} \qquad 7.7a$$

where ρ is a constant of proportionality known as *resistivity*. Resistivity is dependent on the nature of the material. It thus differs from material to material.

From equation 7.7a we see that $R = \rho$ when $l = 1$ and $A = 1$. Hence resistivity is defined as the resistance of unit length of material of unit cross-sectional area. Also from equation 7.7a

$$\rho = \frac{R \times A}{l} \qquad 7.7b$$

Hence when R is in ohms, A in m^2 and l in m, the unit of ρ is ohm-metre (Ω-m).

4 Type of material

The resistance of a conductor depends upon the material of which it is made. Silver is the best electrical conductor followed by copper and aluminium.

Example 7.3

What is the resistivity of a cylindrical material wire whose 1.0 m length has a resistance of 2.0 ohms? The diameter of the wire is 0.5 mm.

Solution

$l = 1.0$ m

$$r = \frac{0.5}{2} \text{ mm} = 0.25 \text{ mm} = 2.5 \times 10^{-4} \text{ m}$$

$$A = \pi r^2 = \frac{22}{7} \times (2.5 \times 10^{-4})^2 \text{ m}^2$$

$R = 2\ \Omega$

From $R = \frac{\rho l}{A}$

$$\rho = \frac{RA}{l} = \frac{2 \times (22/7) \times (2.5 \times 10^{-4})^2}{1}$$

$$= \underline{3.93 \times 10^{-7}} \text{ ohm – metre}$$

Example 7.4

Find the resistance of a wire of length 0.65 m, radius 0.2 mm and resistivity 3×10^{-6} ohm-metre.

Solution

$$R = \frac{\rho l}{A}$$

$$= \frac{3 \times 10^{-6} \times 0.65}{\pi (2 \times 10^{-4})^2}$$

$$= \underline{15.5 \text{ ohms}}$$

7.7 Electrical Work Done in a Given Circuit

Work is done when electricity flows from one point to another of different potential. If Q coulombs of electricity flows between the two points whose differences in potential is V volts, then the work done (W) is given by

$$W = QV \text{ joules} \qquad 7.8$$

(joules = coulomb × volts)

The quantity of electricity Q coulombs is given by

$$Q = It \qquad 7.9$$

where I is the current in amperes and t is the time of flow of current in seconds.

$\therefore\ W = QV = IVt$

$$W = IVt \qquad 7.10$$

(joules = amps × volt × seconds)

But from Ohm's law, $V = IR$

$\therefore\ W = QV = IVt = I \times IR \times t$

$$= I^2Rt$$

$$W = I^2Rt \qquad 7.11$$

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This electrical energy can be converted into other forms of energy as follows:

(i) Heat energy, e.g. in electric irons, electric kettles, immersion heaters. The amount of heat is given by

$$H = IVt = I^2Rt \qquad 7.12$$

(ii) Mechanical energy, e.g. in the movements of the blades of electric fans and in washing machines. Conversion of electric energy to mechanical energy is made possible through the action of electric motors.

(iii) Light energy, e.g. in the glow of filament lamps or bulbs.

(iv) Sound energy, e.g. in telephone ear-piece, in music from record players.

Electric Power

Power (P) is defined as the rate of doing work.

$$P = \frac{\text{work done (joules)}}{\text{time taken (seconds)}}$$

Electric power is the amount of electrical work done per second. From equation 7.10, $W = IVt$. thus

$$P = \frac{IVt}{t} = IV = \frac{V^2}{R} = I^2R$$

$$P = IV = \frac{V^2}{R} = I^2R \qquad 7.13$$

Unit of power is the *watt*.

Hence

watt = joules per second.

One *watt* is the power consumed in an electric circuit when one joule of work is done in one second.

Larger units of power are the kilowatt (kW) and the megawatt (MW).

1 kW = 1000 watts = 10^3 W

1 MW = 10^6 W = 10^3 kW.

Electric appliances usually have some markings on them indicating their rating in watts and in volts. A mark of '100 W, 220 V' means that the appliance consumes a power of 100 watts (or energy of 100 joules per second) when connected to a circuit which maintains a potential difference of 220 V across it. This also implies that the current required to run the appliance can be obtained from the relation:

$P = IV$

$100\ W = I \times 220$

$I = \frac{100}{220} = 0.45$ A

Buying and selling of electric power

Electric energy consumption is measured and sold by the National Electric Power Authority (NEPA) in units of kilowatt-hour (kWh)

One kilowatt-hour (kWh) is the electrical energy consumed by an appliance when power of one kilowatt is used by the appliance for one hour.

$$1\ \text{kWh} = 1\text{kW} \times 1\ \text{hr} = 10^3 \times 60 \times 60\ \text{J}$$

$$= \underline{3.6 \times 10^6\ \text{joules}}$$

Thus when an electric bulb marked 60 W lights up a room for 20 hours the number of kilowatt-hour consumed is given by

$$\text{Energy} = \frac{60}{1000}\ \text{kW} \times 20\ \text{hr}$$

$$= 1.2\ \text{kWh}.$$

If NEPA charges at the rate of 10 kobo per kWh, the cost of lighting up the room for that period is given by:

Cost = 1.2×10 = 12 kobo.

Example 7.5

An electric appliance takes 5 A when operated on a 220 V supply. Find the cost of operating the appliance for 12 hours at 10 kobo per kWh.

Solution

$$\text{Power} = IV = (5 \times 220)\ \text{W} = \frac{5 \times 220}{1000}\ \text{kW}$$

$$\text{Energy consumed} = \text{Power} \times \text{time}$$

$$= \frac{5 \times 220 \times 12}{1000}\ \text{kWh}$$

$$\text{Cost at 10 kobo per kWh} = \frac{5 \times 220 \times 12}{1000} \times 10$$

= 132 kobo = ₦1.32

Heating Effect of Current

When electric current flows through a conductor the resistance of the conductor changes electrical energy into heat energy just as friction changes mechanical energy into heat. It is this heating effect of an electric current that is utilised in such devices as electric pressing iron, electric toaster, electric coil, hair dryer and incandescent lamps.

When electric power is converted into heat, the greater the value of current the greater the heat generated. The heat generated is proportional to the square of the current, the resistance and the time during which the current flows, according to the relation:

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$$H = I^2Rt \qquad 7.14$$

where heat energy H is in joules, I is in amperes, R in ohms and t is in seconds.

Example 7.6

Find the heat in joules produced in an electric coil of resistance 15 ohms when a current of 5 amperes flows through it for 30 minutes.

Solution

$$H = I^2Rt = 5^2 \times 15 \times 30 \times 60 \text{ joules}$$
$$= 675000 \text{ J}$$
$$= 6.75 \times 10^5 \text{ J}$$

Electric filament lamps

Light energy is produced from electrical energy through an electric lamp.

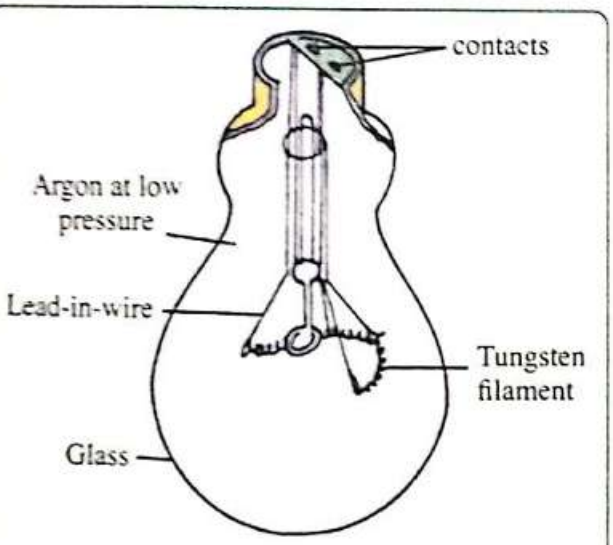


Fig. 7.16 Electric filament lamp

In the electric filament lamp (electric bulb), tungsten filament formed into a small spring-like coil is enclosed in a glass bulb from which air has been evacuated and replaced with an inert gas (e.g. argon) at low pressure. This prevents the hot filament from evaporating and enables the lamps to be run at high temperature.

Current flowing through the filament wire heats it up to such a high temperature that it glows brilliantly and gives light to the surrounding.

Importance of fuses in electric circuit

A *fuse* is simply a short length of wire of low melting point connected to the 'live' wire of an electric circuit. Its purpose is to automatically open the circuit (cut off current flow) whenever excess current might destroy the wiring or the appliance connected to it.

When a current above a predetermined value flows in the circuit to which a fuse is connected, the heat developed in the fuse wire melts the wire and breaks the circuit, thereby cutting off further current flow. The fuse thus acts like a *safety device* protecting electrical components in the circuit from serious damage due to excessive current.

The 'fuse rating' is the maximum safe current permitted to flow in it before the fuse breaks.

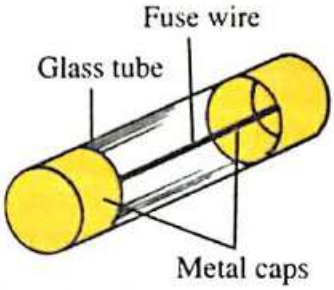


Fig. 7.17 Cartridge fuse

Detecting faults in electric circuit – Simple continuity tester

When there is a fault in a circuit, there will be no current in the circuit. An electric bulb connected in the circuit will not light up. Faults can occur in a circuit as a result of blown fuses leading to a *short circuit* or a broken wire in a burnt-out lamp, leading to an *open circuit*.

To detect faults in a circuit, we use a *continuity tester*. The instrument is used to test whether a circuit is continuous or broken at a certain point. A simple form of continuity tester is as shown in fig. 7.18.

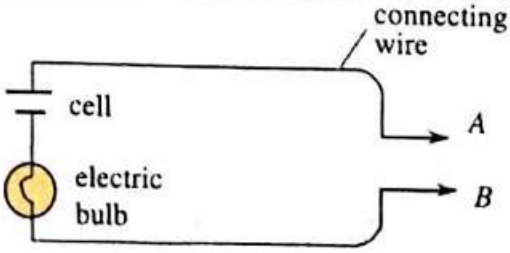


Fig. 7.18 A simple continuity tester

It can be constructed by connecting a 1.5 V cell and an electric bulb as shown in fig. 7.18. When ends A and B are connected, the circuit is complete and the lamp lights up. When A and B are connected across a faulty component of a system or an

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equipment, the lamp will not light up, showing that there is a break in the circuit. When A and B are connected to a good component, the lamp will light up.

In this way, the tester is able to detect faulty apparatus.

Activity 7.1

1 Obtain the following pieces of apparatus from your laboratory: a 2 V electric cell, a rheostat, a plug key, an ammeter, a voltmeter and two resistors of value 3 ohms and 5 ohms.

Make a complete circuit using the resistors

a. in series

b. in parallel.

Measure the current flowing through the circuit and the p.d. across each combination of resistors.

Adjust the rheostat and obtain a fresh set of values of I and V. Repeat the experiment four more times and record your readings on a table.

Plot the graph of I against V for each combination of resistances and see if Ohm's law is obeyed.

2 Pour some quantity of dilute sulphuric acid (tetraoxosulphate (VI) acid) into a glass beaker. Place a copper rod and a zinc rod into the beaker and connect these rods through a wire and a key to a milliammeter. Measure the current flowing through the milliammeter. Remove the plug of the key and note its effect on the milliammeter. Plug back the key and note the effect on the current flow. Replace the milliammeter with a motor car lamp bulb. With the key plugged on, note the glow of the bulb. Leave the plug key on for one hour and note the brightness of the bulb every 15 minutes.

3 Connect the apparatus shown in Fig. 7.19. Consult your teacher if you have any problems. By adjusting the rheostat, measure 5 values of I and V as recorded by the ammeter and the voltmeter respectively. By plotting a graph of I against V, obtain the value of the resistance of the constantine wire from the slope of the graph. Measure the length of the wire using a meter rule, the diameter of the wire (and hence its area) using a micrometer screw gauge.

Calculate the resistivity of the material of the wire from the formula

$$\rho = \frac{RA}{l}$$

using the appropriate units.

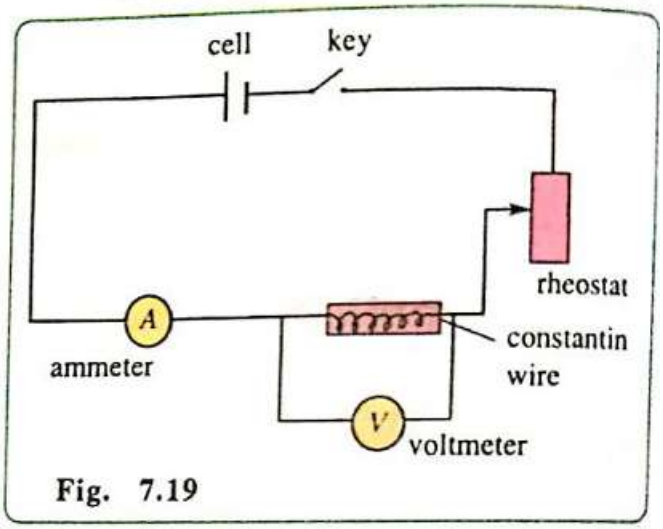


Fig. 7.19

Summary

1 An **electric field** is a region or space in which a charged body or a charge can experience an electric force.

2 An **electric line of force** is an imaginary line drawn in an electric field such that the direction at any point gives the direction of the field at that point or the line giving the path which an isolated small positive charge would follow if placed in the field.

3 Electric current is the rate of flow of charge through a conductor.

4 The **electromotive force** (e.m.f.) of a cell is the p.d. between its terminals when it is not delivering current in an external circuit.

5 A continuous flow of charge can be produced from

a. Chemical energy through electric cells.

b. Heat energy – the thermoelectric effect.

c. Mechanical energy through the dynamo.

d. Solar energy through solar cells.

6 **Conductors** are materials which allow electric currents to flow through them easily e.g. all metals, inorganic acids etc. Insulators are materials which do not allow electric current to flow through them easily.

7 **Resistance** is the opposition to the flow of current. Its unit is the ohm (Ω).

8 In a circuit, ammeters are connected in series and voltmeters in parallel.

9 For series connection of resistances

$$R = R_1 + R_2 + R_3 + \cdots + R_n$$

For parallel connection of resistances

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$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$$

10 **Ohm's law** states that the current flowing through a metallic conductor is directly proportional to the potential difference across its ends provided that temperature and other physical conditions remain constant.

$$\frac{V}{I} = R$$

11 Electric work done or energy consumed in circuit is given by $W = IVt$ or I^2Rt.

12 One kilowatt-hour (1 kWh) is the electrical energy consumed by an appliance when a power of 1 kWh is used for one hour.

13 A fuse is a short length of wire of low melting point connected to the 'live' wire of a circuit to protect an appliance from an excessive current.

14 Electric power $= IV = I^2R = \frac{V^2}{R}$

The unit of power is the *watt*.

Exercise 7

1 Three 3.0 ohm resistors are connected in parallel. What is the equivalent resistance?
A 9.0 ohm B 3.0 ohm
C 1.0 ohm D 0.33 ohm
E 6.0 ohm.

2 Two resistors, 5.0 Ω each can be combined to give an equivalent resistance of
A 10.0 Ω B 25.0 Ω C 2.5 Ω
D 0.25 Ω E 100.0 Ω.
Which of the following alternatives is correct.
(I) A only; (II) A and B
(III) A and C (IV) A and D
(V) A and E.

3

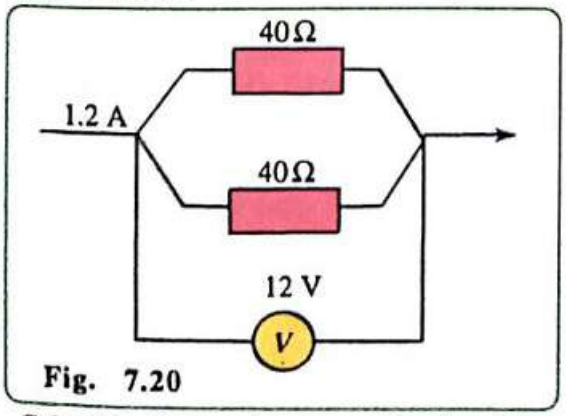


Fig. 7.20

Calculate the total energy developed in 5 minutes by the system above.
A 120 J B 144 J C 240 J
D 740 J E 8640 J

(SSCE)

4 The headlamp of a car takes a current of 0.4 ampere from a 12 volt supply. The energy produced in 5 minutes is
A 24 J B 240 J C 288 J
D 330 J E 1440 J

(SC/GCE)

5 Heat is developed in a wire through which a current is passed. On which of the following does the heat developed depend?
1 The time for which the current passes.
2 The square of the electric current.
3 The resistance of the wire.
4 The melting point of the wire.
A 1 and 2 only B 1, 2, and 3 only
C 1, 2, and 4 only D 2 and 3 only
E 1, 2, 3 and 4.

(SC/GCE)

6 An electric lamp is marked 240 volts, 60 watts. What is its resistance when it is operated at the correct voltage?
A $\frac{1}{960}$ B $\frac{1}{4}$ C 4
D 960 E 14,400

(SC/GCE)

7 What do you understand by an *electric field, electric lines of force, uniform field, variable field*?

8 Sketch the pattern of the lines of force around
(i) an isolated positive charge
(ii) an isolated negative charge
(iii) two positive charges separated from each other
(iv) two negative charges separated from each other·
(v) a positive and a negative charge.

9 Name four properties of electric lines of force. Why do such lines not intersect?

10 Define
a. Electric current
b. Potential difference
c. Electromotive force of a cell
d. Open circuit
e. Closed circuit.

11 State Ohm's law and define resistance. Describe an experiment that demonstrates Ohm's law. How can you obtain the resistance of a short piece of wire using Ohm's law?

12 Four resistors of resistances 1 Ω, 2 Ω, 3 Ω, 5 Ω, are connected in series and afterwards in parallel. Calculate the combined resistance in each case.

13 Calculate the combined resistance of the following combinations shown below

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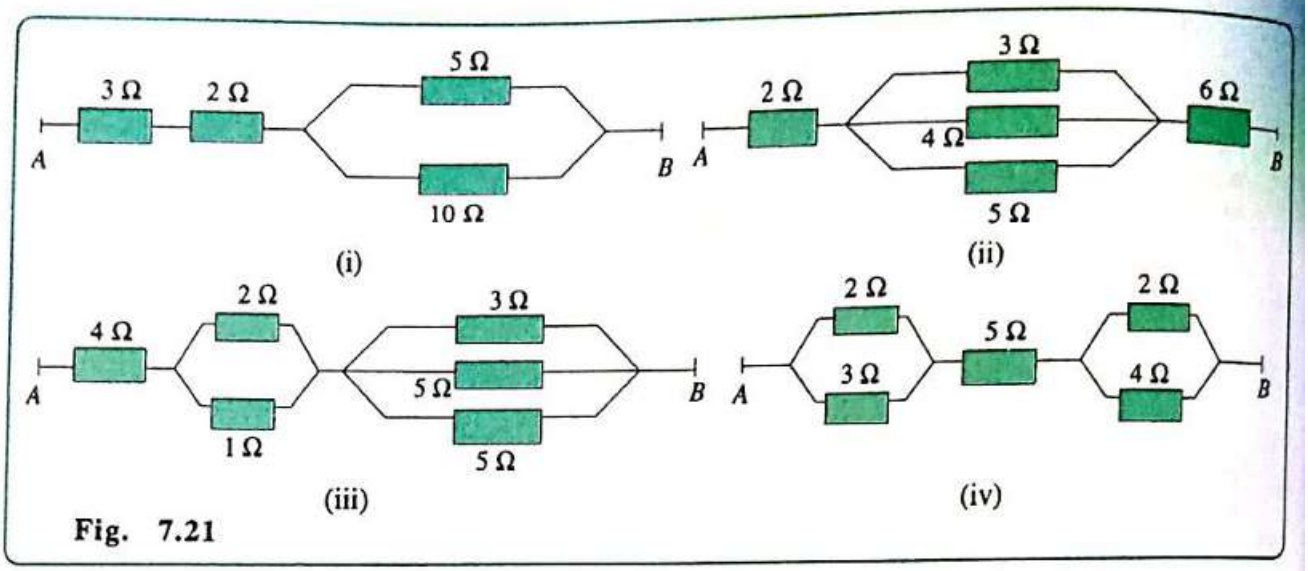


Fig. 7.21

14 Complete the following table assuming that the pieces of apparatus follow Ohm's law.

Apparatus	Current in A	P.d. in V	Resistance in Ω	Power in W
Voltmeter		4	600	
Ammeter	5	0.06		
Power line	40		4	
Torch bulb		2	8	
Electric iron		220		500
Mains lamp	0.5			100

15 Define the volt, the watt and the kilowatt-hour. An electric lamp is marked '100 W, 220 V'. What do you understand by this? What current is taken by the lamp when it is connected to a 220 V mains supply? What is the resistance of the lamp?

16 A television (65 W), a refrigerator (1.2 kW), an electric kettle (650 W) and 10 lamps (40 W each) are connected in series in a house. How much will it cost the housekeeper to switch on all the appliances for 24 hours if the cost of electricity is 15 kobo per kWh?

17 Describe two defects of a simple cell. How can each named effect be minimized?

18 How can electric energy be generated from
(i) chemical energy
(ii) heat energy
(iii) mechanical energy and
(iv) solar energy.

19 Explain the terms *thermoelectric effect*, *thermocouple*. Distinguish between *conductors* and *insulators*. Give 3 examples of each.

20 Define electrical resistance. What factors affect the resistance of a conductor?

21 What is a fuse and what is its use in an electric circuit?

22 List the dams for the production of electricity in Nigeria.

23 Describe how electricity is produced from a dam. Illustrate with a sketch diagram.

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8 PARTICLE NATURE OF MATTER

In this chapter, we shall introduce the kinetic molecular theory of matter which postulates that matter is composed of molecules which are in incessant motion. The chapter discusses the evidence of the particle nature of matter, simple atomic structure, the nature and size of molecules and the three states of matter.

At the end of the chapter the student should be able to:

1 Explain how molecules of a substance move relative to other molecules of the same substance.
2 Describe the atomic structure of matter.
3 State the constituents of the atom.
4 Estimate the size of the molecules of a substance in the form of a solid, liquid or gas.
5 Use molecular theory to explain the three states of matter.

8.1 Structure of Matter

a. Evidence of the particle nature of matter

The idea that matter is made up of minute particles called *atoms* dates back to the ancient Greeks. According to the Greek philosopher Democritus, a given piece of substance, say a piece of yam, can be cut into smaller and smaller bits, until eventually a smallest piece of that substance would be obtained which could not be further subdivided. This smallest, indivisible, piece was called an *atom*. The *atomic theory* of matter assumes that all matter is made up of tiny particles called atoms and that these are at all times in a rapid state of motion. The nature of this motion and its activity depend upon the temperature of matter and other factors.

The experimental evidence of this particle or atomic nature of matter is the *Brownian movement*, named after the Biologist Robert Brown who was credited with its discovery in 1827. While observing tiny pollen grains suspended in water under a microscope, he noticed that the tiny pollen grains moved about in zig-zag paths even though the water appeared to be perfectly still. These pollen grains were supposed to be jostled or knocked about here and there by the vigorously moving molecules of water.

Another evidence in favour of the particle nature of matter was obtained from analysis of chemical reactions. This crucial piece of evidence is known by chemists as the *Law of Definite Proportions* which states that when two or more elements combine to form a compound, they always do so in the same proportions by weight. For example, the compound water (H_2O) is always formed by two parts of hydrogen (H_2) and sixteen parts of oxygen (O) and the common salt (NaCl) is always formed from 23 parts of sodium and 35 parts of chlorine by weight.

A third evidence that matter is composed of extremely small particles is provided by the process of *diffusion*. If we place a few drops of liquid bromine at the bottom of a gas jar on top of which is placed a cover glass, after some time the brown bromine vapour will be seen in the upper part of the gas jar in spite of the fact that bromine vapour is much denser than air and should remain at the bottom of the jar. We can explain this diffusion or spreading out of the bromine by assuming that liquid bromine is composed of particles and that the particles can move about easily.

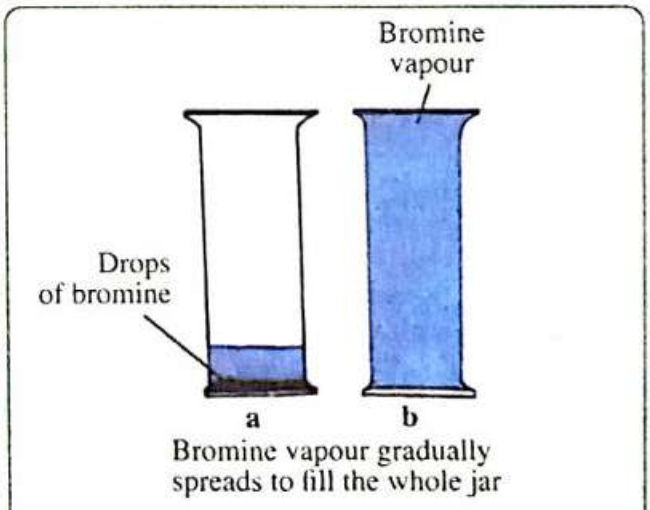


Fig. 8.1 Diffusion of bromine vapour

All the evidence we have given that matter is composed of particles are indirect evidence. We have no direct evidence for the existence of these particles since we cannot observe them with even the most powerful microscopes.

b. Simple atomic structure

An atom is the smallest indivisible particle of an element which can take part in chemical change.

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Activity 8.1

1 Get a tuber of yam and cut out a piece of this tuber about the size of a cube of sugar.
2 Divide this cube into two halves.
3 Take one of these halves and cut it into another half. Continue this process of cutting each successive half into two until you can no longer cut the remaining portion into further bits.

If we suppose the piece of yam to represent an element (e.g. sodium) the smallest bit of the yam represents the atom of the element. The smallest particle of an element, in order to be an atom of that element must be capable of a separate existence.

There are many theories concerning the structure of the atom. We shall discuss them in Book 3. For now we shall consider the atom as simply consisting of two parts held together by electric forces. The two parts of the atom are (1) *the nucleus* and (2) *the electrons*.

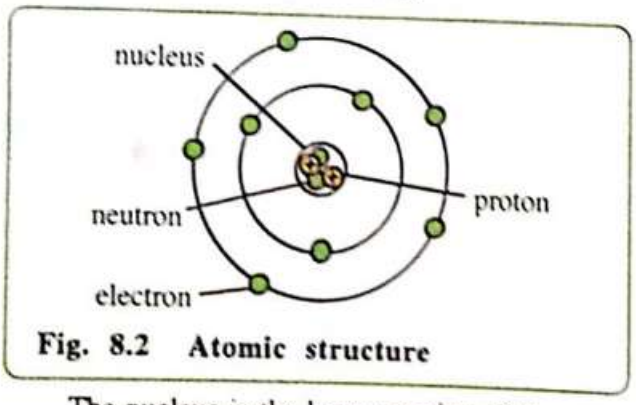


Fig. 8.2 Atomic structure

The nucleus is the heavy portion of the atom and is located at its centre. It consists of two parts – the *protons* and the *neutrons*. The protons carry a positive charge, the neutrons carry no charge.

The second part of the atom is the electron. The electrons are very light (about 1/1840 of the mass of the proton). They are negatively charged. The lightness of electrons makes it easy to transfer them when two materials are made to rub against each other. The electrons circle in orbits around the heavy nucleus and are held in place due to the electrostatic attraction between them and the protons of the nucleus.

In a neutral atom, the total charges due to the electrons must balance the total charges due to the protons.

8.2 Molecules

Most substances cannot exist by themselves as individual atoms. They combine their atoms with themselves or with other atoms to form molecules.

A molecule is the smallest particle of a substance which can have a separate existence and still retain the properties of that substance.

The molecules of any one substance are identical. They have the same structure, the same mass and the same mechanical and electrical properties.

A molecule may be made up of similar atoms of the same element or different atoms of two or more elements. For example a molecule of hydrogen is made up of two atoms of hydrogen but a molecule of water is a combination of two atoms of hydrogen and an atom of oxygen, and a molecule of sodium chloride is a combination of an atom of sodium and an atom of chlorine.

Atoms combine in simple proportions to form molecules. The simplest model of a molecule is that of a rigid sphere, like a small billiard ball, capable of moving and colliding with other molecules or with a wall, and of exerting attractive or repulsive forces on neighbouring molecules.

The size of a molecule

The size of a molecule is extremely small. It is of the order of $10^{-9} - 10^{-10}$ m ($10^{-7} - 10^{-8}$ cm). As a result of this small size, one gram of an element contains several millions of molecules. For example a gram of hydrogen contains about 10^{23} molecules.

To estimate the size of a molecule

It was Lord Raleigh, in 1890, who first estimated the approximate size of a molecule. He argued that when a drop of oil is placed on top of a water surface, the oil will spread out on top of the water until the thickness of the film of oil was one molecule thick.

We can estimate the size of an oil molecule by a similar experiment in the following activity.

Activity 8.2

1 Pour clean water into a shallow dish and sprinkle some lycopodium powder lightly on the water.
2 Obtain a graduated pipette with a fine bore, and with it take up a small volume of an oil (e.g. olive oil) and note the reading on the pipette scale.

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3 Drop a very small quantity of the oil onto the water in the dish. Note again the reading of the pipette scale. Subtract this reading from the first reading to obtain the volume V of oil used in the experiment.

4 Allow the oil to spread, pushing the lycopodium powder outwards and forming a clear thin film of oil on top of the water surface.

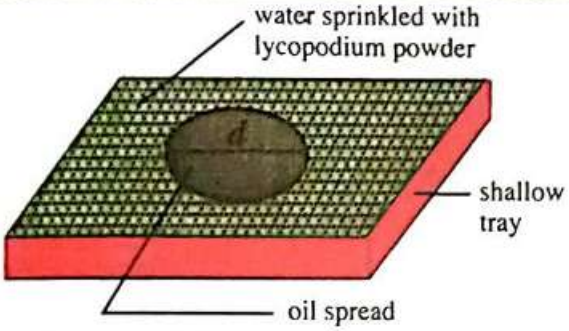


Fig. 8.3 Oil film experiment

5 Measure the diameter of the oil patch to the nearest centimetre using a half-millimetre scale. Calculate the thickness of the oil film as follows.

Let diameter of oil film = d (cm)

Let volume of oil drop = V (cm^3)

$$\text{Area of oil film} = \pi\left(\frac{d}{2}\right)^2 \text{ cm}^2$$

$$\therefore \text{ thickness of film} = \frac{\text{volume}}{\text{area}}$$

$$= \frac{V}{\pi\left(\frac{d}{2}\right)^2}$$

$$= \frac{4V}{\pi d^2} \text{ cm}.$$

Accurate experiments lead to a value of about 2×10^{-7} cm as the thickness of the oil film. Hence we take the size of an oil molecule as about 2×10^{-7} cm.

Incessant motion of molecules

Apart from their small size, another important fact about molecules is their constant motion. We can have an idea of this constant motion of particles by observing the rapid and chaotic motion of dust particles seen in a beam of light entering a dusty room through a window. We can also observe the movement of smoke particles using a smoke cell and a microscope.

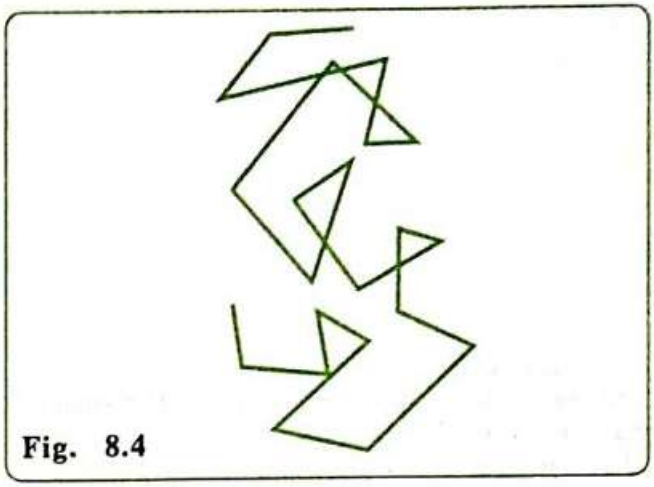


Fig. 8.4

The smoke-cell consists of a small glass cell with a cover. The cell contains a little smoke from a smouldering piece of cord or string. The cell is strongly illuminated from the side by a light which is focused into the interior of the smoke cell using a focusing lens. The smoke cell is then viewed through the microscope. The illuminated smoke particles are observed as bright spots of light undergoing a zig-zag movement like that of a drunken man staggering about.

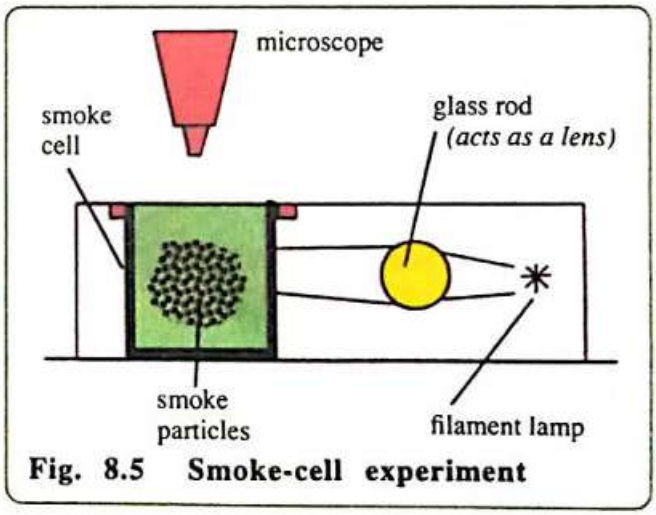


Fig. 8.5 Smoke-cell experiment

The irregular movements of the smoke particles is caused by rapidly moving but invisible air molecules which bombard them from all sides. Because of the smallness of the smoke particles, the number of air molecules striking it on one side is not balanced by the number striking it on the opposite side. There is therefore always a resultant force and the particle moves in the direction of this resultant force. This is repeated when the smoke particle is pushed to another place and hence the zig-zag motion continues. This irregular and constant motion of molecules is known as

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Brownian motion. It was first observed by Robert Brown, a botanist, in 1827.

Brownian motion is the rapid, constant, and irregular motion of tiny particles.

Brownian motion is considered important for two reasons. Firstly, it gives evidence for the existence of the tiny particles of matter called molecules. Secondly, it gives evidence that these molecules are in a constant state of random motion.

Brownian motion also takes place in liquids and in solids. In fact Robert Brown first observed such motion while examining pollen grains in water under a microscope. Here the motion is due to the bombardment of the solid particles by the water molecules. Molecules in solids such as camphor are able to leave the solid, and spread into the air and to our nostrils where their presence can be detected.

The fact that the tiny particles which make up matter are in constant motion is also illustrated in other natural phenomena such as *diffusion* and *osmosis*.

Diffusion When a hydrogen sulphide gas is released in one end of the laboratory, the rotten-egg smell is soon perceived in the entire laboratory space and beyond. The molecules of the gas have travelled throughout the laboratory and filled the whole space with their smell.

We can demonstrate diffusion in liquids by dropping some blue copper sulphate crystals at the bottom of a tall cylinder containing water. A dense blue solution forms at the bottom of the cylinder but the water above this solution still remains clear.

If the cylinder is now put away where it will be undisturbed, after some days it will be observed that the blue colour has spread throughout the entire water. The copper sulphate molecules have sufficient energy to enable them to spread upwards against the force of gravity. The molecules of copper sulphate crystal have spread randomly throughout the liquid through the process of *diffusion*.

In diffusion, the diffusing molecules move from the places where their concentration is high to places where their concentration is low.

Diffusion is the tendency of molecules to migrate and fill an empty space due to their random thermal motions.

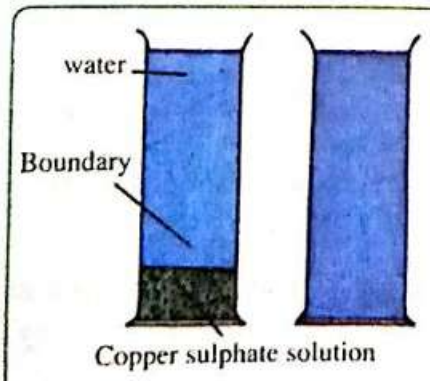
Diffusion is due to movements of molecules and it takes place in gases and liquids and quite slowly in solids. Its rate depends on the density of the gas and its temperature. Light gases at high temperatures diffuse faster than heavy gases at low temperatures.

8.3 States of Matter

Matter can exist in three states – *solids*, *liquids* and *gases*.

Solids

The molecules that constitute a solid are very closely packed. They are held rigidly together by intermolecular forces of cohesion. These molecules are on the average, anchored to fixed positions but can vibrate about these fixed positions. Thus solids have definite shapes and volumes, and cannot be



Spread of colour shows that copper sulphate molecules have diffused throughout the water

Fig. 8.6 Diffusion in liquids

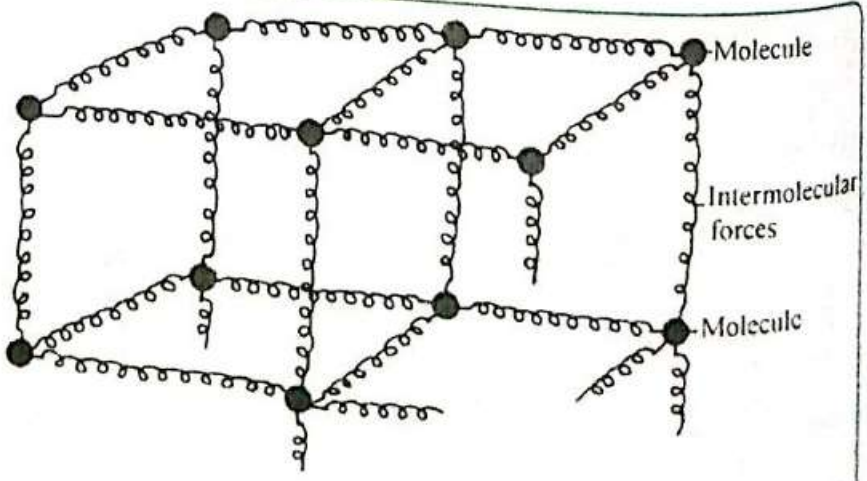


Fig. 8.7 Molecular structure of solids

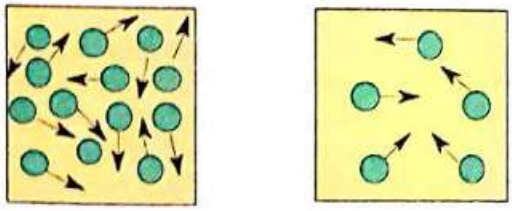
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poured. When a solid is heated, the molecules acquire more energy and vibrate faster and farther away from their average position. At a certain temperature known as the *melting point*, the molecules have acquired enough kinetic energy as to break the intermolecular forces binding them together. The molecular structure of the solid then collapses and the molecules thus freed, move farther apart and become liquids.

Liquids

In liquids, the molecules are freer to move about because the intermolecular forces binding them are weaker than those in solids. The molecules of a liquid are also closely packed together but can move about randomly. A liquid therefore has no definite shape or form but still possesses a fixed volume. It always assumes the shape of its container. Liquids can be poured.

When a liquid is heated, its molecules gain kinetic energy. With increasing temperature the molecules gain enough energy to overcome the intermolecular cohesive forces and then escape from the liquid in the process of *evaporation*.



liquid gas

Fig. 8.8 Molecular structure of liquids and gases

Gases

In gases, the cohesive forces binding the molecules are negligible. These molecules are therefore very free to move about in all directions, restricted only by the walls of the container. A gas has no definite shape and occupies the whole volume of its container. Gases are more easily compressed than liquids and solids. This is because the molecules are relatively far apart.

Summary

1 The evidence of the molecular nature of matter comes from
   a. Brownian motion
   b. law of definite proportion
   c. diffusion
   d. osmosis.
2 The atom consists of a central massive nucleus surrounded by light electrons moving in orbits around the nucleus. The nucleus itself consists of protons and neutrons. Protons are positively charged, electrons are negatively charged and neutrons carry no charge.
3 A molecule is the smallest particle of a substance which can have a separate existence and still retain the properties of that substance.
4 The size of a molecule is about 10^{-9}–10^{-10} m.
5 The three states of matter are solids, liquids and gases. In solids the molecules are closely held together by intermolecular forces and they vibrate about a mean fixed position. In liquids the intermolecular forces are weaker than in solids and the liquid molecules are therefore freer to move. In gases, molecules are very free to move about because the intermolecular forces are very weak.
6 The kinetic energy of the molecules constituting matter increases with temperature.

Exercise 8

1 It is known that a neutron exists in a light atomic nucleus. Which of the following also exists in the nucleus?
   A An electron B A β-particle
   C An α-particle D A proton
   E An atom.

   (JAMB)
2 Which of the following statements is not correct.
   A Matter is made up of molecules
   B The molecules of matter are in constant motion
   C Brownian motion is an evidence of particle nature of matter
   D Molecules of a liquid are stationary
   E Atoms combine to form molecules.
3 A drop of oil of volume 10^{-10} m^3 spreads out on water to make a circular film of radius 10^{-1} m. What is the thickness of the film?
   A 3.2×10^{-9} m B 3.2×10^{-8} m
   C 1.6×10^{-9} m D 1.6×10^{-10} m
   E 3.12×10^{9} m.
4 Describe an experiment you can perform to observe *Brownian motion* in the laboratory. Describe briefly what is observed and the cause of the observed motion.
5 Give a brief description of the structure of the atom.
6 How would you estimate the size of a molecule of an oil?

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7 Distinguish between the three states of matter.

8 What do you understand by *Brownian motion*? What is its significance in the theory of matter. Why is it only exhibited by small particles?

9 Outline the difference between solids, liquids and gases in terms of
   a. the separation of the molecules
   b. the motion of the molecules
   c. the arrangement of the molecules throughout the bulk of the material.

10 What do you understand by *diffusion*? Describe an experiment to demonstrate the phenomenon in the laboratory. What is the significance of diffusion in the molecular theory of matter?

11 What are the three states of matter? How do these states differ from each other? Which in general is the most compact?

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9 ELASTIC PROPERTIES OF SOLIDS

In the last chapter dealing with the particle nature of matter, we stated that the molecules that constitute solids are very closely packed together and held in relatively fixed positions by strong intermolecular forces. As a result of this, it is relatively difficult to displace these molecules because any external force applied on the solid is resisted by these intermolecular forces. It is not so in liquids and gases.

If the external force applied to a solid is strong enough to change or distort the shape of a solid, the solid tends to regain this shape as soon as the applied force is removed. Hence when you stretch a piece of rubber band, it quickly snaps back to its former shape once it is released. A ruler when bent will return to its former shape when the bending force is removed. When some load is suspended from the end of a coil spring of wire, the spring stretches a certain length. When the load is removed, the spring returns to its former length. Such objects which regain their shape and size after the force causing the change is removed are said to be *elastic*, and this property of solids is known as *elasticity*.

Elasticity is the ability of a substance to regain its original shape and size after being distorted by an external force.

An **elastic material** is one that regains its original shape and size after the distorting force has been removed.

We can distort the shape and size of a material by stretching or compressing it. Any elastic material recovers its original size and shape as soon as the stretching or compressing force is removed.

In this chapter we shall study **Hooke's law** – a law that governs the relationship between the stretching force and the extension produced in the elastic material. We shall also give a qualitative treatment of **Young's modulus**, and a quantitative treatment of the work done in stretching or compressing springs and elastic strings.

At the end of the chapter, the student should be able to:

1 Explain the elastic properties of substances.
2 State Hooke's law and show an understanding of the terms stress, strain, elastic limit, Young's modulus.
3 Solve simple problems on work done in stretching or compressing a spring and elastic strings.

9.1 Hooke's Law

When a force is applied to an elastic spring or string, the length of the material increases by a certain amount. If the force is removed, the length returns to its former value. When the stretching force is doubled, the extension of the spring doubles. This relationship between the stretching force and the extension is given by *Hooke's law* which holds until a certain limit of force is reached.

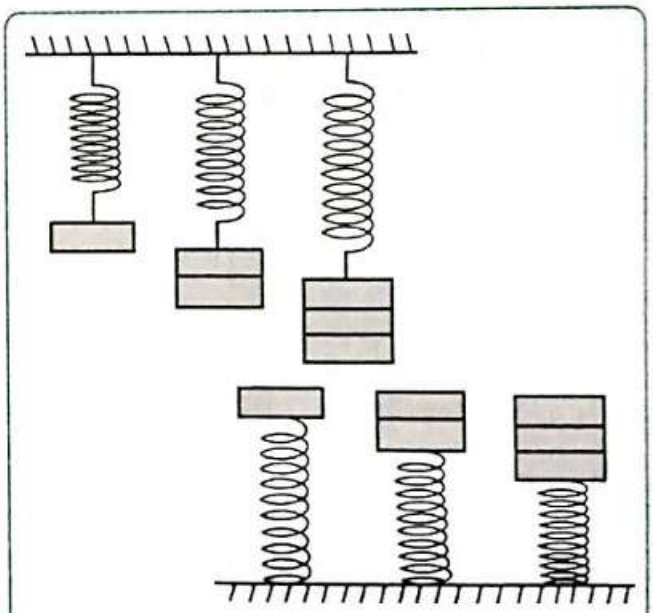


Fig 9 The extension or compression of a spring is directly proportional to the applied force

Hooke's law states that provided the elastic limit of an elastic material is not exceeded, the extension, e, of the material is directly proportional to the load or applied force, F.

We can demonstrate the relationship between the extension and the stretching force by the following experiment.

Activity 9.1: To verify Hooke's law and determine the elastic constant of a spring

1 Obtain a spiral spring from the school laboratory or make one yourself by wrapping about 2 m of a thin bronze or iron wire closely around a biro pen.

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2 Attach one end to a firm support and to the other end attach a scale pan which has been previously weighed.

3 Attach a pointer, P, to the spring with plasticine and arrange a vertical scale over which the pointer movement can be read. Set up the entire apparatus as shown in figure 9.1.

4 Note the initial pointer reading on the scale and call it l.

5 Add weights to the scale pan in equal steps of about 0.05 kg and note and record the pointer reading for each increasing load. Note that the total load stretching the spring in each instance is the weight in the scale pan plus the weight of the scale pan.

6 Remove the load from the scale pan in equal steps also and note and record the corresponding pointer readings.

7 Record your readings as shown in table 9.1

Table 9.1

Load	Scale reading		Average reading	Extension e
	Load increasing	Load decreasing		
	x_1 (cm)	x_2 (cm)	$\frac{x_1 + x_2}{2} = X$ (cm)	$X - l$ (cm)

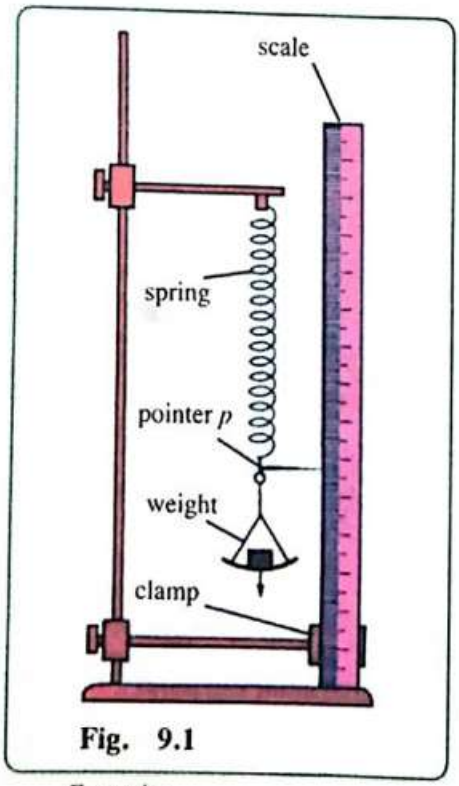


Fig. 9.1

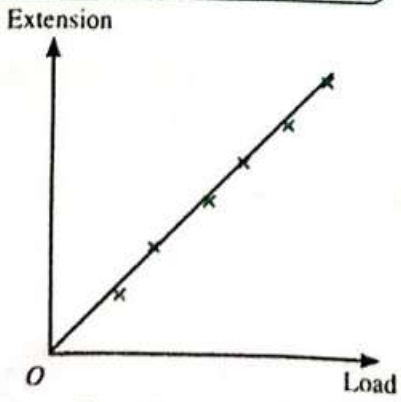
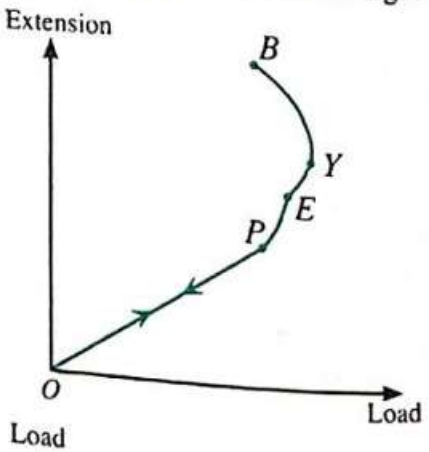


Fig. 9.2 Graph of extension against load. Hooke's law

8 Plot a graph of Extension against load. Provided the total maximum load does not extend the spring permanently, a graph of extension against load will give a straight line graph passing through the origin. This straight line graph shows that the extension of the spring is directly proportional to the load. Thus Hooke's law is verified.

If we continue increasing the load until the spring no longer returns to its original length when the load is removed, a graph of extension against load will be as shown in fig. 9.3

(a)



(b)

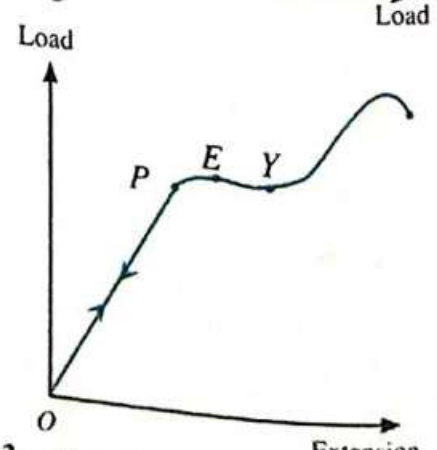


Fig. 9.3 Graph of extension against load up to breaking point

P = Proportionality Limit
E = Elastic Limit
Y = Yield Point

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OE = Elastic Region
EB = Plastic Region
B = Breaking Point

The graph is a straight line OP, followed by a curve $PEYB$ which rises slowly at first and then sharply (fig. 9.3a). Along OP, the straight portion of the graph, Hooke's law is obeyed and the extension is proportional to the force applied. P is known as the *proportionality limit*. Nonetheless, up to a point E (slightly above P) called the *elastic limit*, the object will return to its original length if the applied force is removed.

Elastic limit is the limit of force beyond which the stretched wire does not return to its original length when the stretching force is removed.

That is, *beyond the elastic limit, the material losses its elasticity*. The region OE, from the origin to the elastic limit is called the *elastic region*. Distinguishing between the *proportionality limit* and *elastic limit*, we see that strictly speaking Hooke's law holds up to P, the proportionality limit, but elasticity holds up to the elastic limit.

Beyond the elastic limit E the wire stretches rapidly with increasing load and enters the plastic region. In this region the wire does not return to its original length upon removal of the stretching force. For example at a point 'Y', known as *yield point* the material is said to have 'yielded' or lost all elasticity permanently and has become *plastic*.

Yield point is the point beyond the elastic limit in which the elastic material has yielded all its elasticity permanently and has become plastic.

The change from elastic to plastic region is shown by a sudden and rapid increase in the extension for any slight increase in load. The maximum extension is reached at the *breaking point* B where the material may finally 'snap' or break entirely.

According to Hooke's law, the force F required to stretch or compress an elastic material is directly proportional to the extension, or compression e. Thus:

$$F \propto e$$

or

$$F = ke \qquad 9.1$$

where k is a constant of proportionality called *elastic constant* or *force constant* or *stiffness* of the material. If F is in newtons and e is in metres the elastic constant is given by

$$k = \frac{F}{e} \ (\text{Nm}^{-1}) \qquad 9.2$$

Elastic constant or **stiffness** of an elastic material is the force required to give unit extension.

The working of the spring balance is based on Hooke's law. Here $F = mg$, the weight of the body which is proportional to e, the extension of the spring.

Example 9.1

A force of 0.8 N stretches an elastic spring by 2 cm. Find the elastic constant of the spring.

Solution

From Hooke's law

$$F = ke$$

$$0.8 \text{ N} = \frac{k \times 2 \text{ m}}{100} = k \times 0.02 \text{ m}$$

$$k = \frac{0.8}{0.02} = \underline{40 \text{ Nm}^{-1}}$$

Example 9.2

A force of 2 N stretches an elastic material by 30 mm. What additional force will stretch the material 35 mm? Assume that the elastic limit is not exceeded.

Solution

$$F = ke$$

$$2 = k \times \frac{30}{1000}$$

$$k = \frac{2000}{30} \text{ Nm}^{-1}$$

Let the force stretching the material 35 mm be F

$$F = k \times \frac{35}{1000} = \frac{2000}{30} \times \frac{35}{1000}$$

$$= 2.33 \text{ N.}$$

$\therefore$ additional force = 2.33 N – 2.00 N

$$= \underline{0.33 \text{ N}}$$

Young's modulus

Let a wire of length l (m) and cross-sectional area A (m^2) be extended through e (m) by a force F (N). *The ratio of the force to the area* $\frac{F}{A}$ *is called the stress or tensile stress* of the material.

The unit is Nm^{-2}.

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